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(54) **SMART THERMOSTAT WITH MODEL
PREDICTIVE CONTROL AND DEMAND
RESPONSE INTEGRATION**

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a continuation of application No. 15/625,830, filed on
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(71) Applicant: **Tyco Fire & Security GmbH**,
Neuhausen am Rheinfall (CH)

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28, 2017.

(72) Inventors: **Kerry M. Bell**, Mukwonago, WI (US);
Bridget E. Kapler, New Berlin, WI
(US); **Alan S. Schwegler**, Shorewood,
WI (US); **Leyla Mousavi**, Milwaukee,
WI (US); **Kierstyn R. Robbins**,
Waukesha, WI (US); **Robert D.**
Turney, Watertown, WI (US); **Matthew**
J. Ellis, Milwaukee, WI (US); **Michael**
J. Wenzel, Oak Creek, WI (US);
Mohammad N. ElBsat, Milwaukee, WI
(US); **Juan Esteban Tapiero Bernal**,
Wauwatosa, WI (US); **Brennan H.**
Fentzlaff, Oconomowoc, WI (US)

(73) Assignee: **Tyco Fire & Security GmbH**,
Neuhausen am Rheinfall (CH)

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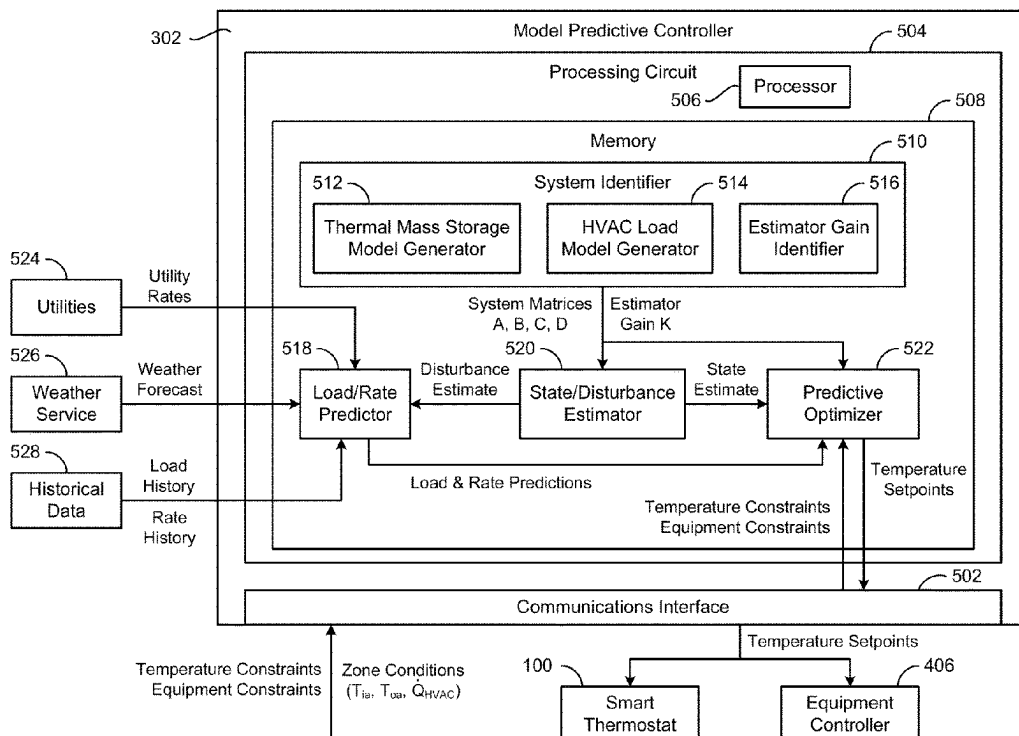
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division of application No. 16/588,467, filed on Sep.
30, 2019, now Pat. No. 11,274,849, which is a con-
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F24F 140/60 (2018.01)
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2130/10 (2018.01); **F24F 2140/60** (2018.01)

(57) **ABSTRACT**

A system includes a plurality of thermostats corresponding
to a plurality of HVAC systems that serve a plurality of
spaces and a computing system communicable with the
plurality of thermostats via a network. The computing
system is configured to, for each space of the plurality of
spaces, obtain a set of training data relating to thermal
behavior of the space, identify a model of thermal behavior
of the space based on the set of training data, perform a
model predictive control process using the model of thermal
behavior of the space to obtain a temperature setpoint for the
space, and provide the temperature setpoint to the thermostat
corresponding to the HVAC system serving the space. The
plurality of thermostats are configured to control the plural-
ity of HVAC systems in accordance with the temperature
setpoints.



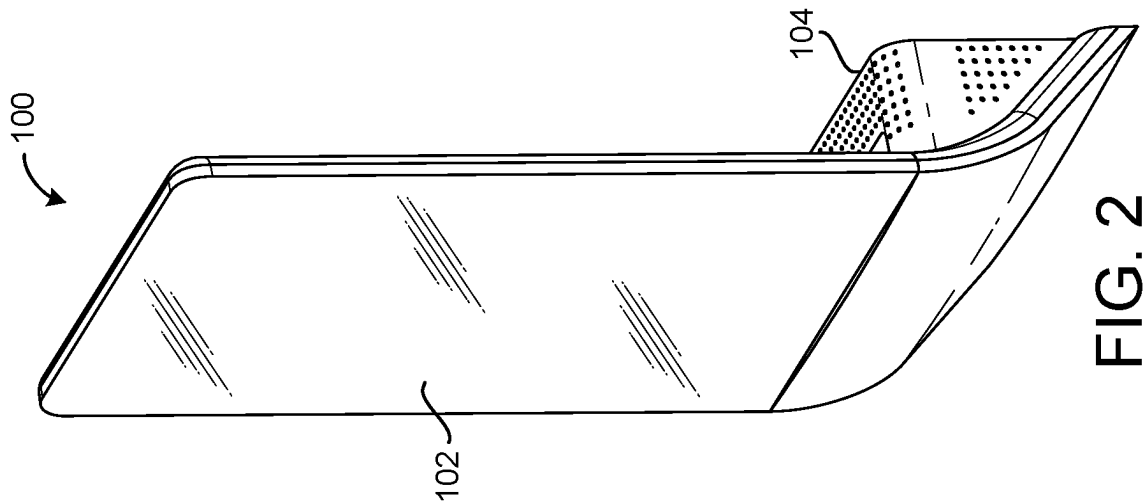


FIG. 2

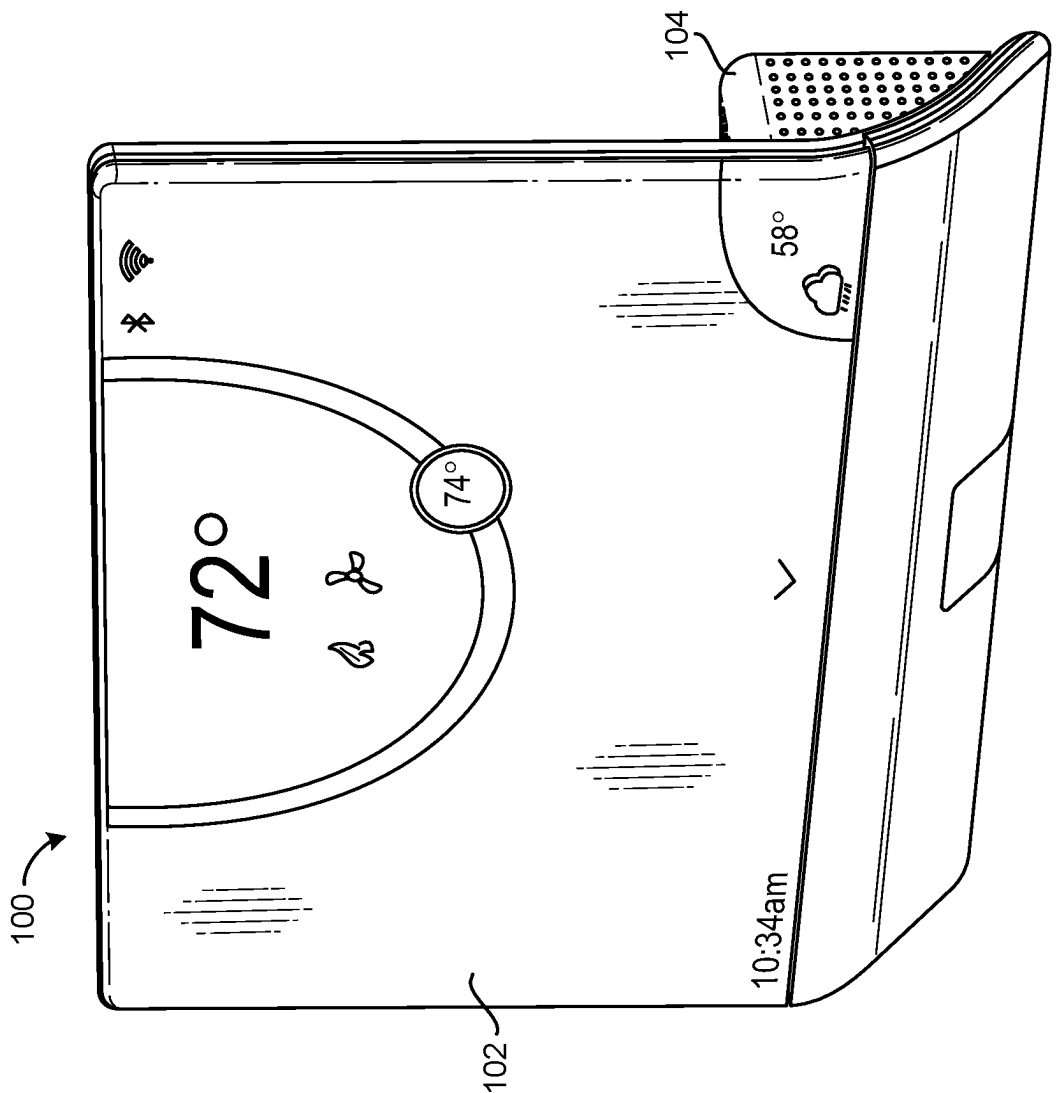


FIG. 1

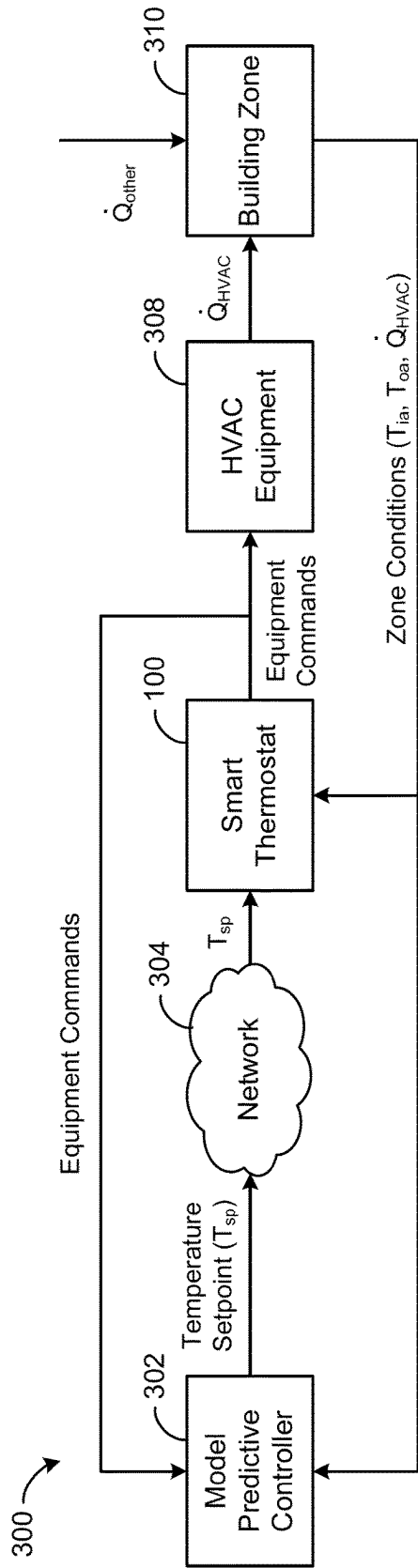


FIG. 3

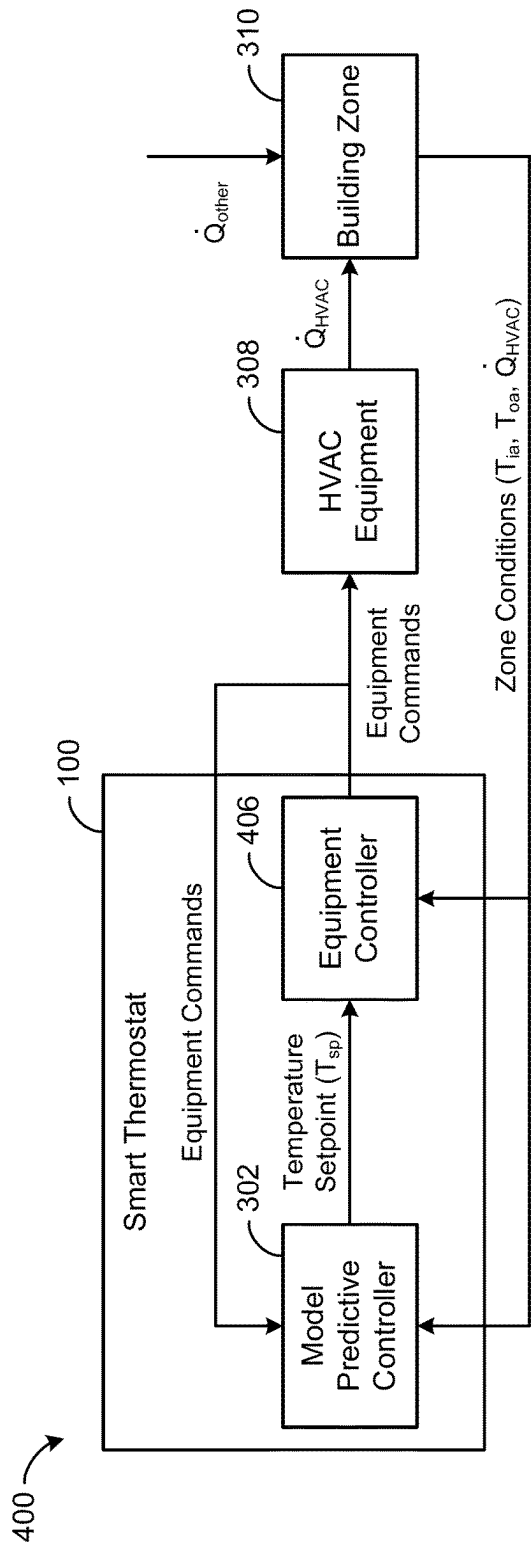
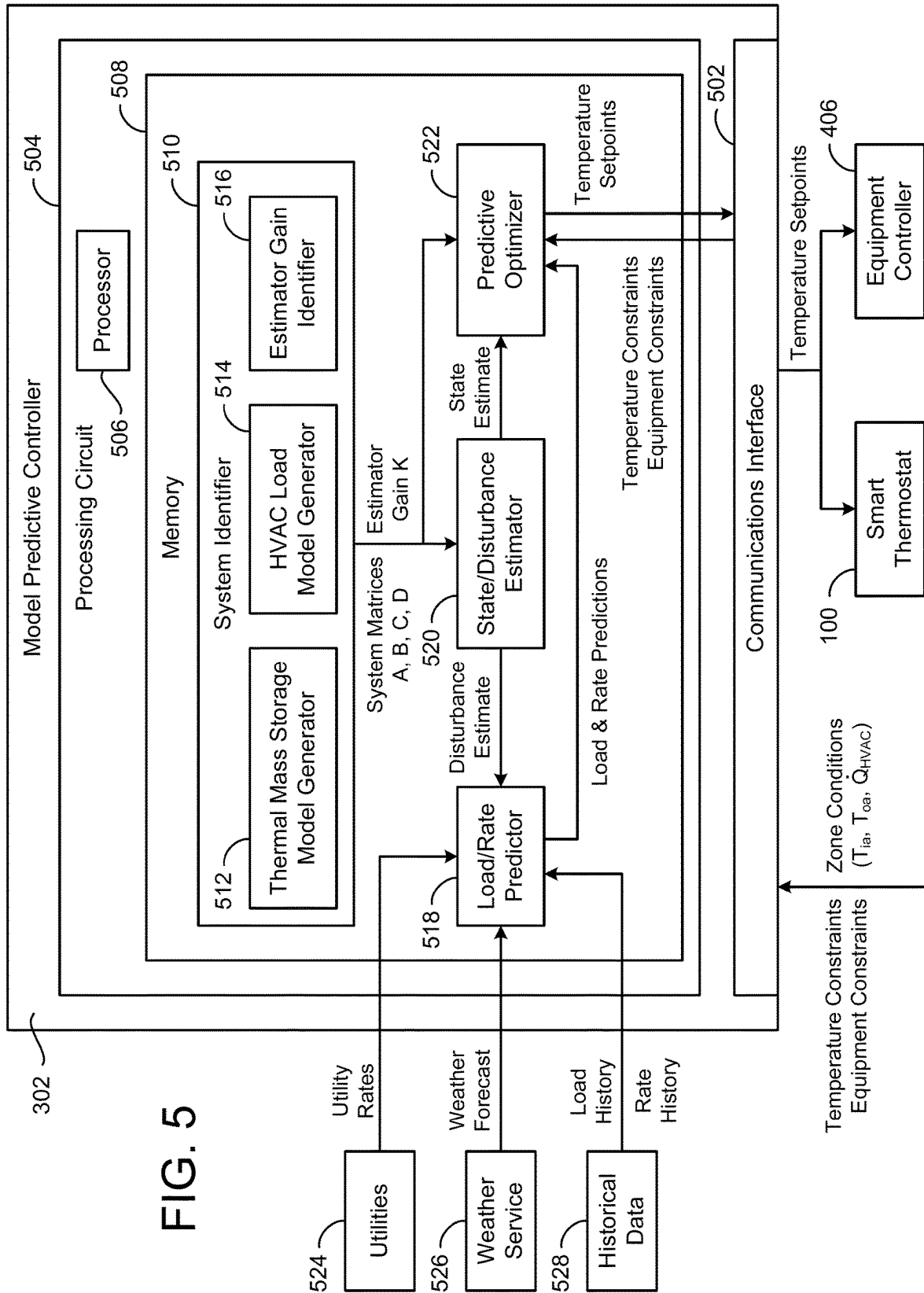


FIG. 4



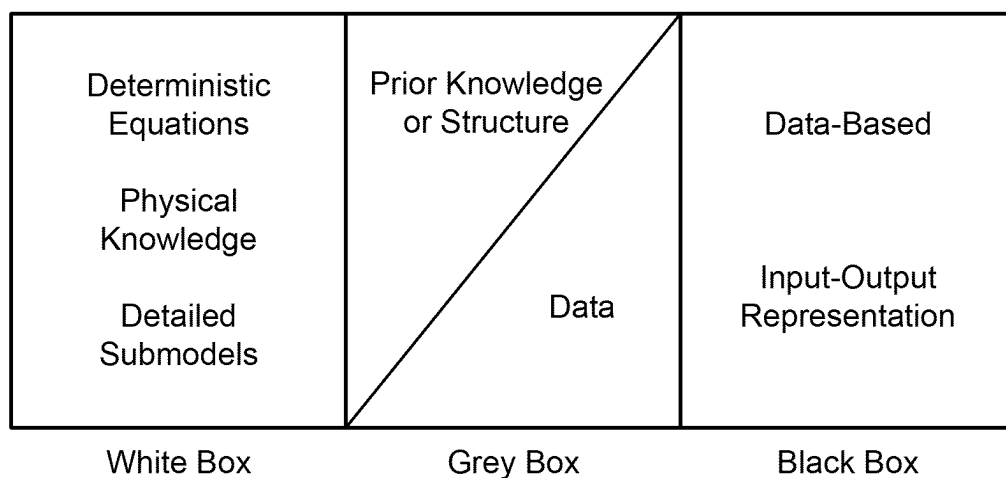


FIG. 6

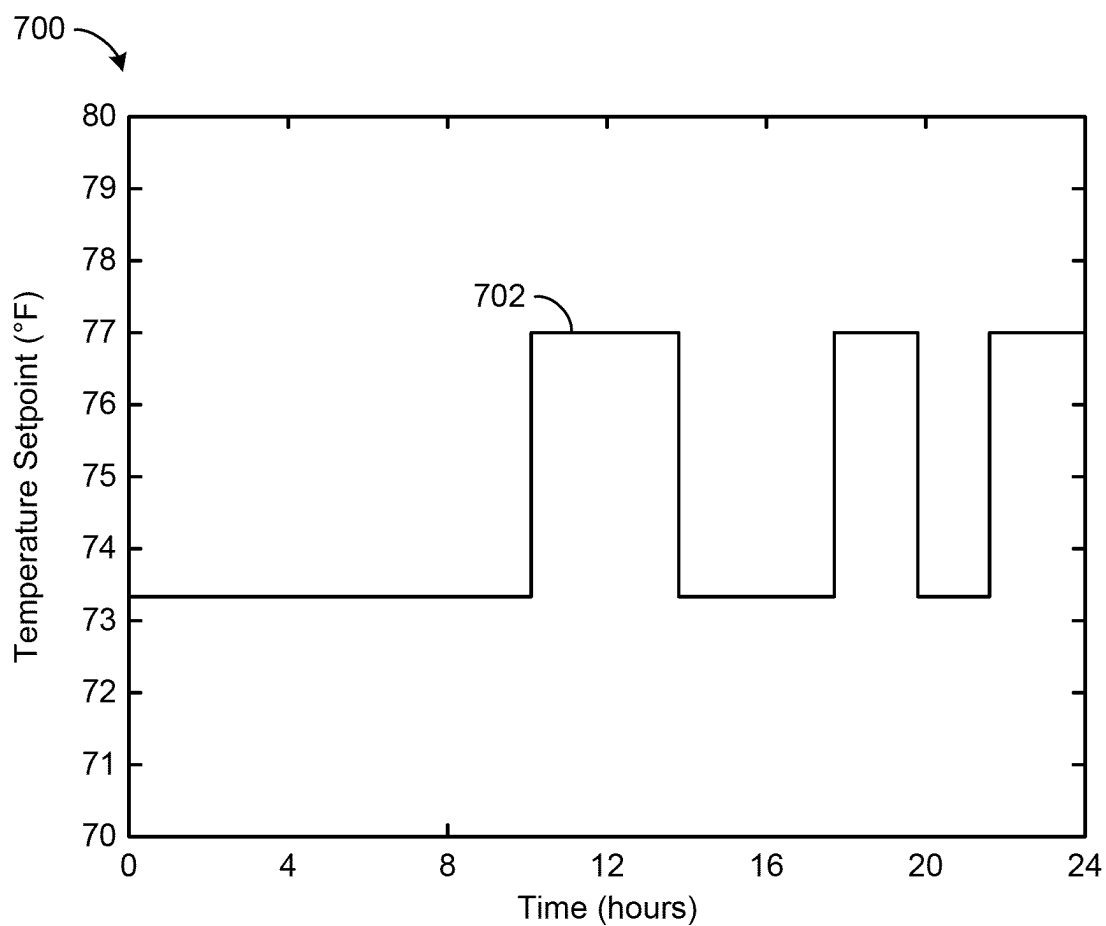


FIG. 7

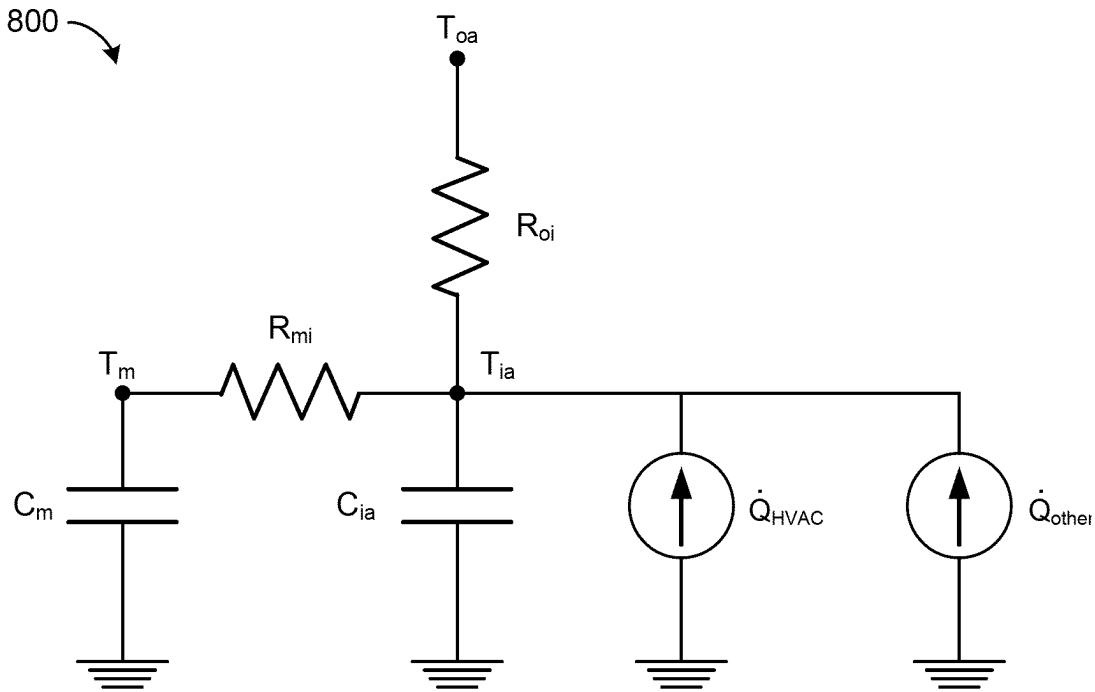


FIG. 8

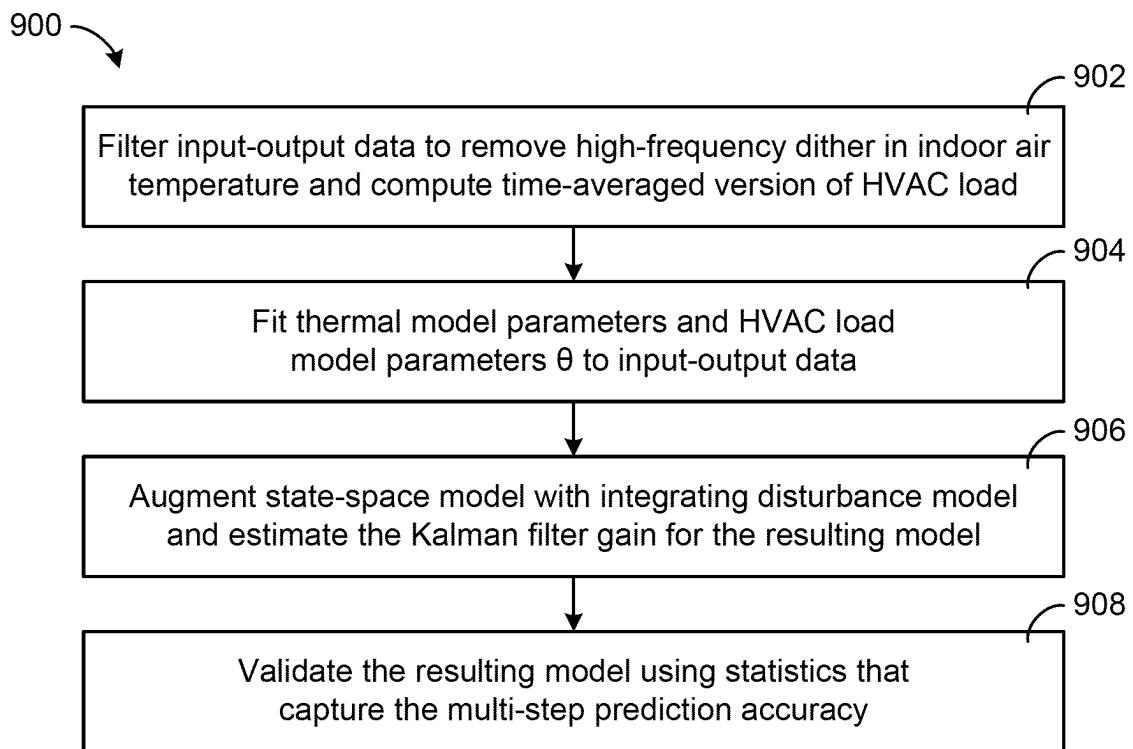


FIG. 9

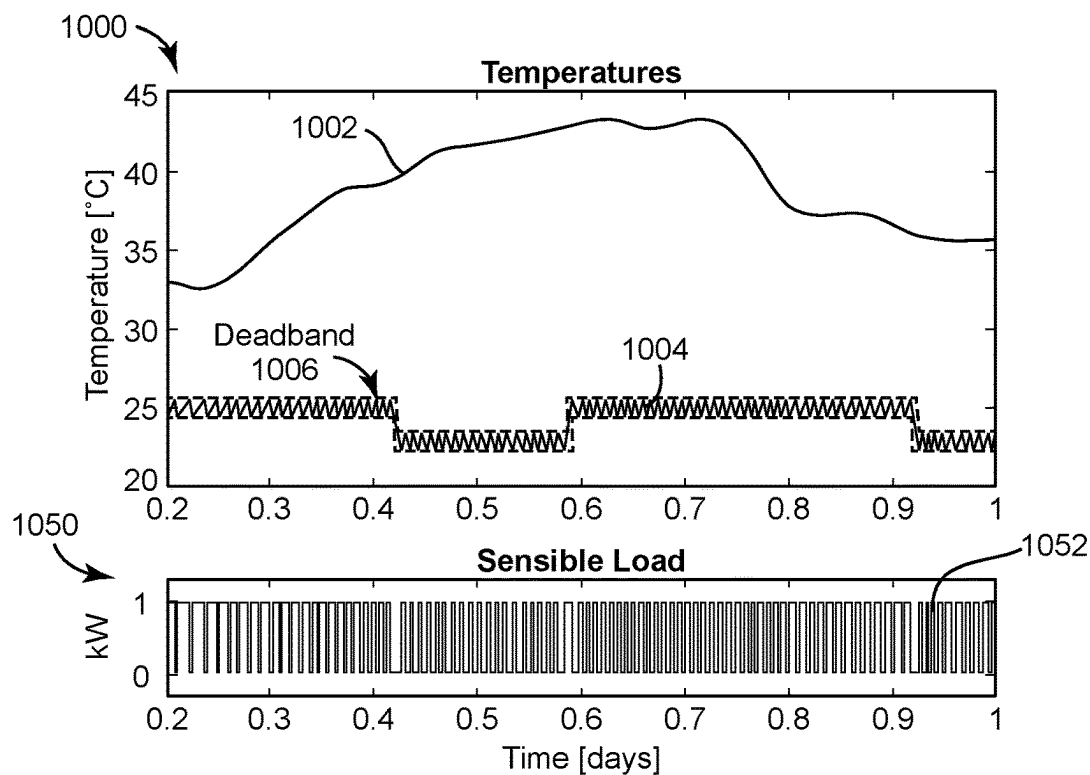


FIG. 10

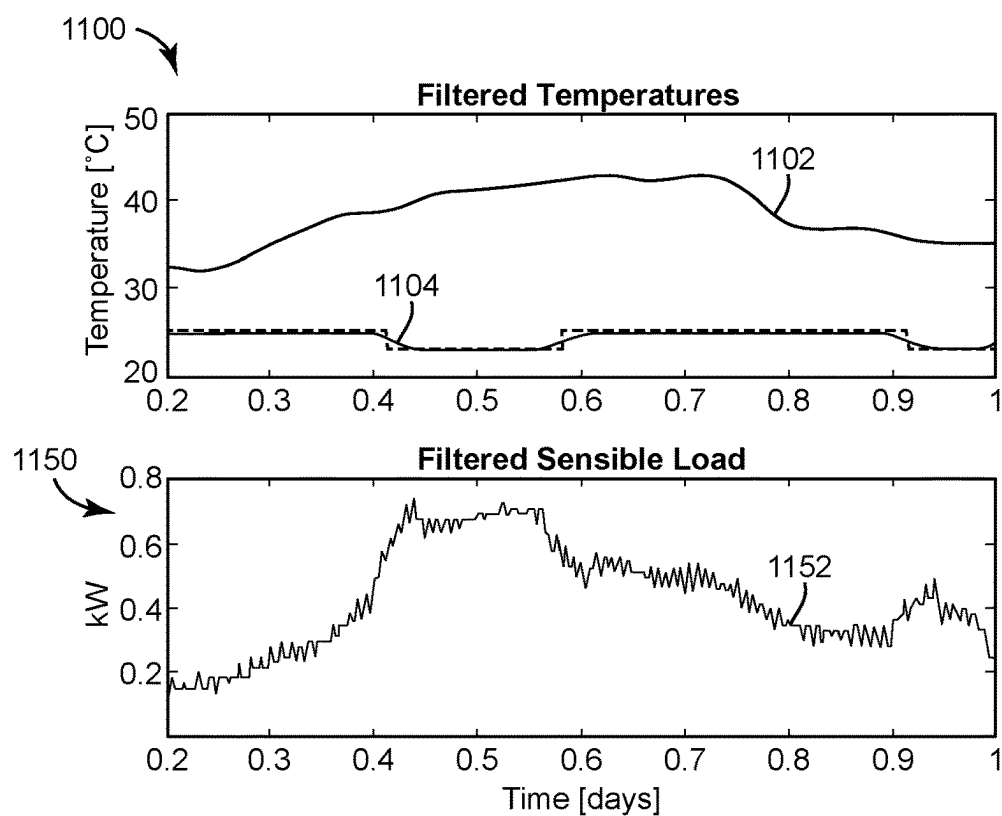


FIG. 11

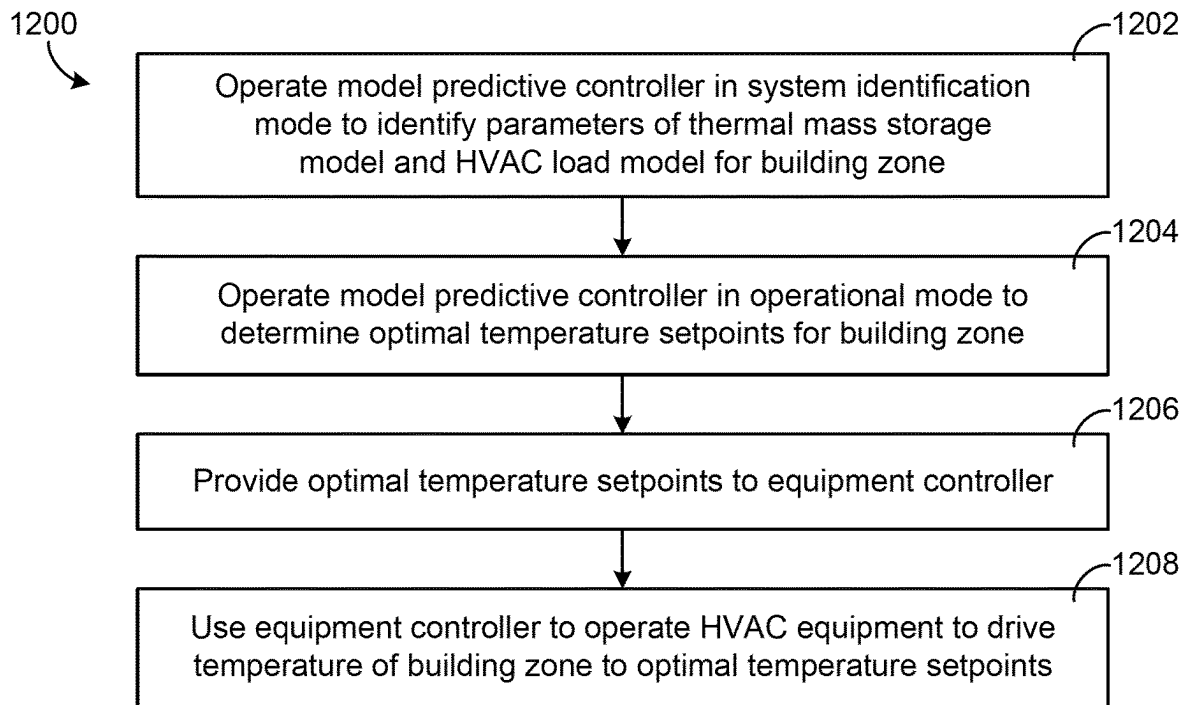


FIG. 12

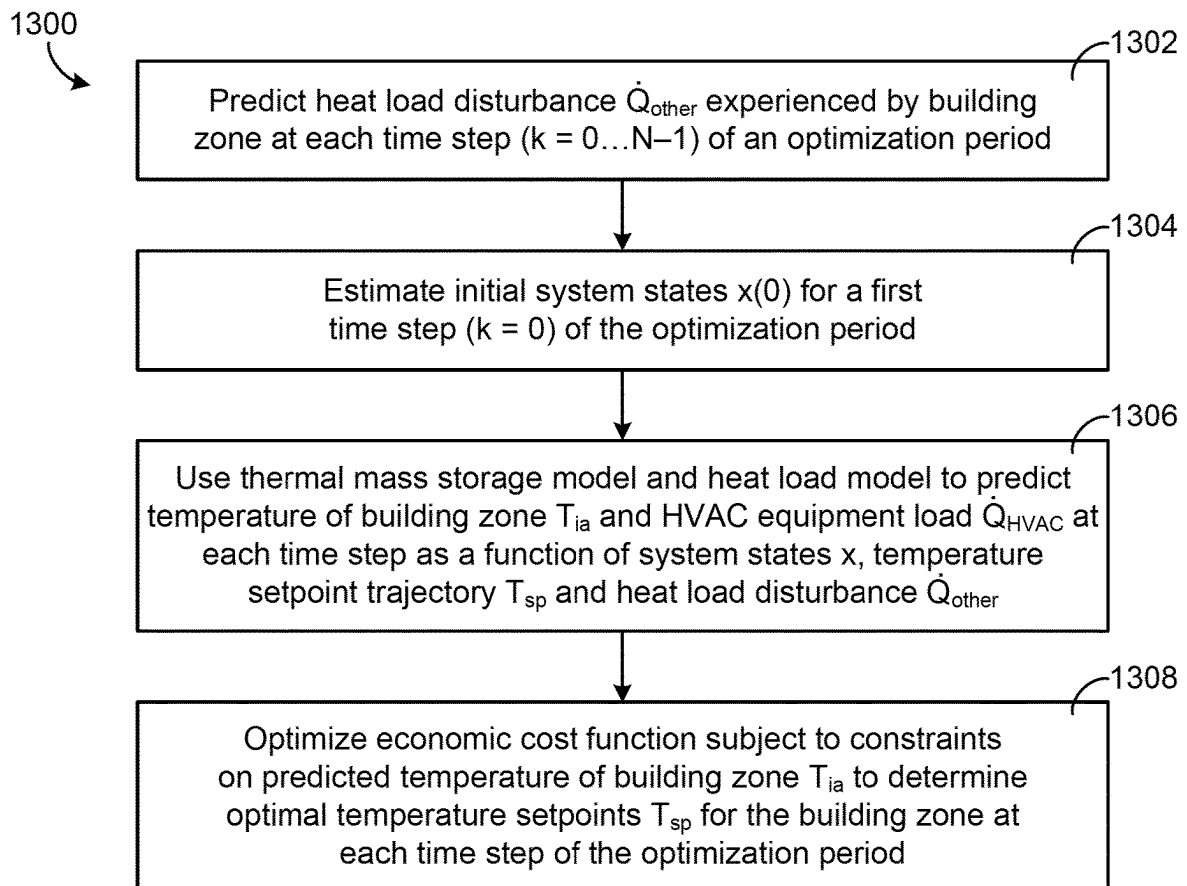


FIG. 13

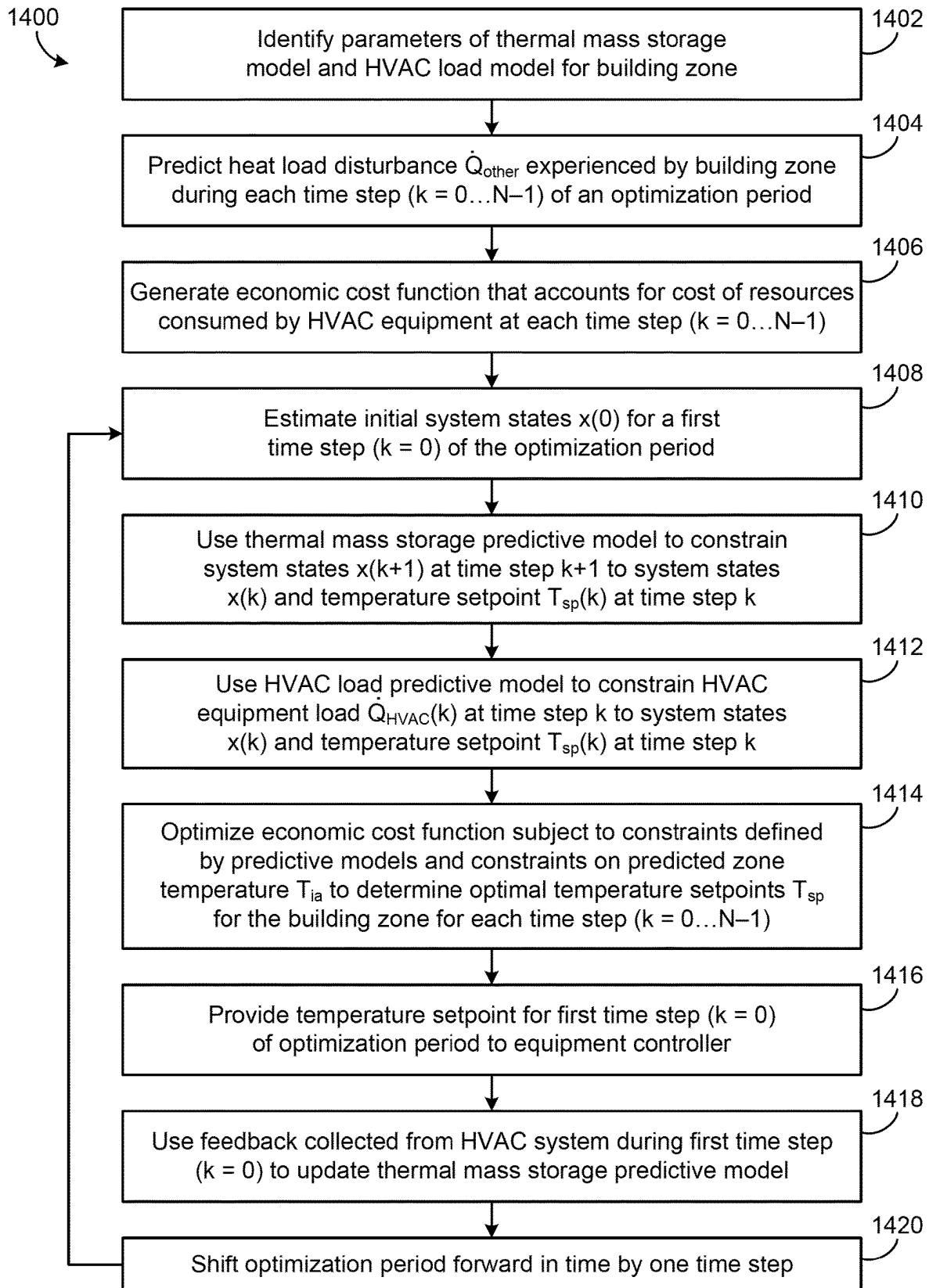


FIG. 14

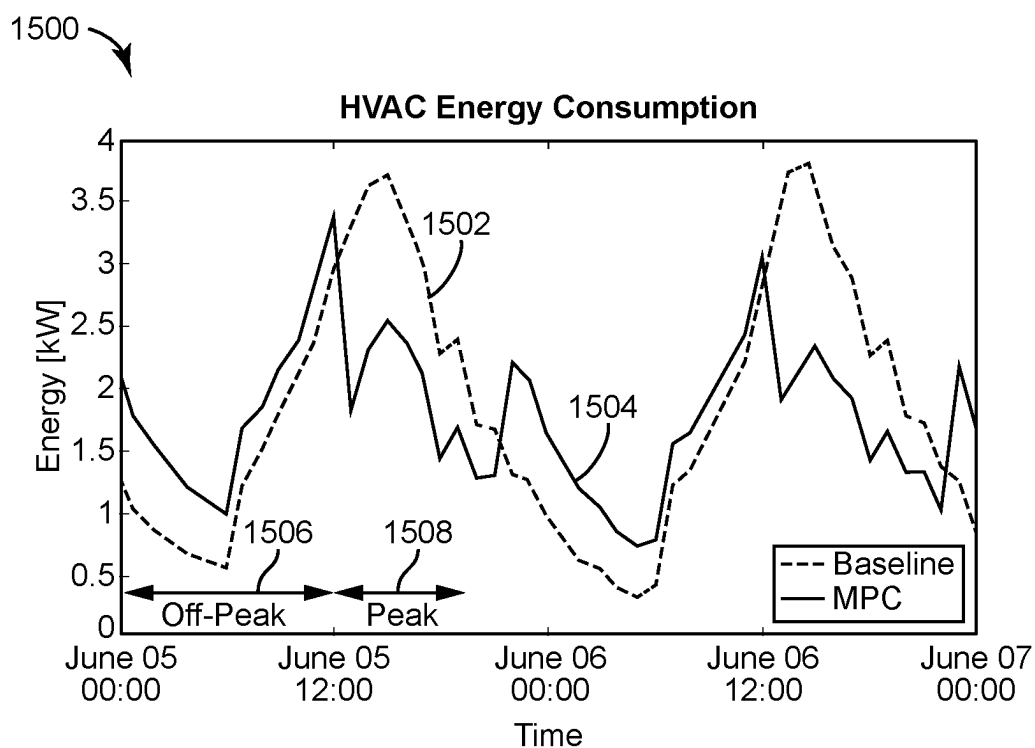


FIG. 15

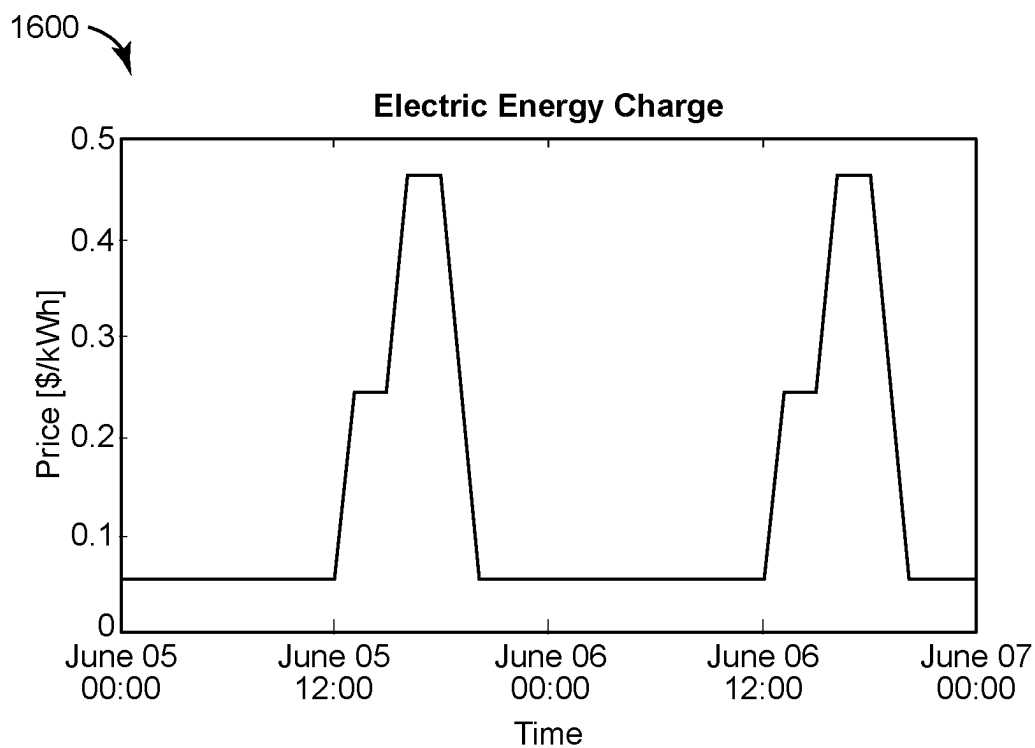


FIG. 16

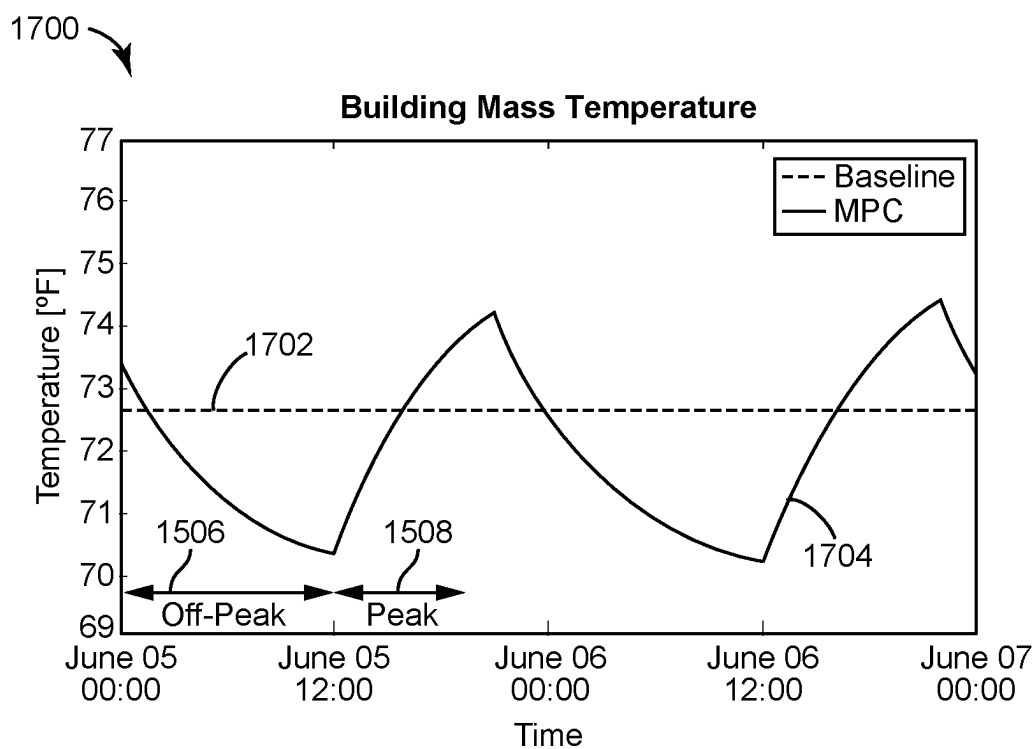


FIG. 17

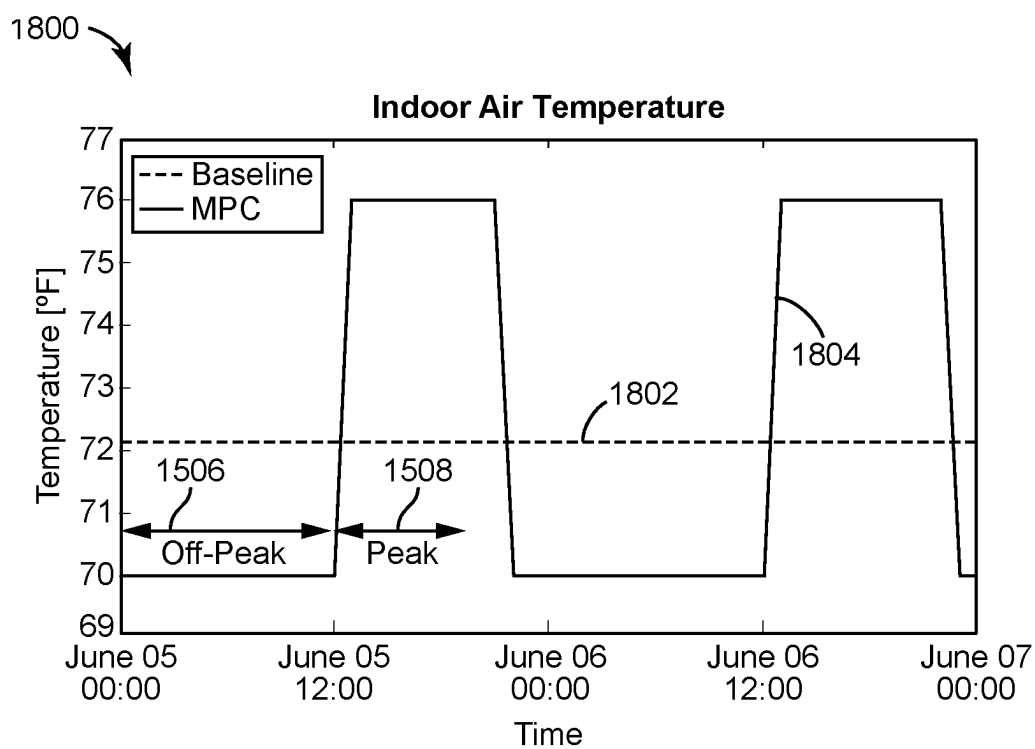


FIG. 18

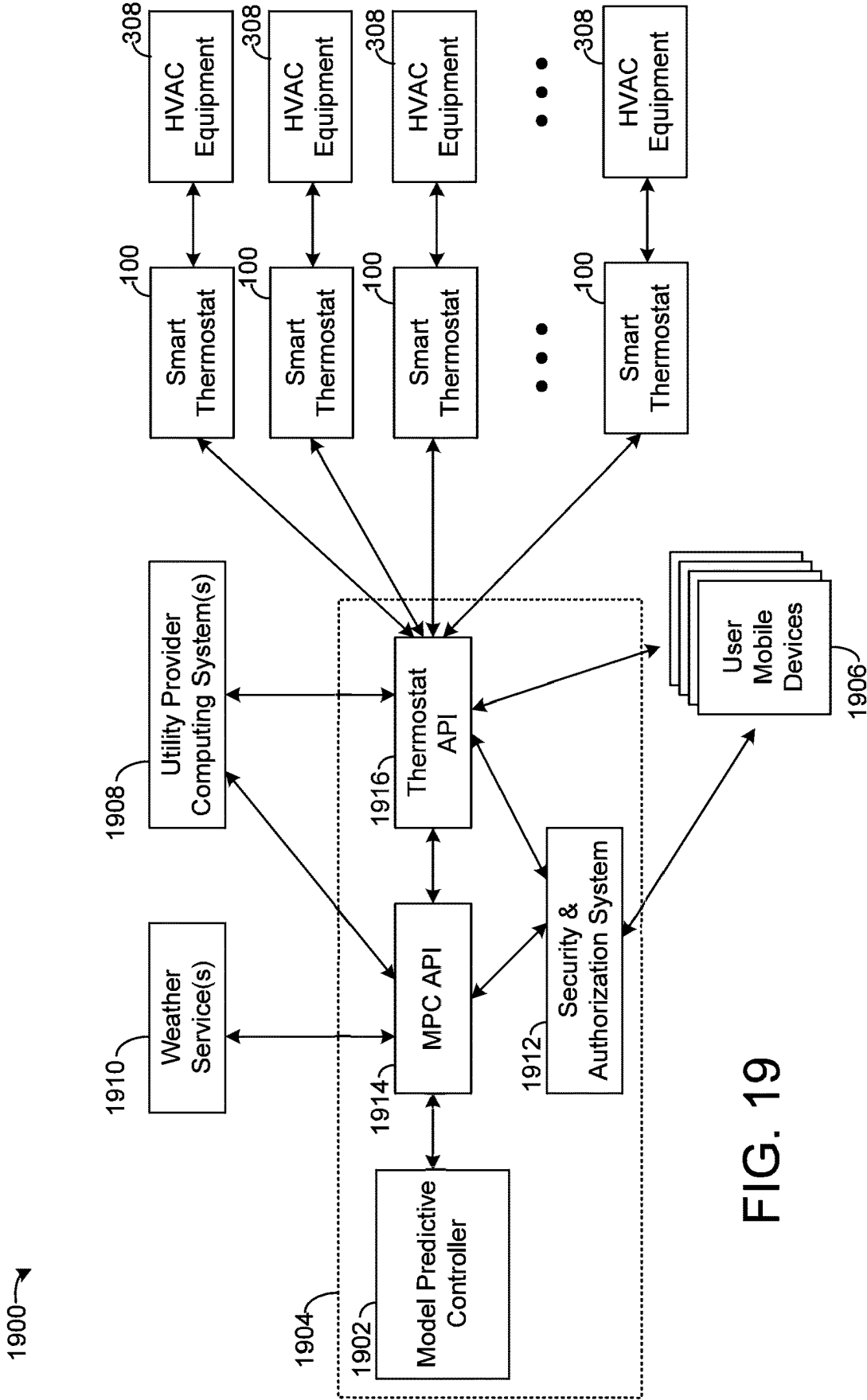


FIG. 19

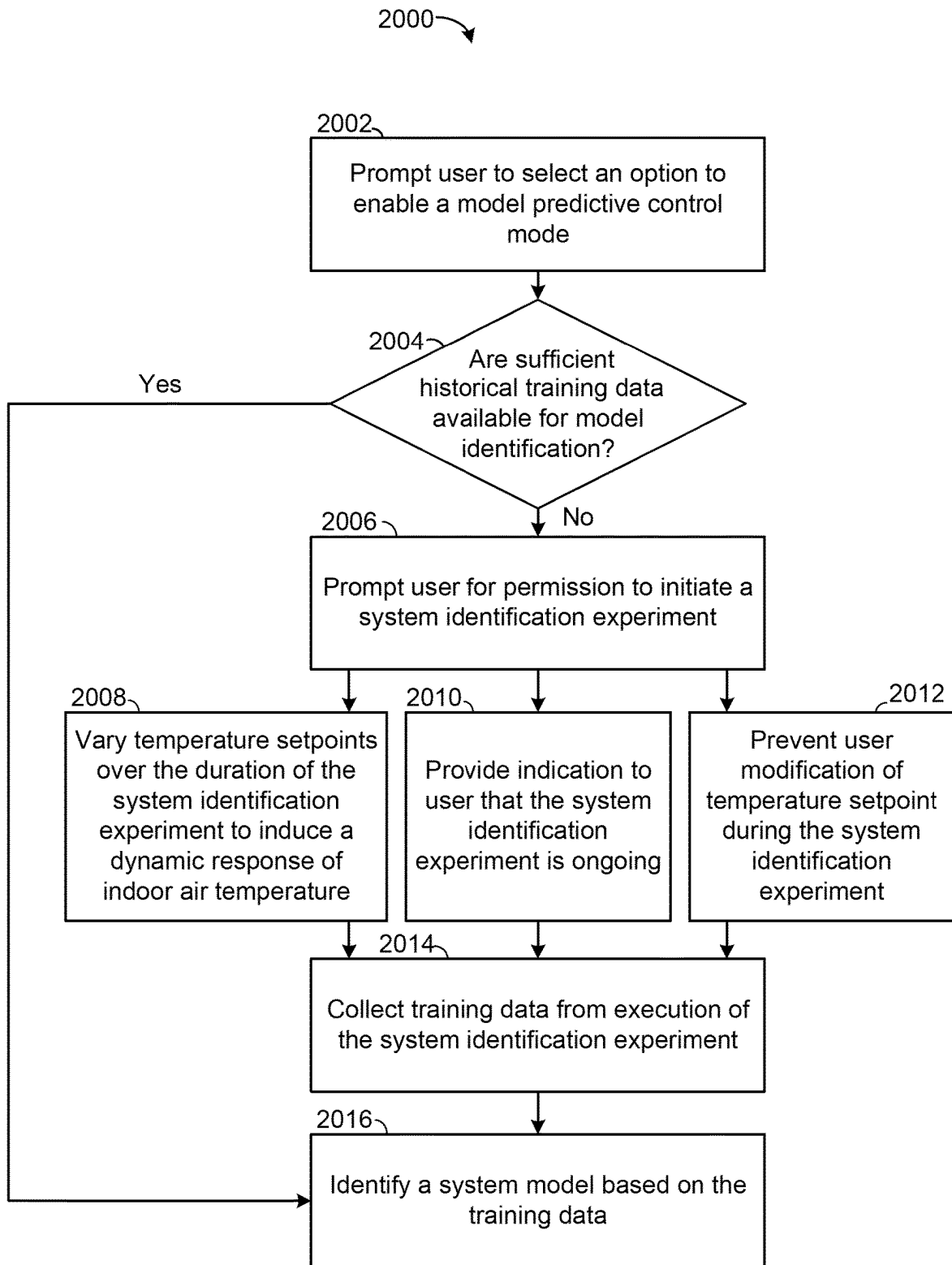


FIG. 20

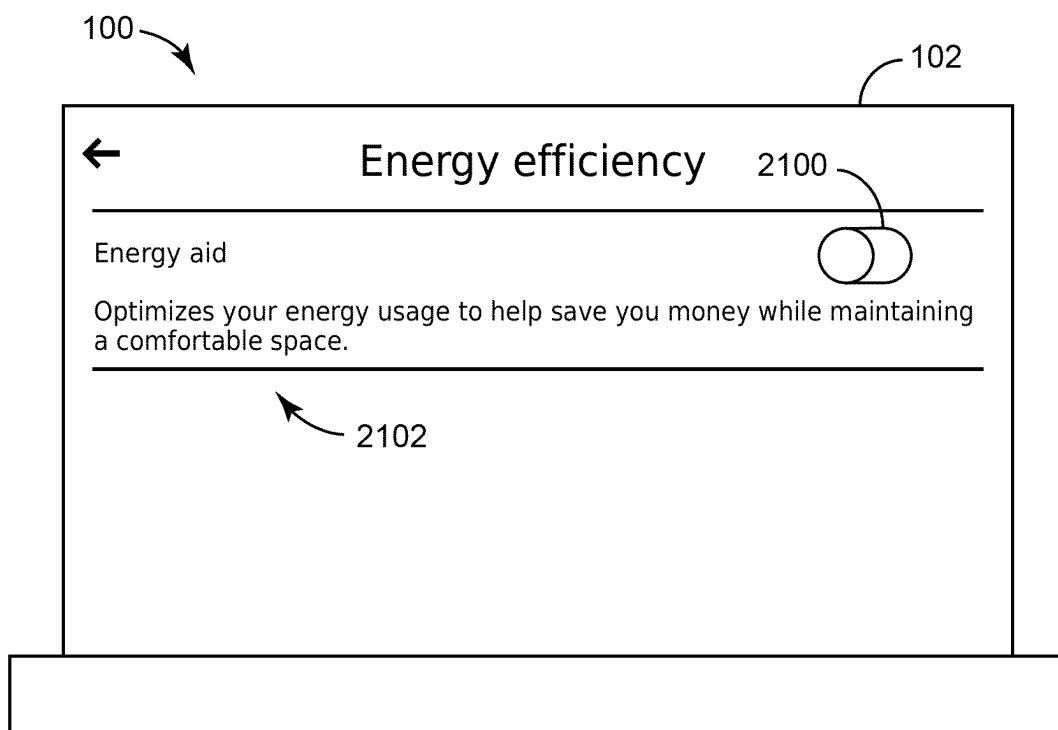


FIG. 21

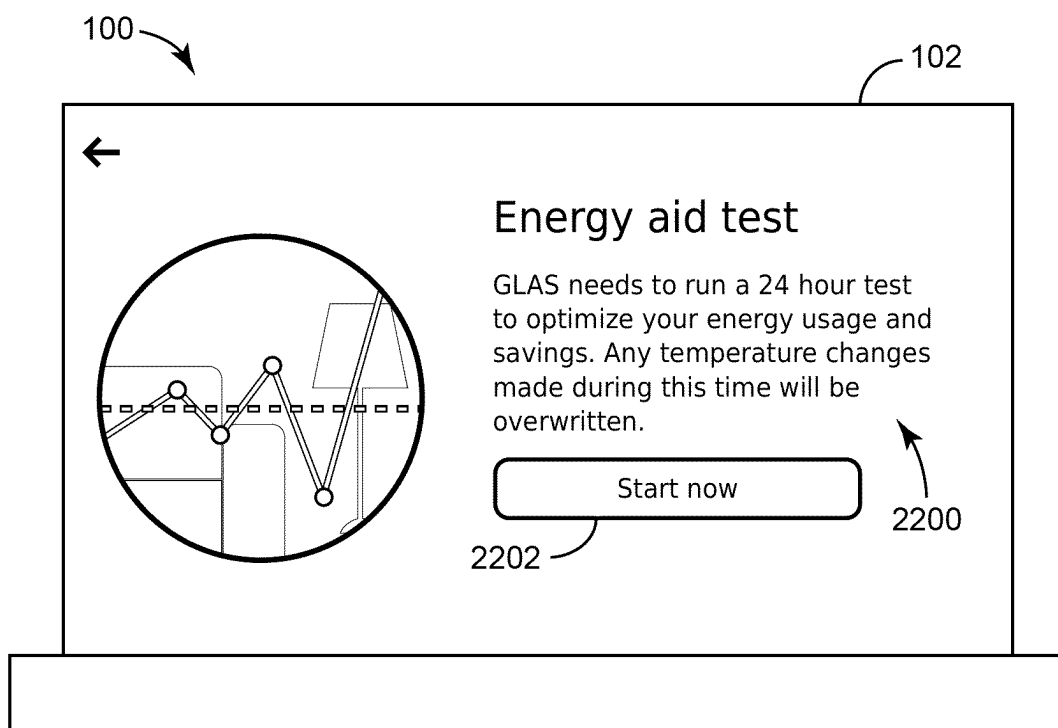


FIG. 22

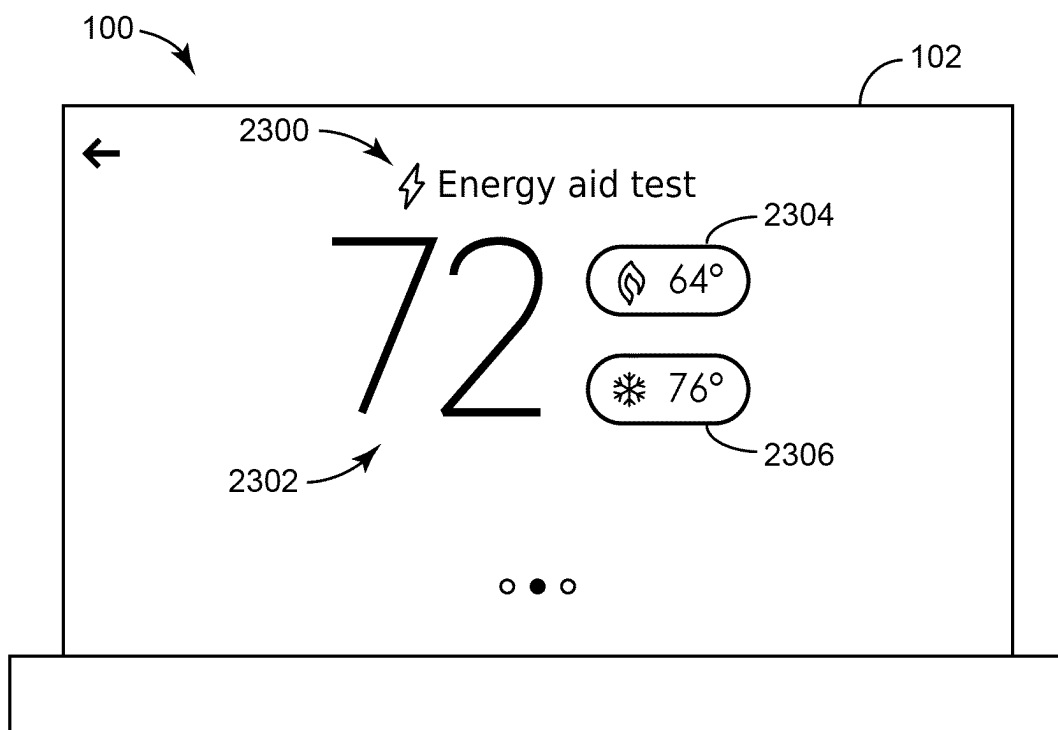


FIG. 23

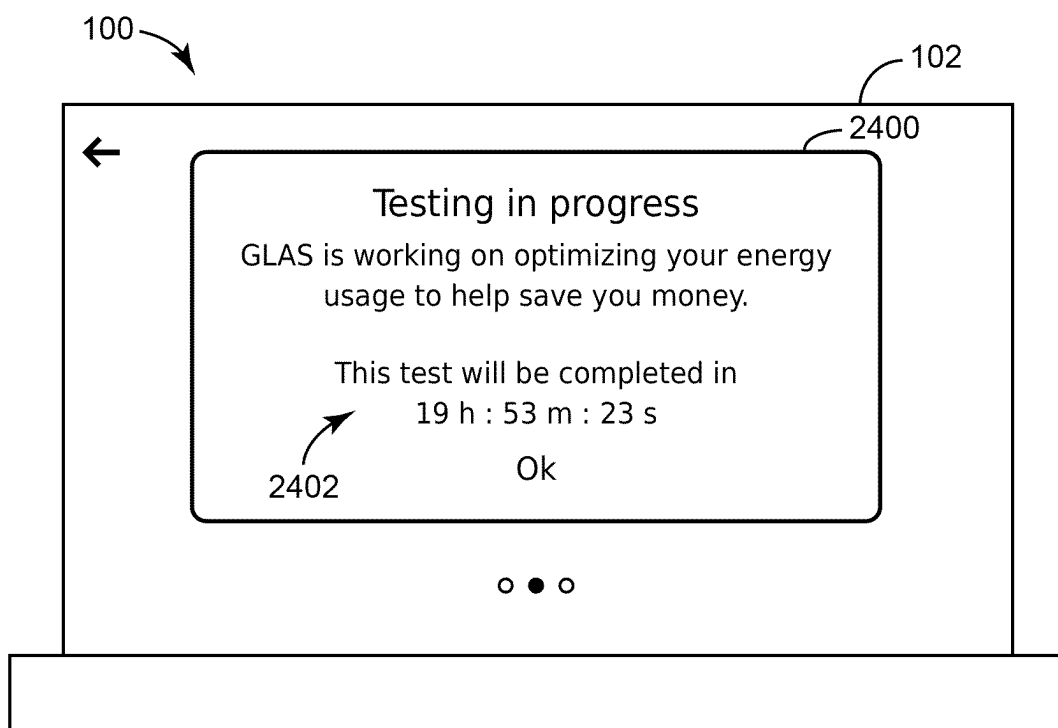


FIG. 24

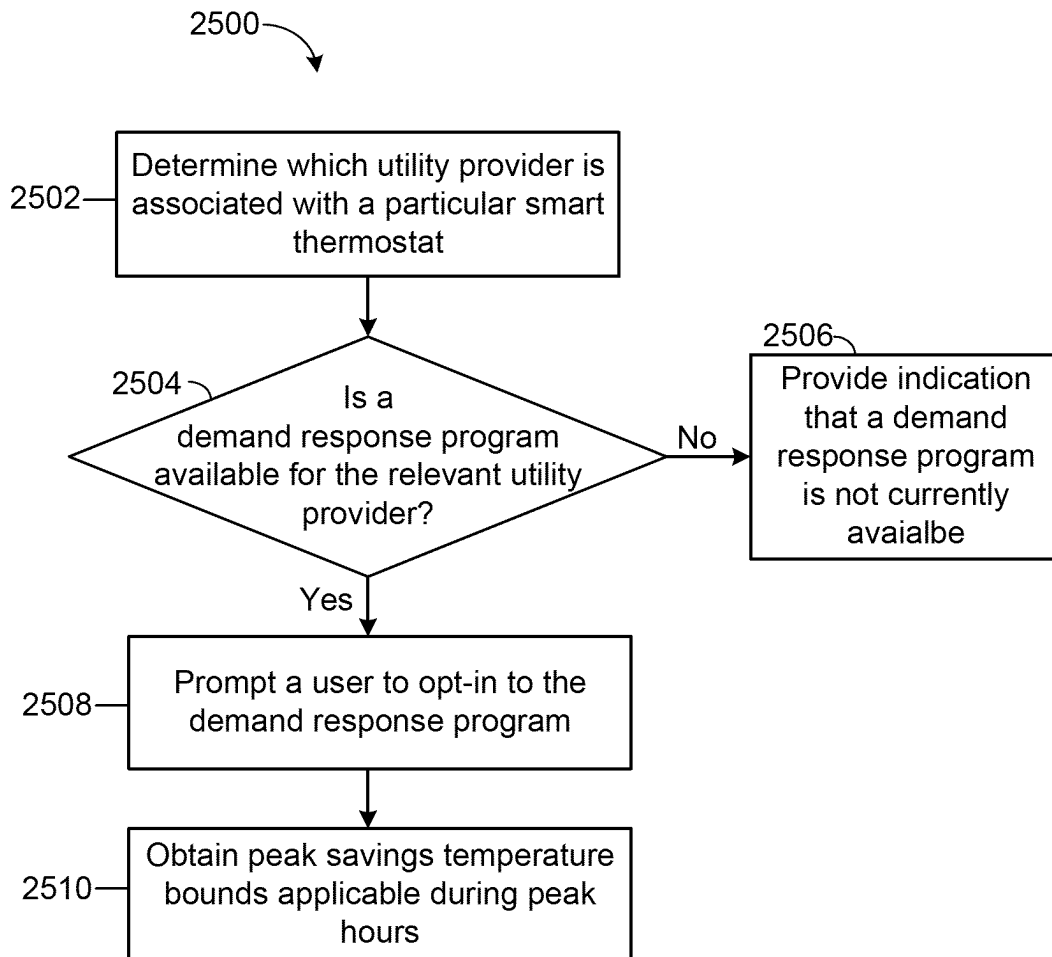


FIG. 25

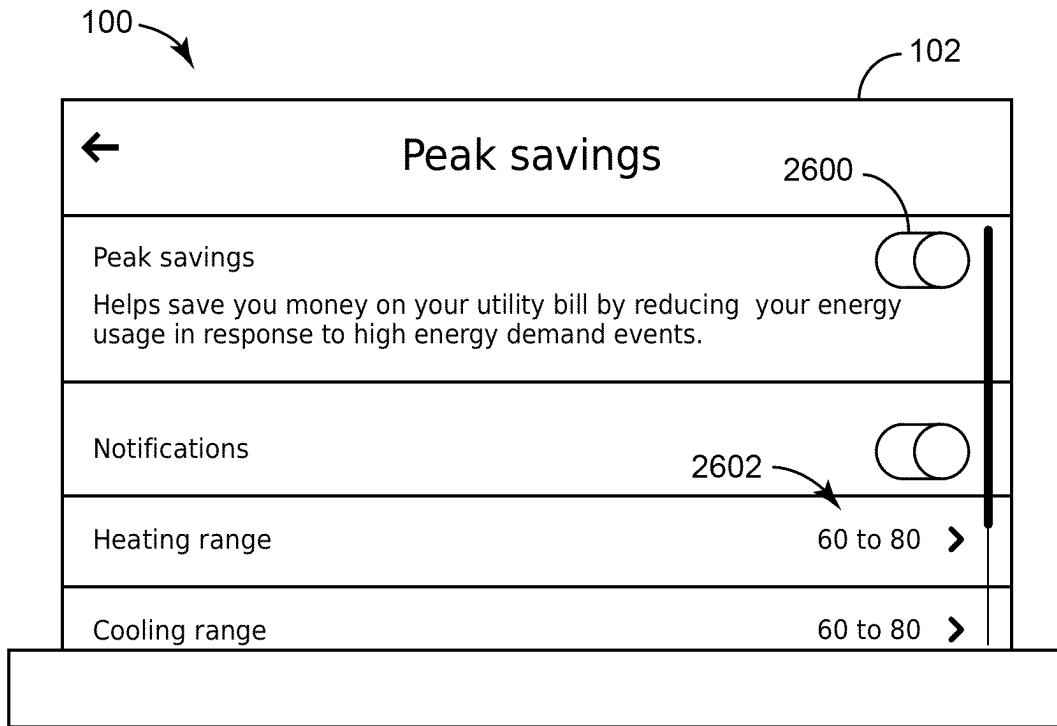


FIG. 26

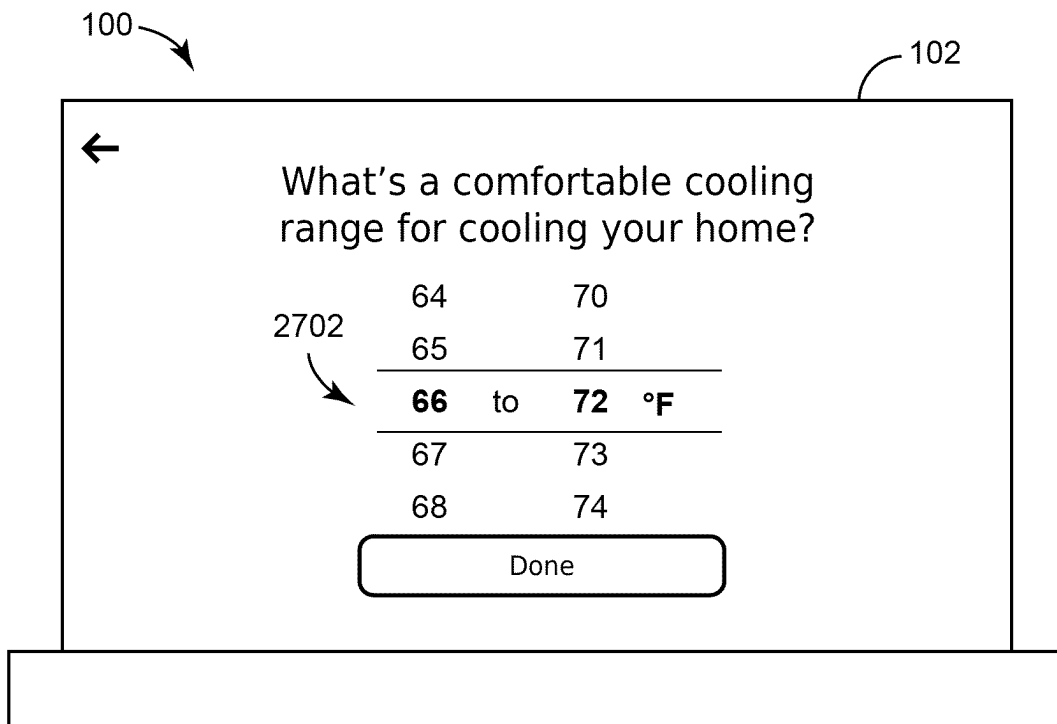


FIG. 27

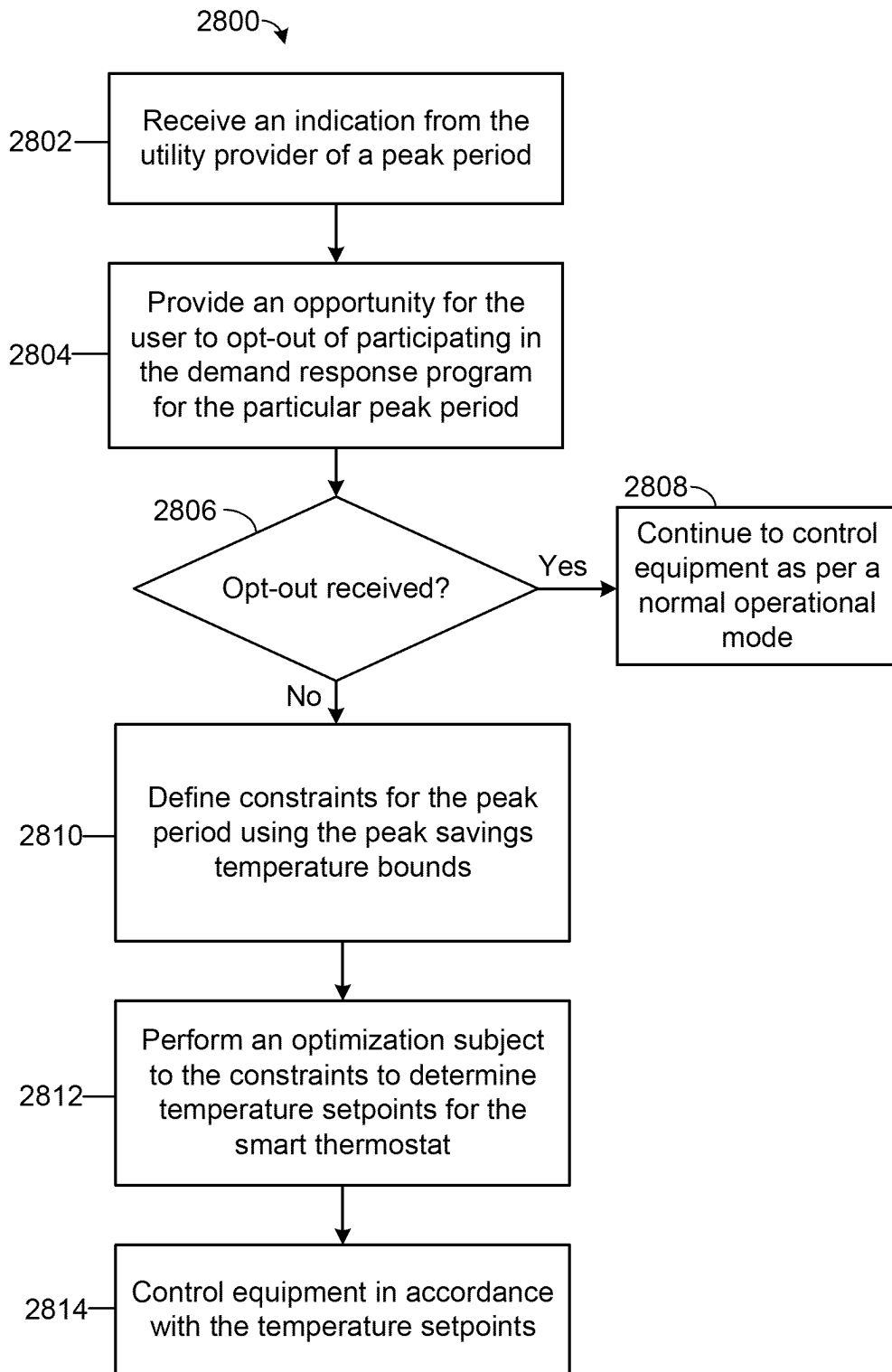


FIG. 28

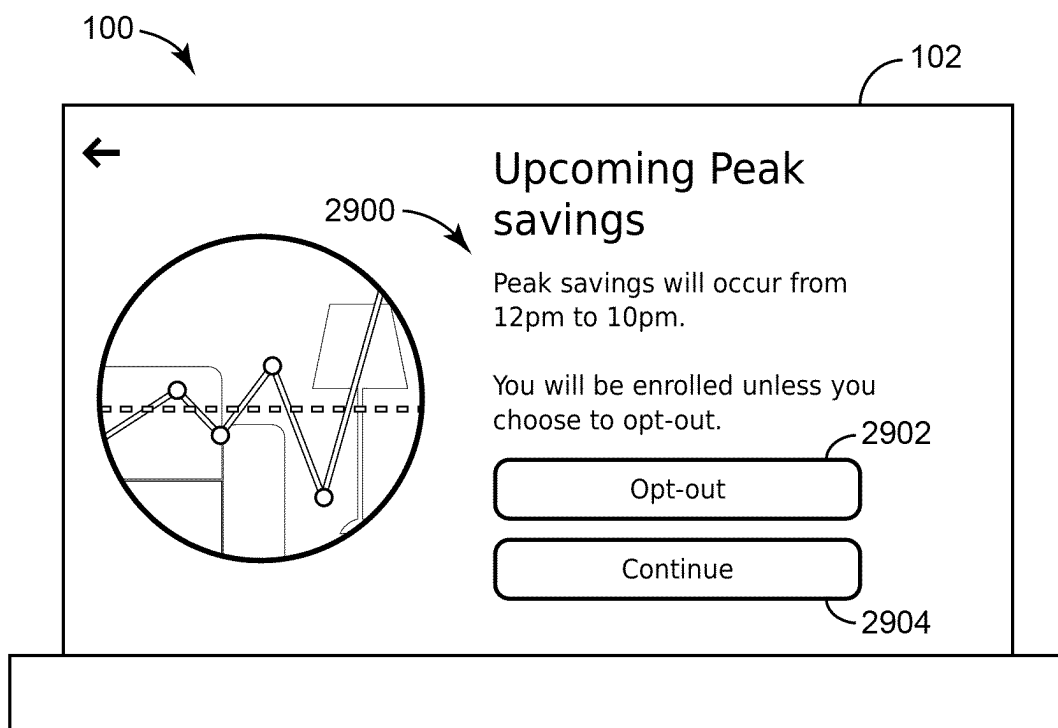


FIG. 29

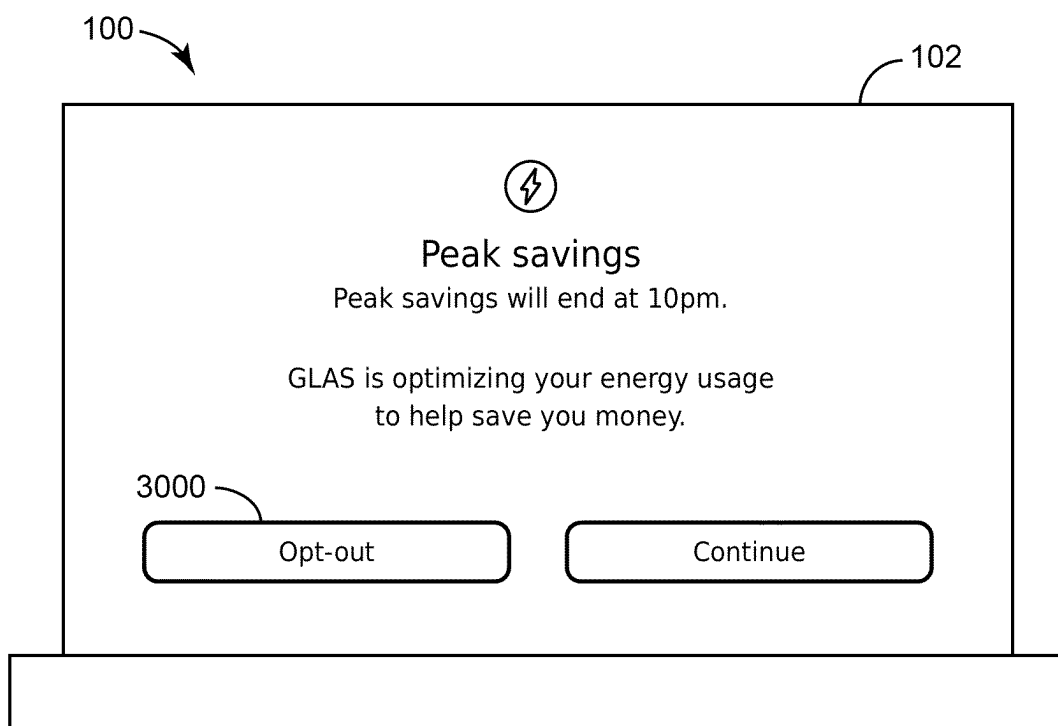


FIG. 30

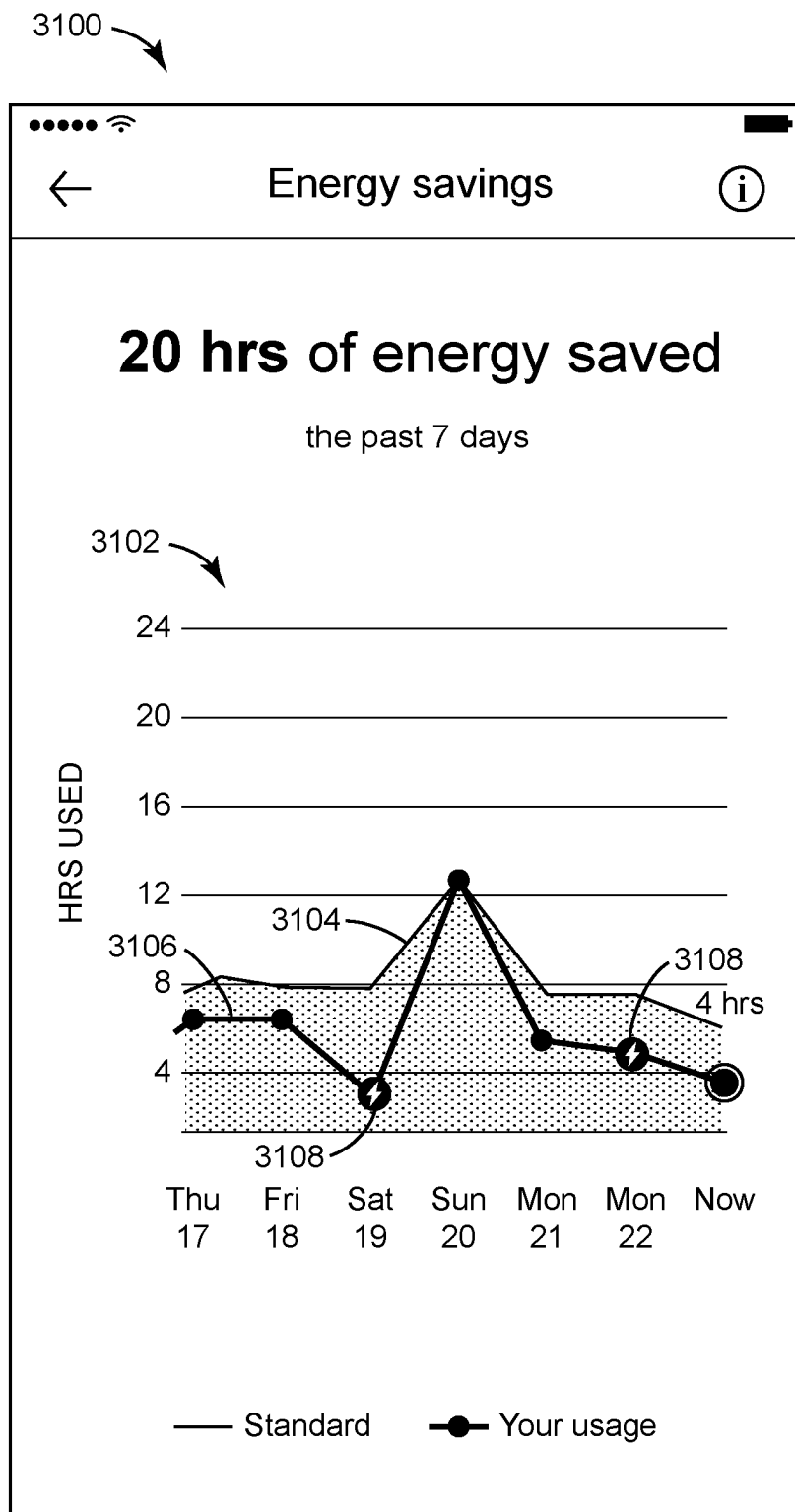


FIG. 31

SMART THERMOSTAT WITH MODEL PREDICTIVE CONTROL AND DEMAND RESPONSE INTEGRATION

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/689,670, filed Mar. 8, 2022, which is a divisional of U.S. patent application Ser. No. 16/588,467 (now U.S. Pat. No. 11,274,849), filed Sep. 30, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 16/185,274 (now U.S. Pat. No. 10,495,337), filed Nov. 9, 2018, which is a continuation of U.S. patent application Ser. No. 15/625,830 (now U.S. Pat. No. 10,146,237) filed Jun. 16, 2017, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/491,545 filed Apr. 28, 2017, all of which are incorporated by reference herein in their entireties.

BACKGROUND

[0002] The present disclosure relates generally to a smart thermostat and more particularly to a smart thermostat with model predictive control. Thermostats are often configured to monitor and control the temperature of a building zone or other space. For example, a thermostat can be mounted on a wall within the building zone and configured to measure the temperature of the building zone. Thermostats typically send commands to HVAC equipment (e.g., on/off commands, heating/cooling commands, etc.) to cause the HVAC equipment to affect the temperature of the building zone.

[0003] Conventional thermostats operate according to a fixed temperature setpoint schedule which defines the temperature setpoints for the thermostat at various times. The temperature setpoint schedule is typically set by a user via a local user interface on the thermostat. In many implementations, a fixed temperature setpoint schedule leads to sub-optimal control of the HVAC equipment, which can increase the cost of heating/cooling the building zone. It would be desirable to automatically determine optimal temperature setpoints for a thermostat in order to take advantage of time-varying energy prices, zone heat transfer characteristics, and/or other factors that can affect the cost of heating/cooling the building zone.

SUMMARY

[0004] One implementation of the present disclosure is a system. The system includes a plurality of thermostats corresponding to a plurality of HVAC systems that serve a plurality of spaces and a computing system communicable with the plurality of thermostats via a network. The computing system is configured to, for each space of the plurality of spaces, obtain a set of training data relating to thermal behavior of the space, identify a model of thermal behavior of the space based on the set of training data, perform a model predictive control process using the model of thermal behavior of the space to obtain a temperature setpoint for the space, and provide the temperature setpoint to the thermostat corresponding to the HVAC system serving the space. The plurality of thermostats are configured to control the plurality of HVAC systems in accordance with the temperature setpoints.

[0005] In some embodiments, the computing system is configured to obtain the set of training data relating to

thermal behavior of the space by executing a system identification experiment for the space. Executing the system identification experiment includes providing a series of temperature setpoint changes to the thermostat corresponding to the HVAC system serving the space. The computing system may be configured to cause a temperature of the space to deviate from a normal comfort range while executing the system identification experiment. The thermostats may be configured to prompt users for permission to perform the system identification experiments. The system identification experiment may include causing the thermostat to display an indication that the system identification experiment is ongoing. Executing the system identification experiment may include preventing a user from adjusting a current temperature setpoint for the space.

[0006] In some embodiments, the computing system is configured to receive data from a utility provider computing system and adjust the model predictive control process based on the data from the utility provider computing system. In some embodiments, the computing system is configured to identify the model of thermal behavior of the space by performing a system identification process to determine values for parameters of the model of thermal behavior.

[0007] Another implementation of the present disclosure is a method for managing temperature in a space. The method includes defining a first temperature range for a normal operating mode, defining a second temperature range for a peak savings mode, operating in the normal operating mode by performing an optimization process to determine temperature setpoints predicted to maintain a temperature of the space within the first temperature range, receiving an indication from a utility company of an occurrence of a peak period, and operating in the peak savings mode for the peak period by performing the optimization process to determine temperature setpoints predicted to maintain the temperature of the space within the second temperature range during the peak period.

[0008] In some embodiments, defining the second temperature range includes prompting a user to input a maximum and minimum temperature for the peak savings mode. In some embodiments, defining the second temperature range includes expanding the first temperature range. In some embodiments, defining the second temperature range includes expanding the first temperature range by an amount prescribed by the utility company.

[0009] In some embodiments, the method includes, in response to the indication from the utility company of the occurrence of the peak period, prompting a user to opt-out of applying the peak savings mode for the peak period. The method may also include, in response to the user opting-out of applying the peak savings mode for the peak period, applying the normal operating mode during the peak period by performing the optimization process to determine temperature setpoints predicted to maintain the temperature of the space within the first temperature range during the peak period.

[0010] In some embodiments, prompting the user to opt-out of applying the peak savings mode for the peak period includes providing an option to opt-out on a graphical user interface of a thermostat. In some embodiments, prompting the user to opt-out of applying the peak savings mode for the peak period includes providing an option to opt-out on a graphical user interface of a smartphone. In some embodi-

ments, prompting the user to opt-out of applying the peak savings mode for the peak period includes providing an estimate of cost savings associated with applying the peak savings mode for the peak period.

[0011] In some embodiments, the method includes operating heating or cooling equipment to drive the temperature of the space toward the temperature setpoints. In some embodiments, the optimization process includes a model predictive control process configured to minimize at least one of energy usage or energy cost over a time horizon subject to a temperature constraint. The temperature constraint may be defined by the first temperature range in the normal operating mode and by the second temperature range in the peak savings mode.

[0012] Another implementation of the present disclosure is a thermostat for equipment that operate to affect a temperature of a space. The thermostat includes a processing circuit configured to prompt a user for permission to initiate a training experiment, execute the training experiment by controlling the equipment over a duration of the training experiment to induce a dynamic response of the temperature of the space while collecting training data relating to a thermal behavior of the space, and control the equipment in a model predictive control mode in accordance with an output of an optimization process. The optimization process is based on a thermal model for the space identified using the training data.

[0013] In some embodiments, the processing circuit is configured to prevent the user from adjusting a temperature setpoint for the space during execution of the training experiment. In some embodiments, controlling the equipment over the duration of the training experiment to induce a dynamic response of the temperature of the space comprises varying a temperature setpoint for the space away from a user-selected temperature. In some embodiments, the processing circuit is configured to cause an indication that the training experiment is ongoing to be displayed during execution of the training experiment.

[0014] Another implementation of the present disclosure is a thermostat for monitoring and controlling temperature of a building zone. The thermostat includes an equipment controller and a model predictive controller. The equipment controller is configured to drive the temperature of the building zone to an optimal temperature setpoint by operating HVAC equipment to provide heating or cooling to the building zone. The model predictive controller is configured to determine the optimal temperature setpoint by generating a cost function that accounts for a cost operating the HVAC equipment during each of a plurality of time steps in an optimization period, using a predictive model to predict the temperature of the building zone during each of the plurality of time steps, and optimizing the cost function subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of the plurality of time steps.

[0015] In some embodiments, the model predictive controller is configured to determine the cost of operating the HVAC equipment during each of the plurality of time steps using a set of time-varying utility rates comprising a utility rate value for each time step. The time-varying utility rates may be received from a utility provider or predicted by the model predictive controller.

[0016] In some embodiments, the model predictive controller is configured to predict the temperature of the build-

ing zone during each of the plurality of time steps as a function of a temperature setpoint trajectory comprising a temperature setpoint for each of the plurality of time steps.

[0017] In some embodiments, the model predictive controller is configured to optimize the cost function subject to a constraint on the optimal temperature setpoints that limits the optimal temperature setpoints within a temperature setpoint range.

[0018] In some embodiments, the model predictive controller is configured to generate the predictive model by performing a system identification process. The system identification process may include modulating the temperature setpoint within a constrained temperature setpoint range, collecting a set of input-output data, and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of the temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint during each of a plurality of time steps during a learning period.

[0019] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of at least one of heat transfer between air within the building zone and solid mass within the building zone, heat transfer between the building zone and the HVAC equipment, and an unmeasured heat load disturbance. In some embodiments, the model predictive controller is configured to predict a value of the unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period.

[0020] In some embodiments, the predictive model includes an HVAC load model that defines the heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0021] In some embodiments, the model predictive controller is configured to predict the cost of operating the HVAC equipment as a function of an amount of the heating or cooling provided by the HVAC equipment at each time step of the optimization period.

[0022] In some embodiments, the constraint on the predicted temperature of the building zone requires the model predictive controller to maintain the predicted temperature of the building zone within a first zone temperature range during a first time step of the optimization period and within a second zone temperature range, different from the first zone temperature range, during another time step of the optimization period subsequent to the first time step.

[0023] Another implementation of the present disclosure is a method performed by a thermostat for a building zone for monitoring and controlling temperature of the building zone. The method includes generating a cost function that accounts for a cost operating HVAC equipment during each of a plurality of time steps in an optimization period, using a predictive model to predict the temperature of the building zone during each of the plurality of time steps, optimizing the cost function subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of the plurality of time steps, and operating HVAC equipment to provide heating or cooling to the building zone to drive the temperature of the building zone to the optimal temperature setpoints.

[0024] In some embodiments, the method includes receiving a set of time-varying utility rates from a utility provider or predicting the time-varying utility rates. The set of time-varying utility rates may include a utility rate value for each time step. The method may include determining the cost of operating the HVAC equipment during each of the plurality of time steps using the set of time-varying utility rates.

[0025] In some embodiments, using the predictive model to predict the temperature of the building zone includes predicting the temperature of the building zone during each of the plurality of time steps as a function of a temperature setpoint trajectory that includes a temperature setpoint for each of the plurality of time steps.

[0026] In some embodiments, optimizing the cost function includes optimizing the cost function subject to a constraint on the optimal temperature setpoints that limits the optimal temperature setpoints within a temperature setpoint range.

[0027] In some embodiments, the method includes generating the predictive model by performing a system identification process. The system identification process may include modulating the temperature setpoint within a constrained temperature setpoint range, collecting a set of input-output data, and fitting parameters of the predictive model to the set of input-output data. The input-output data include values of the temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint during each of a plurality of time steps during a learning period.

[0028] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of at least one of heat transfer between air within the building zone and solid mass within the building zone, heat transfer between the building zone and the HVAC equipment, and an unmeasured heat load disturbance. In some embodiments, the method includes predicting a value of the unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period.

[0029] In some embodiments, the predictive model includes an HVAC load model that defines the heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0030] In some embodiments, the method includes predicting the cost of operating the HVAC equipment as a function of an amount of the heating or cooling provided by the HVAC equipment at each time step of the optimization period.

[0031] Another implementation of the present disclosure is a thermostat for monitoring and controlling temperature of a building zone. The thermostat includes an equipment controller and a model predictive controller. The equipment controller is configured to drive the temperature of the building zone to a zone temperature setpoint by operating HVAC equipment to provide heating or cooling to the building zone. The model predictive controller is configured to determine the zone temperature setpoint by generating a cost function that accounts for a cost operating the HVAC equipment during each of a plurality of time steps in an optimization period, using a predictive model to predict the temperature of the building zone during each of the plurality of time steps, and performing an optimization of the cost function subject to a constraint on the predicted temperature

of the building zone to determine a temperature setpoint trajectory including a temperature setpoint value for each of the plurality of time steps.

[0032] Another implementation of the present disclosure is a model predictive control system for monitoring and controlling temperature of a building zone. The model predictive control system includes a thermostat and a model predictive controller. The thermostat is configured to drive the temperature of the building zone to an optimal temperature setpoint by operating HVAC equipment to provide heating or cooling to the building zone. The model predictive controller is configured to determine the optimal temperature setpoint and provide the optimal temperature setpoint to the thermostat via a communications network. The model predictive controller determines the optimal temperature setpoint by generating a cost function that accounts for a cost operating the HVAC equipment during each of a plurality of time steps in an optimization period, using a predictive model to predict the temperature of the building zone during each of the plurality of time steps, and optimizing the cost function subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of the plurality of time steps.

[0033] In some embodiments, the model predictive controller is configured to determine the cost of operating the HVAC equipment during each of the plurality of time steps using a set of time-varying utility rates that includes a utility rate value for each time step. The time-varying utility rates may be received from a utility provider or predicted by the model predictive controller.

[0034] In some embodiments, the model predictive controller is configured to predict the temperature of the building zone during each of the plurality of time steps as a function of a temperature setpoint trajectory comprising a temperature setpoint for each of the plurality of time steps.

[0035] In some embodiments, the model predictive controller is configured to optimize the cost function subject to a constraint on the optimal temperature setpoints that limits the optimal temperature setpoints within a temperature setpoint range.

[0036] In some embodiments, the model predictive controller is configured to generate the predictive model by performing a system identification process. The system identification process may include modulating the temperature setpoint within a constrained temperature setpoint range, collecting a set of input-output data, and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of the temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint during each of a plurality of time steps during a learning period.

[0037] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of at least one of heat transfer between air within the building zone and solid mass within the building zone, heat transfer between the building zone and the HVAC equipment, and an unmeasured heat load disturbance. In some embodiments, the model predictive controller is configured to predict a value of the unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period.

[0038] In some embodiments, the predictive model includes an HVAC load model that defines the heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0039] In some embodiments, the model predictive controller is configured to predict the cost of operating the HVAC equipment as a function of an amount of the heating or cooling provided by the HVAC equipment at each time step of the optimization period.

[0040] In some embodiments, the constraint on the predicted temperature of the building zone requires the model predictive controller to maintain the predicted temperature of the building zone within a first zone temperature range during a first time step of the optimization period and within a second zone temperature range, different from the first zone temperature range, during another time step of the optimization period subsequent to the first time step.

[0041] Another implementation of the present disclosure is a method for monitoring and controlling temperature of a building zone. The method includes generating a cost function that accounts for a cost operating HVAC equipment during each of a plurality of time steps in an optimization period, using a predictive model to predict the temperature of the building zone during each of the plurality of time steps, optimizing the cost function at a model predictive controller subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of the plurality of time steps, and providing an optimal temperature setpoint from the model predictive controller to a thermostat for the building zone via a communications network. The method further includes, at the thermostat, driving the temperature of the building zone to the optimal temperature setpoint by operating HVAC equipment to provide heating or cooling to the building zone.

[0042] In some embodiments, the method includes receiving a set of time-varying utility rates from a utility provider or predicting the time-varying utility rates. The set of time-varying utility rates may include a utility rate value for each time step. The method may include determining the cost of operating the HVAC equipment during each of the plurality of time steps using the set of time-varying utility rates.

[0043] In some embodiments, using the predictive model to predict the temperature of the building zone includes predicting the temperature of the building zone during each of the plurality of time steps as a function of a temperature setpoint trajectory that includes a temperature setpoint for each of the plurality of time steps.

[0044] In some embodiments, optimizing the cost function includes optimizing the cost function subject to a constraint on the optimal temperature setpoints that limits the optimal temperature setpoints within a temperature setpoint range.

[0045] In some embodiments, the method includes generating the predictive model by performing a system identification process. The system identification process may include modulating the temperature setpoint within a constrained temperature setpoint range, collecting a set of input-output data, and fitting parameters of the predictive model to the set of input-output data. The input-output data include values of the temperature setpoint and values of the temperature of the building zone that result from modulating

the temperature setpoint during each of a plurality of time steps during a learning period.

[0046] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of at least one of heat transfer between air within the building zone and solid mass within the building zone, heat transfer between the building zone and the HVAC equipment, and an unmeasured heat load disturbance. In some embodiments, the method includes predicting a value of the unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period.

[0047] In some embodiments, the predictive model includes an HVAC load model that defines the heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0048] In some embodiments, the method includes predicting the cost of operating the HVAC equipment as a function of an amount of the heating or cooling provided by the HVAC equipment at each time step of the optimization period.

[0049] Another implementation of the present disclosure is a model predictive control system for monitoring and controlling temperature of a building zone. The model predictive control system includes a thermostat and a model predictive controller. The thermostat is configured to drive the temperature of the building zone to a zone temperature setpoint by operating HVAC equipment to provide heating or cooling to the building zone. The model predictive controller is configured to determine the zone temperature setpoint and provide the zone temperature setpoint to the thermostat via a communications network. The model predictive controller determines the zone temperature setpoint by generating a cost function that accounts for a cost operating the HVAC equipment during each of a plurality of time steps in an optimization period, using a predictive model to predict the temperature of the building zone during each of the plurality of time steps, and performing an optimization of cost function subject to a constraint on the predicted temperature of the building zone to determine the temperature setpoint trajectory.

[0050] Another implementation of the present disclosure is a model predictive controller for monitoring and controlling temperature of a building zone. The model predictive controller includes a system identifier and a predictive optimizer. The system identifier is configured to operate the model predictive controller in a system identification mode. Operating in the system identification mode includes performing a system identification process to automatically generate a predictive model based on heat transfer characteristics of the building zone. The predictive optimizer is configured to operate the model predictive controller in an operational mode. Operating in the operational mode includes using the predictive model to predict the temperature of the building zone. The model predictive controller is configured to automatically transition from the operational mode to the system identification mode in response to a determination that a prediction error of the predictive model exceeds a threshold value.

[0051] In some embodiments, operating in the operational mode includes optimizing a cost function subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of a

plurality of time steps in an optimization period. In some embodiments, the cost function accounts for a cost of operating HVAC equipment to provide heating or cooling to the building zone during each of the plurality of time steps in the optimization period.

[0052] In some embodiments, performing the system identification process includes collecting a set of input-output data and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of a temperature setpoint for the building zone and the temperature of the building zone during each of a plurality of time steps during a learning period.

[0053] In some embodiments, the set of input-output data includes a discrete HVAC staging trajectory for staged HVAC equipment. The HVAC staging trajectory may include a discrete HVAC equipment load at each of the plurality of time steps during the learning period. In some embodiments, performing the system identification process includes filtering the input-output data to generate a continuous HVAC equipment load signal from the discrete HVAC staging trajectory.

[0054] In some embodiments, collecting the set of input-output data includes modulating a temperature setpoint within a constrained temperature setpoint range and recording values of the modulated temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint. In some embodiments, performing the system identification process comprises filtering the input-output data to remove oscillations in the temperature of the building zone around the temperature setpoint resulting from activating and deactivating staged HVAC equipment.

[0055] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of heat transfer between air within the building zone and solid mass within the building zone.

[0056] In some embodiments, the model predictive controller includes a load/rate predictor configured to predict a value of an unmeasured heat load disturbance experienced by the building zone at each of a plurality of time steps in an optimization period. In some embodiments, the predictive model defines the temperature of the building zone as a function of the unmeasured heat load disturbance.

[0057] In some embodiments, the predictive model includes an HVAC load model that defines an amount of heating or cooling provided by HVAC equipment controlled by the model predictive controller as a function of the temperature of the building zone and a temperature setpoint for the building zone.

[0058] Another implementation of the present disclosure is a method for monitoring and controlling temperature of a building zone. The method includes operating a model predictive controller in a system identification mode. Operating in the system identification mode includes performing a system identification process to automatically generate a predictive model based on heat transfer characteristics of the building zone. The method further includes operating the model predictive controller in an operational mode. Operating in the operational mode includes using the predictive model to predict the temperature of the building zone. The method further includes automatically transitioning from the operational mode to the system identification mode in

response to a determination that a prediction error of the predictive model exceeds a threshold value.

[0059] In some embodiments, operating in the operational mode includes optimizing a cost function subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of a plurality of time steps in an optimization period. In some embodiments, the cost function accounts for a cost of operating HVAC equipment to provide heating or cooling to the building zone during each of the plurality of time steps in the optimization period.

[0060] In some embodiments, performing the system identification process includes collecting a set of input-output data and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of a temperature setpoint for the building zone and the temperature of the building zone during each of a plurality of time steps during a learning period.

[0061] In some embodiments, the set of input-output data includes a discrete HVAC staging trajectory for staged HVAC equipment. The HVAC staging trajectory may include a discrete HVAC equipment load at each of the plurality of time steps during the learning period. In some embodiments, performing the system identification process includes filtering the input-output data to generate a continuous HVAC equipment load signal from the discrete HVAC staging trajectory.

[0062] In some embodiments, collecting the set of input-output data comprises modulating a temperature setpoint within a constrained temperature setpoint range and recording values of the modulated temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint. In some embodiments, performing the system identification process includes filtering the input-output data to remove oscillations in the temperature of the building zone around the temperature setpoint resulting from activating and deactivating staged HVAC equipment.

[0063] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of heat transfer between air within the building zone and solid mass within the building zone.

[0064] In some embodiments, the method includes predicting a value of an unmeasured heat load disturbance experienced by the building zone at each of a plurality of time steps in an optimization period. In some embodiments, the predictive model defines the temperature of the building zone as a function of the unmeasured heat load disturbance.

[0065] In some embodiments, the predictive model includes an HVAC load model that defines an amount of heating or cooling provided by HVAC equipment controlled by the model predictive controller as a function of the temperature of the building zone and a temperature setpoint for the building zone.

[0066] Another implementation of the present disclosure is a model predictive controller for monitoring and controlling temperature of a building zone. The model predictive controller includes a state/disturbance estimator and a predictive optimizer. The state/disturbance estimator is configured to estimate an initial state of the building zone at a beginning of an optimization period. The state of the building zone includes a temperature of air within the building zone and a temperature of solid mass within the building

zone. The predictive optimizer is configured to use a predictive model of the building zone to predict the state of the building zone at each of a plurality of time steps of the optimization period based on the estimated initial state of the building zone and a temperature setpoint trajectory comprising a temperature setpoint for each of the plurality of time steps. The predictive optimizer is configured to generate optimal temperature setpoints for each of the plurality of time steps by optimizing a cost function that accounts for a cost operating HVAC equipment during each of the plurality of time steps.

[0067] In some embodiments, the predictive optimizer is configured to generate the cost function and determine the cost operating the HVAC equipment during each time step using a set of time-varying utility rates that includes a utility rate value for each time step. The set of time-varying utility rates can be received from a utility provider or predicted by the model predictive controller.

[0068] In some embodiments, the predicted state of the building zone includes a predicted temperature of the building zone. The predictive optimizer may be configured to optimize the cost function subject to a constraint on the predicted temperature of the building zone.

[0069] In some embodiments, the model predictive controller includes a load/rate predictor configured to predict a value of an unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period. In some embodiments, the predictive model defines the temperature of the building zone as a function of the unmeasured heat load disturbance experienced by the building zone at each time step.

[0070] In some embodiments, the predictive model includes a mass storage model that defines the temperature of the building zone as a function of heat transfer between air within the building zone and solid mass within the building zone. In some embodiments, the predictive model defines the temperature of the building zone as a function of heat transfer between the building zone and the HVAC equipment. In some embodiments, the predictive model includes an HVAC load model that defines an amount of heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0071] In some embodiments, the predictive optimizer is configured to predict the cost of operating the HVAC equipment as a function of an amount of heating or cooling provided by the HVAC equipment at each time step of the optimization period.

[0072] In some embodiments, the model predictive controller includes a system identifier configured to generate the predictive model by performing a system identification process. The system identification process may include collecting a set of input-output data and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of the temperature setpoint and the temperature of the building zone during each of a plurality of time steps during a learning period.

[0073] In some embodiments, performing system identification process includes modulating the temperature setpoint within a constrained temperature setpoint range and recording values of the modulated temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint.

[0074] Another implementation of the present disclosure is a method for monitoring and controlling temperature of a building zone. The method includes estimating an initial state of the building zone at a beginning of an optimization period. The state of the building zone includes a temperature of air within the building zone and a temperature of solid mass within the building zone. The method includes using a predictive model of the building zone to predict the state of the building zone at each of a plurality of time steps of the optimization period based on the estimated initial state of the building zone and a temperature setpoint trajectory including a temperature setpoint for each of the plurality of time steps. The method includes generating optimal temperature setpoints for each of the plurality of time steps by optimizing a cost function that accounts for a cost operating HVAC equipment during each of the plurality of time steps.

[0075] In some embodiments, the method includes generating the cost function and determining the cost operating the HVAC equipment during each time step using a set of time-varying utility rates comprising a utility rate value for each time step. The set of time-varying utility rates can be received from a utility provider or predicted as part of the method.

[0076] In some embodiments, the predicted state of the building zone includes a predicted temperature of the building zone. The method may include optimizing the cost function subject to a constraint on the predicted temperature of the building zone.

[0077] In some embodiments, the method includes predicting a value of an unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period. In some embodiments, the predictive model defines the temperature of the building zone as a function of the unmeasured heat load disturbance experienced by the building zone at each time step.

[0078] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of heat transfer between air within the building zone and solid mass within the building zone. In some embodiments, the predictive model defines the temperature of the building zone as a function of heat transfer between the building zone and the HVAC equipment. In some embodiments, the predictive model includes an HVAC load model that defines an amount of heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0079] In some embodiments, the method includes predicting the cost of operating the HVAC equipment as a function of an amount of heating or cooling provided by the HVAC equipment at each time step of the optimization period.

[0080] In some embodiments, the method includes generating the predictive model by performing a system identification process. The system identification process includes collecting a set of input-output data and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of the temperature setpoint and the temperature of the building zone during each of a plurality of time steps during a learning period.

[0081] Another implementation of the present disclosure is a model predictive controller for monitoring and controlling temperature of a building zone. The model predictive controller includes a state/disturbance estimator and a pre-

dictive optimizer. The state/disturbance estimator is configured to estimate an initial state of the building zone at a beginning of an optimization period. The state of the building zone includes a temperature of air within the building zone and a temperature of solid mass within the building zone. The predictive optimizer is configured to use a predictive model of the building zone to predict the state of the building zone at each of a plurality of time steps of the optimization period based on the estimated initial state of the building zone and a temperature setpoint trajectory including a temperature setpoint for each of the plurality of time steps. The predictive optimizer is configured to generate temperature setpoints for each of the plurality of time steps by optimizing a cost function that accounts for a cost operating HVAC equipment during each of the plurality of time steps.

[0082] Another implementation of the present disclosure is a thermostat for monitoring and controlling temperature of a building zone. The thermostat includes a load/rate predictor and a predictive optimizer. The load/rate predictor is configured to predict a price of one or more resources consumed by HVAC equipment to generate heating or cooling for the building zone at each of a plurality of time steps of an optimization period. The predictive optimizer is configured to generate a cost function that accounts for a cost of operating the HVAC equipment during each time step as a function of the predicted prices at each time step, use a predictive model to predict the temperature of the building zone during each of the plurality of time steps as a function of a temperature setpoint trajectory including a temperature setpoint for each of the plurality of time steps, and optimize the cost function subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of the plurality of time steps.

[0083] In some embodiments, the thermostat includes an equipment controller configured to drive the temperature of the building zone to an optimal temperature setpoint by operating the HVAC equipment to provide heating or cooling to the building zone.

[0084] In some embodiments, the predictive model defines the temperature of the building zone as a function of an unmeasured heat load disturbance. In some embodiments, the load/rate predictor is configured to predict a value of the unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period as a function of at least one of day type, time of day, building occupancy, outdoor air temperature, and weather forecasts.

[0085] In some embodiments, the thermostat includes a system identifier configured to generate the predictive model by performing a system identification process. The system identification process includes collecting a set of input-output data and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of the temperature setpoint and the temperature of the building zone during each of a plurality of time steps during a learning period.

[0086] In some embodiments, the system identification process includes modulating the temperature setpoint within a constrained temperature setpoint range and recording values of the modulated temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint.

[0087] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of heat transfer between air within the building zone and solid mass within the building zone. In some embodiments, the predictive model defines the temperature of the building zone as a function of heat transfer between the building zone and the HVAC equipment. In some embodiments, the predictive model includes an HVAC load model that defines the heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0088] In some embodiments, the predictive optimizer is configured to predict the cost of operating the HVAC equipment as a function of an amount of the heating or cooling provided by the HVAC equipment at each time step of the optimization period.

[0089] Another implementation of the present disclosure is a method performed by a thermostat for a building zone for monitoring and controlling temperature of the building zone. The method includes predicting a price of one or more resources consumed by HVAC equipment to generate heating or cooling for the building zone at each of a plurality of time steps of an optimization period, generating a cost function that accounts for a cost of operating the HVAC equipment during each time step as a function of the predicted prices at each time step, using a predictive model to predict the temperature of the building zone during each of the plurality of time steps as a function of a temperature setpoint trajectory including a temperature setpoint for each of the plurality of time steps, and optimizing the cost function subject to a constraint on the predicted temperature of the building zone to determine optimal temperature setpoints for each of the plurality of time steps.

[0090] In some embodiments, the method includes driving the temperature of the building zone to an optimal temperature setpoint by operating the HVAC equipment to provide heating or cooling to the building zone.

[0091] In some embodiments, the predictive model defines the temperature of the building zone as a function of an unmeasured heat load disturbance. In some embodiments, the method includes predicting a value of the unmeasured heat load disturbance experienced by the building zone at each of the plurality of time steps in the optimization period as a function of at least one of day type, time of day, building occupancy, outdoor air temperature, and weather forecasts.

[0092] In some embodiments, the method includes generating the predictive model by performing a system identification process. The system identification process may include collecting a set of input-output data and fitting parameters of the predictive model to the set of input-output data. The input-output data may include values of the temperature setpoint and the temperature of the building zone during each of a plurality of time steps during a learning period.

[0093] In some embodiments, performing the system identification process includes modulating the temperature setpoint within a constrained temperature setpoint range and recording values of the modulated temperature setpoint and values of the temperature of the building zone that result from modulating the temperature setpoint.

[0094] In some embodiments, the predictive model includes a thermal mass storage model that defines the temperature of the building zone as a function of heat

transfer between air within the building zone and solid mass within the building zone. In some embodiments, the predictive model defines the temperature of the building zone as a function of heat transfer between the building zone and the HVAC equipment. In some embodiments, the predictive model includes an HVAC load model that defines the heating or cooling provided by the HVAC equipment as a function of the temperature of the building zone and the temperature setpoint.

[0095] Another implementation of the present disclosure is a thermostat for monitoring and controlling temperature of a building zone. The thermostat includes a load/rate predictor and a predictive optimizer. The load/rate predictor is configured to predict a price of one or more resources consumed by HVAC equipment to generate heating or cooling for the building zone at each of a plurality of time steps of an optimization period. The predictive optimizer is configured to generate a cost function that accounts for a cost of operating the HVAC equipment during each time step as a function of the predicted prices at each time step, use a predictive model to predict the temperature of the building zone during each of the plurality of time steps as a function of a temperature setpoint trajectory including a temperature setpoint for each of the plurality of time steps, and perform an optimization of the cost function subject to a constraint on the predicted temperature of the building zone to determine temperature setpoints for each of the plurality of time steps.

[0096] Those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices and/or processes described herein, as defined solely by the claims, will become apparent in the detailed description set forth herein and taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0097] FIG. 1 is a drawing of a smart thermostat having a landscape aspect ratio, according to some embodiments.

[0098] FIG. 2 is a drawing of a smart thermostat having a portrait aspect ratio, according to some embodiments.

[0099] FIG. 3 is a block diagram of a model predictive control system including a model predictive controller and a smart thermostat which communicates with the model predictive controller via a network, according to some embodiments.

[0100] FIG. 4 is a block diagram of a model predictive control system including a smart thermostat that includes a model predictive controller and an equipment controller, according to some embodiments.

[0101] FIG. 5 is a block diagram illustrating the model predictive controller of FIGS. 3-4 in greater detail, according to some embodiments.

[0102] FIG. 6 is a block diagram illustrating various types of predictive models, according to some embodiments.

[0103] FIG. 7 is a graph of a temperature excitation signal over a one-day period, according to some embodiments.

[0104] FIG. 8 is a thermal circuit diagram that models the heat transfer characteristics of a building zone, according to some embodiments.

[0105] FIG. 9 is a flowchart of a process which can be performed by the model predictive controller of FIG. 5 to identify parameters of a predictive model, according to some embodiments.

[0106] FIG. 10 is a graph of an unfiltered temperature signal and an unfiltered sensible load signal which can be received as feedback from a controlled HVAC system, according to some embodiments.

[0107] FIG. 11 is a graph of a filtered temperature signal and a filtered sensible load signal which can be generated by filtering the signals shown in FIG. 10, according to some embodiments.

[0108] FIG. 12 is a flowchart of a process for operating the model predictive controller of FIG. 5 in a system identification mode and in an operational mode, according to some embodiments.

[0109] FIG. 13 is a flowchart of a process which can be performed by the model predictive controller of FIG. 5 to determine optimal temperature setpoints when operating in the operational mode, according to some embodiments.

[0110] FIG. 14 is a flowchart of a process which can be performed by the model predictive controller of FIG. 5 to generate a predictive model and use the predictive model to determine optimal temperature setpoints for a building zone, according to some embodiments.

[0111] FIG. 15 is a graph comparing the energy consumption of the model predictive control systems of FIGS. 3-4 with a baseline system that does not use model predictive control, according to some embodiments.

[0112] FIG. 16 is a graph of time-varying electricity prices illustrating different electricity prices at different times, according to some embodiments.

[0113] FIG. 17 is a graph comparing the building mass temperature in the model predictive control systems of FIGS. 3-4 with a baseline system that does not use model predictive control, according to some embodiments.

[0114] FIG. 18 is a graph comparing the zone air temperature in the model predictive control systems of FIGS. 3-4 with a baseline system that does not use model predictive control, according to some embodiments.

[0115] FIG. 19 is a block diagram of the computing architecture for a network of smart thermostats with model predictive control and demand response integration, according to some embodiments.

[0116] FIG. 20 is a flowchart of a process for obtaining a system model for a space served by a smart thermostat of FIG. 19, according to some embodiments.

[0117] FIG. 21 is a view of a graphical user interface displayed on a smart thermostat for prompting a user to opt-in to an energy aid mode, according to some embodiments.

[0118] FIG. 22 is a view of a graphical user interface displayed on a smart thermostat for prompting a user for permission to initiate a system identification experiment, according to some embodiments.

[0119] FIG. 23 is a view of a graphical user interface displayed on a smart thermostat for providing an indication that a system identification experiment is ongoing, according to some embodiments.

[0120] FIG. 24 is a view of a graphical user interface displayed on a smart thermostat for providing information to a user regarding an ongoing system identification experiment, according to some embodiments.

[0121] FIG. 25 is a flowchart of a process for activate a peak savings mode for the smart thermostat, according to some embodiments.

[0122] FIG. 26 is a view of a graphical user interface displayed on a smart thermostat for prompting a user to opt-in to a peak savings mode, according to some embodiments.

[0123] FIG. 27 is a view of a graphical user interface displayed on a smart thermostat for prompting a user to set temperature bounds for a peak savings mode, according to some embodiments.

[0124] FIG. 28 is a flowchart of a process for providing a peak savings mode for the smart thermostat, according to some embodiments.

[0125] FIG. 29 is a view of a graphical user interface displayed on a smart thermostat for informing a user of an upcoming peak savings period and allowing the user to opt-out for the upcoming period, according to some embodiments.

[0126] FIG. 30 is a view of a graphical user interface displayed on a smart thermostat for informing a user of a current peak savings period and allowing the user to opt-out for the current period, according to some embodiments.

[0127] FIG. 31 is a view of a graphical user interface that includes a graph showing energy savings over time, according to some embodiments.

DETAILED DESCRIPTION

Overview

[0128] Referring generally to the FIGURES, a smart thermostat with model predictive control and components thereof are shown according, to some embodiments. The smart thermostat can be configured to monitor and control one or more environmental conditions of a building zone (e.g., temperature, humidity, air quality, etc.). In some embodiments, the smart thermostat is mounted on a wall within the building zone and configured to measure the temperature, humidity, and/or other environmental conditions of the building zone. The smart thermostat can be configured to communicate with HVAC equipment that operates to affect the measured environmental conditions.

[0129] In some embodiments, the smart thermostat includes a model predictive controller and an equipment controller. In other embodiments, the model predictive controller is separate from the smart thermostat and communicates with the smart thermostat via a communications network (e.g., the Internet, a building network, etc.). The model predictive controller can be configured to determine optimal temperature setpoints T_{sp} for the building zone for each of a plurality of time steps in an optimization period. The equipment controller can be configured to receive the optimal temperature setpoints T_{sp} from the model predictive controller and can operate HVAC equipment to drive the temperature of the building zone to the optimal temperature setpoints.

[0130] To determine the optimal temperature setpoints T_{sp} , the model predictive controller can optimize an objective function (i.e., a cost function) that accounts for the cost of operating the HVAC equipment over the duration of the optimization period. The costs of operating the HVAC equipment can include, for example, the costs of resources consumed by the HVAC equipment during operation (e.g., electricity, natural gas, water, etc.), demand charges imposed by an electric utility, peak load contribution charges, equipment degradation/replacement costs, and/or other costs associated with the operation of the HVAC equipment. The

optimization performed by the model predictive controller is described in greater detail below.

[0131] The model predictive controller can optimize the objective function subject to a set of constraints. The constraints may include temperature constraints (e.g., a maximum temperature limit and a minimum temperature limit for the building zone), equipment capacity constraints, load change constraints, thermal mass storage constraints, HVAC load constraints, and/or other constraints that limit the operation of the HVAC equipment and/or describe the temperature evolution of the building zone. The model predictive controller can automatically generate the optimization constraints by performing a system identification process and generating predictive models that describe the controlled system (i.e., the HVAC equipment and the building zone).

[0132] In some embodiments, the model predictive controller generates a thermal mass storage model that describes the temperature of the air within the building zone T_{ia} as a function of several physical parameters and variables. The physical parameters in the thermal mass storage model may include, for example, the thermal capacitance C_{ia} of the air in the building zone, the thermal capacitance C_m of the solid mass within the building zone, the thermal resistance R_{mi} between the air within the building zone and the solid mass, the thermal resistance R_{oi} between the air within the building zone and the ambient environment, etc. The variables in the thermal mass storage model may include the temperature of air within the building zone T_{ia} , the temperature of the solid mass within the building zone T_m , the outside air temperature T_{oa} , the amount of heating or cooling provided by the HVAC equipment $\dot{Q}_{HVAC}$, and/or a thermal energy load disturbance (i.e., a heat load disturbance) $\dot{Q}_{other}$.

[0133] In some embodiments, the model predictive controller generates an HVAC load model. The HVAC load model may describe the amount of heating or cooling $\dot{Q}_{HVAC}$ provided by the HVAC equipment as a function of the temperature setpoint T_{sp} and the temperature of the air T_{ia} within the building zone. The model predictive controller can use the thermal mass storage model and the HVAC load model to predict system states (i.e., building mass temperature T_m , building air temperature T_{ia} , etc.), predict the load disturbance $\dot{Q}_{other}$, and establish constraints on the optimization performed by the model predictive controller.

[0134] In some embodiments, the model predictive controller uses the thermal mass storage model and the HVAC load model to take advantage of the building mass as a thermal energy storage medium. For example, thermal energy can be stored in the solid building mass (i.e., pre-cooling or preheating the building zone) to allow the model predictive controller to take advantage of time-varying utility rates and demand charges when determining the optimal temperature setpoints T_{sp} . To precool the building zone, thermal energy may be removed from the thermal mass during periods when energy prices are lowest. During periods when energy prices are largest, thermal energy may be moved back into the solid mass from the zone air, which reduces the sensible HVAC load $\dot{Q}_{HVAC}$ needed to maintain the zone air within a comfortable temperature range. Similarly, the model predictive controller can preheat the solid mass within the building zone when energy prices are low and use the stored thermal energy to warm the zone air when energy prices are higher.

[0135] In some embodiments, the model predictive controller is configured to operate in two distinct modes: (1) parameter identification mode and (2) operational mode. In the parameter identification mode, the model predictive controller can manipulate the temperature setpoint T_{sp} provided to the smart thermostat and/or the equipment controller to induce a dynamic response in the zone temperature T_{ia} . This procedure of inducing a dynamic response is referred to as the parameter identification experiment or system identification process and can be used to identify the values of the parameters in the thermal mass storage model and the HVAC load model. In the operational mode, the model predictive controller can solve an optimal control problem to determine the optimal temperature setpoint trajectory T_{sp} (i.e., a time series of temperature setpoints) that minimizes the cost of energy consumed by the HVAC equipment over the duration of the optimization period. The optimal temperature setpoints T_{sp} can be provided to the smart thermostat and/or the equipment controller for use in generating operating commands for the HVAC equipment.

[0136] The setpoint T_{sp} computed for the first time step of the optimization period can be sent to the smart thermostat and/or the equipment controller to be implemented during the first time step. The smart thermostat and/or the equipment controller can turn stages of HVAC equipment on and/or off such that the temperature T_{ia} of the zone is forced to and then maintained at the temperature setpoint T_{sp} . At the next time step, the model predictive controller receives updated information (i.e., feedback on the zone conditions), resolves the optimal control problem, and sends the temperature setpoint T_{sp} for the next time step to the smart thermostat and/or the equipment controller.

[0137] The model predictive controller can collect feedback on the zone conditions (e.g., zone temperature T_{ia} , outdoor air temperature T_{oa} , heating/cooling load $\dot{Q}_{HVAC}$) when operating in the parameter identification mode. The zone conditions can be monitored and recorded over the identification experiment, along with the commanded zone temperature setpoint T_{sp} , as input-output data (i.e., training data). The model predictive controller can use the input-output data to compute the zone thermal model parameters by performing a system identification process. The feedback received from the smart thermostat may include the current minimum and maximum allowable setpoint to limit the adjustments to the zone temperature setpoint T_{sp} made during the identification experiment. These and other features of the smart thermostat and/or the model predictive controller are described in greater detail below.

[0138] Throughout this disclosure, the terms “optimal,” “optimized,” “optimum,” and the like are used to refer to values (e.g., temperature setpoint values, HVAC load values, etc.) that are determined by performing an optimization process. Similarly, the term “optimizing” is used to refer to the process of performing an optimization. In some instances, the values resulting from the optimization process are true optimal values (i.e., values that achieve the minimum possible value or maximum possible value for a performance variable given the constraints on the optimization process). In other instances, the values resulting from the optimization process are not true optimal values. This can result from imperfect information used to perform the optimization, inaccuracies in the predictive model used to constrain the optimization, and/or various other factors that can prevent the optimization from converging on the true

optimal values. The terms “optimal,” “optimized,” “optimum,” and the like should be interpreted to include any values that result from performing an optimization, regardless of whether the values are true optimal values. Similarly, the term “optimizing” should be interpreted to include the process of performing an optimization, regardless of whether that optimization converges on the true optimal values.

Smart Thermostat

[0139] Referring now to FIGS. 1-2, a smart thermostat **100** is shown, according to some embodiments. Smart thermostat **100** can be configured to monitor and control one or more environmental conditions of a building zone (e.g., temperature, humidity, air quality, etc.). In some embodiments, smart thermostat **100** is mounted on a wall within the building zone and configured to measure the temperature, humidity, and/or other environmental conditions of the building zone. Smart thermostat **100** can be configured to communicate with HVAC equipment that operates to affect the measured environmental conditions.

[0140] Smart thermostat **100** can be installed in a home, office building, school, hospital, or any other environment-controlled space. In residential implementations, the HVAC equipment controlled by smart thermostat **100** may include a home furnace, air conditioning unit, and/or other types of residential HVAC equipment. In commercial implementations, the HVAC equipment may include one or more chillers, boilers, air handling units, rooftop units, dampers, or other types of HVAC equipment configured to affect the environment of a building zone. It is contemplated that smart thermostat **100** can be configured to control any type of HVAC equipment.

[0141] In some embodiments, smart thermostat **100** includes a communications interface that enables smart thermostat **100** to connect to a communications network (e.g., a local area network, the Internet, a cellular network, etc.) and communicate with various external systems and devices (e.g., user devices, remote servers, remote controllers, etc.). For example, smart thermostat **100** can be configured to receive weather forecasts from a remote weather service via the Internet. In some embodiments, smart thermostat **100** receives temperature setpoints from a model predictive controller via the communications network. In other embodiments, the model predictive controller is a component of smart thermostat **100**.

[0142] Smart thermostat **100** is shown to include a display screen **102** and a base **104**. Base **104** can attach to a wall or other surface upon which smart thermostat **100** is mounted. Display screen **102** may be transparent or semi-transparent (e.g., an organic LED display) and can be configured to display text, graphics, and other information for presentation to a user. In some embodiments, display screen **102** is touch-sensitive (e.g., a capacitive or resistive touch screen) and configured to receive user input. Display screen **102** may have a landscape aspect ratio as shown in FIG. 1 (i.e., the width of display screen **102** may exceed the height of display screen **102**) or a portrait aspect ratio as shown in FIG. 2 (i.e., the height of display screen **102** may exceed the width of display screen **102**). Display screen **102** may be cantilevered from base **104** (i.e., attached to base **104** along a single edge of display screen **102**) and may extend from base **104** in a direction substantially parallel to the mounting surface.

[0143] Smart thermostat **100** can include some of all of the features of the thermostats described in U.S. patent application Ser. No. 15/143,373 filed Apr. 29, 2016, U.S. patent application Ser. No. 15/146,763 filed May 4, 2016, U.S. patent application Ser. No. 15/146,749 filed May 4, 2016, U.S. patent application Ser. No. 15/146,202 filed May 4, 2016, U.S. patent application Ser. No. 15/146,134 filed May 4, 2016, U.S. patent application Ser. No. 15/146,649 filed May 4, 2016, U.S. Provisional Patent Application No. 62/331,863 filed May 4, 2016, U.S. Provisional Patent Application No. 62/352,955 filed Jun. 21, 2016, U.S. patent application Ser. No. 15/298,191 filed Oct. 19, 2016, U.S. patent application Ser. No. 15/336,793 filed Oct. 28, 2016, U.S. patent application Ser. No. 15/336,792 filed Oct. 28, 2016, U.S. patent application Ser. No. 15/336,789 filed Oct. 28, 2016, U.S. patent application Ser. No. 15/338,221 filed Oct. 28, 2016, U.S. patent application Ser. No. 15/338,215 filed Oct. 28, 2016, U.S. patent application Ser. No. 15/336,791 filed Oct. 28, 2016, U.S. patent application Ser. No. 15/397,722 filed Jan. 3, 2017, and/or U.S. Provisional Patent Application No. 62/446,296 filed Jan. 13, 2017. The entire disclosure of each of these patent applications is incorporated by reference herein.

Model Predictive Control Systems

[0144] Referring now to FIGS. 3-4, a pair of model predictive control (MPC) systems **300** and **400** are shown, according to some embodiments. MPC system **300** is shown to include a model predictive controller **302**, a communications network **304**, smart thermostat **100**, HVAC equipment **308**, and a building zone **310**. In MPC system **300**, model predictive controller **302** is separate from smart thermostat **100** and configured to provide a temperature setpoint T_{sp} to smart thermostat **100** via communications network **304**. In MPC system **400**, model predictive controller **302** is a component of smart thermostat **100** along with an equipment controller **406**.

[0145] Building zone **310** can include one or more rooms or zones within a home, office building, school, hospital, or any other environment-controlled space. HVAC equipment **308** can include any type of equipment operable to affect the temperature, humidity, and/or other environmental conditions of building zone **310**. For example, HVAC equipment **308** can include a home furnace, air conditioning unit, one or more chillers, boilers, air handling units, rooftop units, dampers, or other types of HVAC equipment configured to affect the environment of building zone **310**. In some embodiments, other building devices (e.g., lighting, window shades) can be controlled by the smart thermostat **100** to affect heat disturbances at the building zone **310**.

[0146] Model predictive controller **302** can be configured to determine an optimal temperature setpoint T_{sp} for smart thermostat **100** for each of a plurality of time steps during an optimization period. Smart thermostat **100** can use the temperature setpoints T_{sp} provided by model predictive controller **302** to generate equipment commands for HVAC equipment **308**. The equipment commands can be generated by smart thermostat **100** and/or equipment controller **406** (e.g., an on/off controller, an equipment staging controller, a proportional-integral (PI) controller, a proportional-integral-derivative (PID) controller, etc.). HVAC equipment **308** operate according to the equipment commands to provide variable amount of heating or cooling $\dot{Q}_{HVAC}$ to building zone **310**. By controlling the temperature setpoint T_{sp} , model

predictive controller **302** can modulate the amount of heating or cooling $\dot{Q}_{HVAC}$ provided by HVAC equipment **308**, thereby affecting the temperature of building zone **310**.

[0147] To determine the optimal temperature setpoints T_{sp} , model predictive controller **302** can optimize an objective function (i.e., a cost function) that accounts for the cost of operating HVAC equipment **308** over the duration of the optimization period. The costs of operating HVAC equipment **308** can include, for example, the costs of resources consumed by HVAC equipment **308** during operation (e.g., electricity, natural gas, water, etc.), demand charges imposed by an electric utility, peak load contribution charges, equipment degradation/replacement costs, and/or other costs associated with the operation of HVAC equipment **308**. The optimization performed by model predictive controller **302** is described in greater detail with reference to FIG. 5.

[0148] Model predictive controller **302** can optimize the objective function subject to a set of constraints. The constraints may include temperature constraints (e.g., a maximum temperature limit and a minimum temperature limit for building zone **310**), equipment capacity constraints, load change constraints, thermal mass storage constraints, HVAC load constraints, and/or other constraints that limit the operation of HVAC equipment **308** and/or describe the temperature evolution of building zone **310**. Model predictive controller **302** can automatically generate the optimization constraints by performing a system identification process and generating predictive models that describe the controlled system (i.e., HVAC equipment **308** and building zone **310**).

[0149] In some embodiments, model predictive controller **302** generates a thermal mass storage model that describes the temperature of the air within building zone **310** T_{ia} as a function of several physical parameters and variables. The physical parameters in the thermal mass storage model may include, for example, the thermal capacitance C_{ia} of the air in building zone **310**, the thermal capacitance C_m of the solid mass within building zone **310**, the thermal resistance R_{mi} between the air within building zone **310** and the solid mass, the thermal resistance R_{oi} between the air within building zone **310** and the ambient environment, etc. The variables in the thermal mass storage model may include the temperature of air within building zone **310** T_{ia} , the temperature of the solid mass within building zone **310** T_m , the outside air temperature T_{oa} , the amount of heating or cooling provided by HVAC equipment **308** $\dot{Q}_{HVAC}$, and/or a thermal energy load disturbance $\dot{Q}_{other}$.

[0150] In some embodiments, model predictive controller **302** generates a HVAC load model. The HVAC load model may describe the amount of heating or cooling $\dot{Q}_{HVAC}$ provided by HVAC equipment **308** as a function of the temperature setpoint T_{sp} and the temperature of the air T_{ia} within building zone **310**. Model predictive controller **302** can use the thermal mass storage model and the HVAC load model to predict system states (i.e., building mass temperature T_m , building air temperature T_{ia} , etc.), predict the load disturbance $\dot{Q}_{other}$, and establish constraints on the optimization performed by model predictive controller **302**.

[0151] In some embodiments, model predictive controller **302** uses the thermal mass storage model and the HVAC load model to take advantage of the building mass as a thermal energy storage medium. For example, thermal energy can be stored in the solid building mass (i.e., precooling or preheating building zone **310**) to allow model predictive con-

troller **302** to take advantage of time-varying utility rates and demand charges when determining the optimal temperature setpoints T_{sp} . To precool building zone **310**, thermal energy may be removed from the thermal mass during periods when energy prices are lowest. During periods when energy prices are largest, thermal energy may be moved back into the solid mass from the zone air, which reduces the sensible HVAC load $\dot{Q}_{HVAC}$ needed to maintain the zone air within a comfortable temperature range. Similarly, model predictive controller **302** can preheat the solid mass within building zone **310** when energy prices are low and use the stored thermal energy to warm the zone air when energy prices are higher. In some embodiments, the model predictive controller **302** is configured to provide setpoints for window shades and/or lighting devices, allowing the model predictive controller **302** to affect an amount of heat provided to the building zone by solar irradiance and/or by the operation of lighting devices in the building zone.

[0152] In some embodiments, model predictive controller **302** is configured to operate in two distinct modes: (1) parameter identification mode and (2) operational mode. In the parameter identification mode, model predictive controller **302** can manipulate the temperature setpoint T_{sp} provided to smart thermostat **100** and/or equipment controller **406** to induce a dynamic response in the zone temperature T_{ia} . This procedure of inducing a dynamic response is referred to as the parameter identification experiment or system identification process and can be used to identify the values of the parameters in the thermal mass storage model and the HVAC load model. In the operational mode, model predictive controller **302** can solve an optimal control problem to determine the optimal temperature setpoint trajectory T_{sp} (i.e., a time series of temperature setpoints) that minimizes the cost of energy consumed by HVAC equipment **308** over the duration of the optimization period. The optimal temperature setpoints T_{sp} can be provided to smart thermostat **100** and/or equipment controller **406** for use in generating operating commands for HVAC equipment **308**.

[0153] The setpoint T_{sp} computed for the first time step of the optimization period can be sent to smart thermostat **100** and/or equipment controller **406** to be implemented during the first time step. Smart thermostat **100** and/or equipment controller **406** can turn stages of HVAC equipment **308** on and/or off such that the temperature T_{ia} of the zone is forced to and then maintained at the temperature setpoint T_{sp} . At the next time step, model predictive controller **302** receives updated information (i.e., feedback on the zone conditions), resolves the optimal control problem, and sends the temperature setpoint T_{sp} for the next time step to smart thermostat **100** and/or equipment controller **406**.

[0154] Model predictive controller **302** can collect feedback on the zone conditions (e.g., zone temperature T_{ia} , outdoor air temperature T_{oa} , heating/cooling load $\dot{Q}_{HVAC}$) when operating in the parameter identification mode. The zone conditions can be monitored and recorded over the identification experiment, along with the commanded zone temperature setpoint T_{sp} , as input-output data (i.e., training data). Model predictive controller **302** can use the input-output data to compute the zone thermal model parameters by performing a system identification process. The feedback received from smart thermostat **100** may include the current minimum and maximum allowable setpoint to limit the adjustments to the zone temperature setpoint T_{sp} made during the identification experiment.

[0155] In some embodiments, a pseudo-random binary signal (PRBS) is generated that is subsequently used to produce a signal that takes values in the set $\{T_{sp,1}, T_{sp,2}\}$, where $T_{sp,1}$ and $T_{sp,2}$ are maximum and minimum allowable temperature setpoints depending on mode of operation of smart thermostat **100** (e.g., heating, cooling, home, sleep, or away mode). The minimum and maximum temperature setpoints may be time-varying to account for the different operational modes of smart thermostat **100** (e.g., home, sleep, and away). In some embodiments, smart thermostat **100** uses occupancy detection and/or user feedback to change the mode.

[0156] To determine the optimal temperature setpoints T_{sp} in the operational mode, model predictive controller **302** can use the thermal model of building zone **310** to predict the zone temperature T_{ia} and HVAC fuel consumption over the optimization period, given a forecast of the weather and heat disturbance load on the zone (e.g., heat generated from people and electrical equipment and gained through solar radiation). The zone temperature T_{ia} may be subject to comfort constraints. The predicted trajectories of the zone temperature T_{ia} , temperature setpoint T_{sp} , and HVAC fuel consumption can be sent to smart thermostat **100** to be displayed to the user allowing the user to better understand the setpoint decision made by model predictive controller **302**.

[0157] In some embodiments, smart thermostat **100** allows a user to override the optimal temperature setpoints computed by model predictive controller **302**. For example, the smart thermostat **100** may operate the equipment **308** using the user-input temperature setpoint for a preset amount of time after the user input is received before switching back to the optimal temperature setpoints. As another example, a user-input temperature setpoint or schedule remains in control until an instruction is received from the user to use the optimal temperature setpoints from the model predictive controller **302**. As another example, the user-input temperature setpoint may be used to define one or more constraints on an optimization performed by the model predictive controller **302**. For example, a predicted temperature may be constrained to be within a preset error of the temperature setpoint input by the user while the model predictive controller **302** continues to operate as described in greater detail below.

[0158] Smart thermostat **100** may include a savings estimator configured to determine a potential cost savings resulting from the optimal temperature setpoints relative to the user-specified temperature setpoints. Smart thermostat **100** can be configured to display the potential cost savings via display screen **102** to inform the user of the economic cost predicted to result from overriding the optimal temperature setpoints. These and other features of model predictive controller **302** are described in greater detail below.

Model Predictive Controller

[0159] Referring now to FIG. 5, a block diagram illustrating model predictive controller **302** in greater detail is shown, according to some embodiments. Model predictive controller **302** is shown to include a communications interface **502** and a processing circuit **504**. Communications interface **502** may facilitate communications between model predictive controller **302** and external systems or devices. For example, communications interface **502** may receive measurements of zone conditions from smart thermostat

100, HVAC equipment **308**, and/or building zone **310**. Zone conditions may include the temperature T_{ia} of the air within building zone **310**, the outside air temperature T_{oa} , and/or the amount of heating/cooling $\dot{Q}_{HVAC}$ provided to building zone **310** at each time step of the optimization period. Communications interface **502** may also receive temperature constraints for building zone **310** and/or equipment constraints for HVAC equipment **308**. In some embodiments, communications interface **502** receives utility rates from utilities **524** (e.g., time-varying energy prices, demand charges, etc.), weather forecasts from a weather service **526**, and/or historical load and rate data from historical data **528**. In some embodiments, model predictive controller **302** uses communications interface **502** to provide temperature setpoints T_{sp} to smart thermostat **100** and/or equipment controller **406**.

[0160] Communications interface **502** may include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications external systems or devices. In various embodiments, the communications may be direct (e.g., local wired or wireless communications) or via a communications network (e.g., a WAN, the Internet, a cellular network, etc.). For example, communications interface **502** can include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, communications interface **502** can include a Wi-Fi transceiver for communicating via a wireless communications network or cellular or mobile phone communications transceivers.

[0161] Processing circuit **504** is shown to include a processor **506** and memory **508**. Processor **506** may be a general purpose or specific purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable processing components. Processor **506** is configured to execute computer code or instructions stored in memory **508** or received from other computer readable media (e.g., CDROM, network storage, a remote server, etc.).

[0162] Memory **508** may include one or more devices (e.g., memory units, memory devices, storage devices, etc.) for storing data and/or computer code for completing and/or facilitating the various processes described in the present disclosure. Memory **508** may include random access memory (RAM), read-only memory (ROM), hard drive storage, temporary storage, non-volatile memory, flash memory, optical memory, or any other suitable memory for storing software objects and/or computer instructions. Memory **508** may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. Memory **508** may be communicably connected to processor **506** via processing circuit **504** and may include computer code for executing (e.g., by processor **506**) one or more processes described herein. When processor **506** executes instructions stored in memory **508** for completing the various activities described herein, processor **506** generally configures controller **302** (and more particularly processing circuit **504**) to complete such activities.

[0163] Still referring to FIG. 5, model predictive controller **302** is shown to include a system identifier **510**, a load/rate predictor **518**, a state/disturbance estimator **520**, and a predictive optimizer **522**. In brief overview, system identifier **510** can be configured to perform a system identification process to identify the values of the parameters in the building thermal mass model and the HVAC load model. The parameters identified by system identifier **510** may define the values of the system matrices A, B, C, and D and the estimator gain K. State/disturbance estimator **520** can be configured to compute an estimate of the current state $\hat{x}(k)$ and unmeasured disturbance $\hat{d}(k)$. The estimation performed by state/disturbance estimator **520** can be performed at each sample time and may use the system matrices A, B, C, D and estimator gain K identified by system identifier **510**.

[0164] Load/rate predictor **518** can generate predictions or forecasts of the internal heat load $\dot{Q}_{other}$, the non-HVAC building electricity consumption (used to calculate electrical demand charges), and the utility rates at each time step of the optimization period as a function of historical data **528** and several key predictor variables (e.g., time-of-day, outdoor air conditions, etc.). The models used by load/rate predictor **518** may be different than that identified by system identifier **510**. In some embodiments, load/rate predictor **518** uses the prediction techniques described in U.S. patent application Ser. No. 14/717,593 filed May 20, 2015, the entire disclosure of which is incorporated by reference herein.

[0165] Predictive optimizer **522** can manipulate the temperature setpoint T_{sp} of building zone **310** to minimize the economic cost of operating HVAC equipment **308** over the duration of the optimization period using the building thermal mass storage model, the HVAC load model, and state and disturbance estimates and forecasts. In some embodiments, predictive optimizer **522** uses open-loop predictions to optimize the temperature setpoint T_{sp} trajectory over the optimization period. At each time step of the optimization period, the predictions can be updated based on feedback from the controlled system (e.g., measured temperatures, measured HVAC loads, etc.).

[0166] Before discussing system identifier **510**, load/rate predictor **518**, state/disturbance estimator **520**, and predictive optimizer **522** in detail, the notation used throughout the remainder of the present disclosure and the class of system model used by model predictive controller **302** are explained. The set of integers is denoted by $\mathbb{I}$ and the set of positive integers is denoted by $\mathbb{I}_{\geq 0}$, and the set of integers contained in the interval $[a, b]$ is denoted by $\mathbb{I}_{a:b}$. For a time-dependent vector $x(k) \in \mathbb{R}^n$, $\hat{x}(ik) \in \mathbb{R}^n$ denotes the estimated value of x at time step $i \in \mathbb{I}_{\geq 0}$ given the measurement at time step $k \in \mathbb{I}_{\geq 0}$ where $i \geq k$ and $\hat{\tilde{x}}(ik) \in \mathbb{R}^n$ denotes the (open-loop) predicted value of x at time step $i \in \mathbb{I}_{\geq 0}$ with the prediction initialized at time $k \in \mathbb{I}_{\geq 0}$ ($i \geq k$). For notational simplicity, the notation of the predicted value of x at time i starting from time k is abbreviated to $\hat{\tilde{x}}(i)$. Boldface letters are used to represent a sequence with cardinality $N \in \mathbb{I}_{\geq 0}$ (i.e., $x := \{x(0), \dots, x(N-1)\}$). The notation $\cdot^*$ (e.g., x^*) denotes an optimal quantity with respect to some optimization problem.

[0167] In some embodiments, the model generated and used by model predictive controller **302** is a discrete-time linear time-invariant model, as shown in the following equation:

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned} \quad (\text{Equation 1})$$

where $k \in \mathbb{N}_{\geq 0}$ is the time index, $x(k) \in \mathbb{R}^n$ is the state vector, $y(k) \in \mathbb{R}^p$ is the measured output, and $u(k) \in \mathbb{R}^m$ is the input vector. The matrix A can be assumed to be stable; that is the real parts of the eigenvalues of A lie within the unit circle. The pair (A, B) can be assumed to be controllable and the pair (A, C) can be assumed to be observable. Some HVAC equipment **308** and devices are only operated at finite operating stages. In the building models considered, this gives rise to discrete outputs whereby the measured output takes values within a finite set of numbers: $y_i(k) \in \{y_{i,stage,1}, y_{i,stage,2}, \dots, y_{i,stage,m}\}$ for some i . In this case, the output does not evolve according to Equation 1.

[0168] In some embodiments, the model generated and used by model predictive controller **302** is a grey box continuous-time models derived from thermal resistance-capacitance (RC) modeling principals. Model predictive controller **302** can be configured to parameterize the resulting continuous-time thermal models and identify the continuous-time model parameters. The continuous-time version of the model of Equation 1 can be represented as shown in the following equation

$$\begin{aligned} \dot{x}(t) &= A_c x(t) + B_c u(t) \\ y(t) &= C_c x(t) + D_c u(t) \end{aligned} \quad (\text{Equation 2})$$

where $t \geq 0$ is continuous time (the initial time is taken to be zero) and A_c, B_c, C_c, D_c are the continuous-time versions of the discrete-time system matrices A, B, C , and D . The values of the parameters in the matrices A, B, C , and D as well as an estimator gain K can be determined by system identifier **510** by performing a system identification process (described in greater detail below)

State/Disturbance Estimator

[0169] State/disturbance estimator **520** can be configured to compute an estimate of the current state $\hat{x}(k)$ and unmeasured disturbance $\hat{d}(k)$. The estimation performed by state/disturbance estimator **520** can be performed at each sample time (i.e., when operating in the operational mode) and may use the system matrices A, B, C, D and estimator gain K identified by system identifier **510**. In some embodiments, state/disturbance estimator **520** modifies the system model to account for measurement noise and process noise, as shown in the following equation:

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) + w(k) \\ y(k) &= Cx(k) + Du(k) + v(k) \end{aligned} \quad (\text{Equation 3})$$

where $w(k)$ is the process noise and $v(k)$ is the measurement noise.

[0170] State/disturbance estimator **520** can use the modified system model of Equation 3 to estimate the system state $\hat{x}(k)$ as shown in the following equation:

$$\begin{aligned} \hat{x}(k+1|k) &= A\hat{x}(k|k-1) + Bu(k) + K(y(k) - \hat{y}(k|k-1)) \\ \hat{y}(k|k-1) &= C\hat{x}(k|k-1) + Du(k) \end{aligned} \quad (\text{Equation 4})$$

where $\hat{x}(k+1|k)$ is the estimated/predicted state at time step $k+1$ given the measurement at time step k , $\hat{y}(k|k-1)$ is the predicted output at time step k given the measurement at time step $k-1$, and K is the estimator gain. Equation 4 describes the prediction step of a Kalman filter. The Kalman filter may be the optimal linear filter under some ideal assumptions (e.g., the system is linear, no plant-model mismatch, the process and measurement noise are follow a white noise distribution with known covariance matrices).

[0171] In some embodiments, the estimated state vector $\hat{x}(k+1|k)$, the output vector $\hat{y}(k|k-1)$, and the input vector $u(k)$ are defined as follows:

$$\begin{aligned} \hat{x}(k+1|k) &= \begin{bmatrix} \hat{T}_{ia}(k+1|k) \\ \hat{T}_m(k+1|k) \\ \hat{I}(k+1|k) \end{bmatrix}, \\ \hat{y}(k|k-1) &= \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{Q}_{HVAC}(k|k-1) \end{bmatrix}, \quad u(k) = \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} \end{aligned}$$

where $\hat{T}_{ia}$ is an estimate of the zone air temperature T_{ia} , $\hat{T}_m$ is an estimate of the zone mass temperature T_m , $\hat{I}$ is an estimate of the integrating disturbance I , $\hat{Q}_{HVAC}$ is an estimate of the heating or cooling load provided by HVAC equipment **308**, T_{sp} is the temperature setpoint, and T_{oa} is the outdoor air temperature.

[0172] The state estimation described by Equation 4 is effective in addressing stationary noise. However, unmeasured correlated disturbances are also common in practice. Correlated disturbances can lead to biased estimates/predictions. In some embodiments, state/disturbance estimator **520** accounts for disturbances by adding a disturbance state to the system model, as shown in the following equation:

$$\begin{aligned} \begin{bmatrix} \hat{x}(k+1|k) \\ \hat{d}(k+1|k) \end{bmatrix} &= \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + \\ &\quad \begin{bmatrix} B \\ 0 \end{bmatrix} u(k) + \begin{bmatrix} K_x \\ K_d \end{bmatrix} (y(k) - \hat{y}(k|k-1)) \\ \hat{y}(k|k-1) &= \begin{bmatrix} C & C_d \end{bmatrix} \begin{bmatrix} \hat{x}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + Du(k) \end{aligned} \quad (\text{Equation 5})$$

where $\hat{d}(k)$ is the unmeasured disturbance at time step k . The effect of adding the disturbance state is similar to that of the integral action of a proportional-integral controller. State/disturbance estimator **520** may select values of B_d and C_d and modify the estimation problem as shown in Equation 5. In some embodiments, state/disturbance estimator **520** selects values of B_d and C_d such that the system model is an integrating input and/or output disturbance model.

[0173] In some embodiments, the choice of disturbance models may have a significant impact on the ability of model predictive controller **302** to adequately estimate the lumped thermal building mass temperature T_m . Accurate estimation

of the thermal mass temperature T_m enables model predictive controller **302** to effectively use the solid building mass to store thermal energy when preheating or precooling building zone **310** in order to time-shift the thermal energy load and minimize the cost of operating HVAC equipment **308**.

System Identifier

[0174] System identifier **510** can be configured to perform a system identification process to identify the values of the parameters in the building thermal mass model and the HVAC load model. The functions performed by system identifier **510** may be performed when operating in the parameter identification mode. The parameters identified by system identifier **510** may define the values of the system matrices A, B, C, and D and the estimator gain K. System identifier **510** can perform the system identification process during the commissioning phase of the control system and/or after the detection of a substantial change in the model parameters that has resulted in significant closed-loop performance deterioration. In some embodiments, the system identification process performed by system identifier **510** is the same or similar to the system identification process described in U.S. Pat. No. 9,235,657 titled "System Identification and Model Development" and granted Jan. 12, 2016. The entire disclosure of U.S. Pat. No. 9,235,657 is incorporated by reference herein.

[0175] System identification (SI) is the art and science of building mathematical models of dynamic systems from observed input-output data. In general, a model for a system may be developed using three paradigms: white box, grey box or black box models. These three system identification modeling paradigms are summarized in FIG. 6.

[0176] For applications that require high fidelity models, white box models may be derived via detailed first-principles modeling approaches. This approach often requires a high degree of knowledge of the physical system and may result in a high-order nonlinear dynamic model. While the resulting model may capture most of the relevant physical behavior a system, there is usually a very high engineering investment required to develop such a model. Given the complexities of the resulting models, often white box models cannot be used in real-time predictive controllers owing to real-time computational restrictions. Nevertheless, white box models may help to evaluate control approaches in simulation to identify and mitigate potential pitfalls of the approach.

[0177] In a black box modeling paradigm, little a priori knowledge about the model structure is assumed except for standard assumptions like linearity and stability of the model. Black box methods fit the best model to the input-output data. While there is little to no assumed a priori knowledge about the model, which offers a deal of flexibility, there are a few drawbacks to the approach. Given that little about the model structure is assumed, black box models often tend to have a large amount of parameters, which requires a large amount of data to properly select the model order and appropriate input-output representation. To capture the necessary amount of input-output data needed for black box methods, an undesirable length for the system identification experiment may be required.

[0178] Finally, grey box modeling assumes that there is a known model structure often derived through simplified first-principals or semi-physical modeling. The model struc-

ture is parameterized and parameter estimation techniques are used to estimate the parameters. Owing to the incorporation of more a priori knowledge into grey box models, the resulting number of parameters for these types of models are often less than black box models. As a result of fewer parameters as well as the fact that the model structure (given it is valid) already encodes important relationships, less input-output data is typically needed to fit a good model with respect to that needed to fit a good model for black box methods.

[0179] In some embodiments, system identifier **510** uses a grey box model to represent HVAC equipment **308** and building zone **310** (i.e., the controlled system). In a grey box model system identification approach, a model structure $\mathcal{M}$ is selected, which is subsequently parameterized with a parameter $\theta \in D_{\mathcal{M}} \subset \mathbb{R}^d$ where $D_{\mathcal{M}}$ is the set of admissible model parameter values. The resulting possible models are given by the set:

$$\mathcal{M} = \{\mathcal{M}(\theta) | \theta \in D_{\mathcal{M}}\} \quad (\text{Equation 6})$$

[0180] To fit a proper parameter vector to the controlled system, system identifier **510** can collect input-output data. The input-output data may include samples of the input vector $u(k)$ and corresponding samples of the output vector $y(k)$, as shown in the following equation:

$$Z^N = [y(1), u(1), y(2), u(2), \dots, y(N), u(N)] \quad (\text{Equation 7})$$

where $N > 0$ is the number of samples collected. Parameter estimation is the problem of selecting the proper value $\hat{\theta}_N$ of the parameter vector given the input-output data Z^N (i.e., it is the mapping: $Z^N \rightarrow \hat{\theta}_N \in D_{\mathcal{M}}$).

[0181] In some embodiments, system identifier **510** uses a prediction error method to select the values of the parameter vector $\hat{\theta}_N$. A prediction error method (PEM) refers to a particular family of parameter estimation methods that is of interest in the context of the present disclosure. The prediction error method used by system identifier **510** may include fitting the parameter vector $\hat{\theta}_N$ by minimizing some function of the difference between the predicted output and observed output. For example, let $\hat{y}(k, \theta)$ be the predicted output at time step k given the past input-output sequence Z^{k-1} and the model $\mathcal{M}(\theta)$ for $\theta \in D_{\mathcal{M}}$. The prediction error at time step k is given by:

$$\varepsilon(k, \theta) := y(k) - \hat{y}(k, \theta) \quad (\text{Equation 8})$$

[0182] In some embodiments, system identifier **510** filters the prediction error sequence through a stable linear filter and defines the following prediction performance metric:

$$V_N(\theta, Z^N) = \frac{1}{N} \sum_{k=1}^N \ell(\varepsilon_F(k, \theta)) \quad (\text{Equation 9})$$

where $\ell(\bullet)$ is the cost function (e.g., a positive definite function) and $\varepsilon_f(k, \theta)$ is the filtered prediction error. The parameter estimate $\hat{\theta}_N$ of a prediction error method is then given by:

$$\hat{\theta}_N = \hat{\theta}_N(Z^N) = \operatorname{argmin}_{\theta \in D_M} V_N(\theta, Z^N) \quad (\text{Equation 10})$$

[0183] In various embodiments, system identifier **510** can use any parameter estimation method that corresponds to Equation 10. For example, system identifier **510** can define a quadratic cost function that is the square of the prediction errors, as shown in the following equation:

$$\ell(\varepsilon(k, \theta)) = \ell(y(k) - \hat{y}(k|k-1, \theta)) = \|y(k) - \hat{y}(k|k-1, \theta)\|_2^2 \quad (\text{Equation 11})$$

where $\hat{y}(k|k-1, \theta)$ denotes the one-step ahead prediction of the output using the model $\mathcal{M}(\theta)$. When the prediction errors are independently and identically distributed random variables from a normal distribution (i.e., the process and measurement noise is Gaussian white noise) and the model being identified is linear, the cost function of Equation 11 is optimal from a statistical point-of-view.

[0184] Still referring to FIG. 5, system identifier **510** is shown to include a thermal mass storage model generator **512**, a HVAC load model generator **514**, and an estimator gain identifier **516**. Thermal mass storage model generator **512** can be configured to estimate or identify the parameters of a thermal mass storage model that describes the relationship between indoor air temperature T_{ia} , building mass temperature T_m , outdoor air temperature T_{oa} , internal load $\dot{Q}_{other}$, and sensible HVAC load $\dot{Q}_{HVAC}$. The indoor air temperature T_{ia} represents the temperature of the air within building zone **310**. The building mass temperature T_m represents the temperature of the solid objects within building zone **310**. The internal load $\dot{Q}_{other}$ represents heat generated internally within building zone **310** by building occupants, electronics, resistive loads, and other sources of heat. The sensible HVAC load $\dot{Q}_{HVAC}$ represents the amount of heating or cooling applied to building zone **310** by HVAC equipment **308**.

[0185] HVAC load model generator **514** can be configured to generate a HVAC load model that describes the relationship between indoor air temperature T_{ia} , the temperature setpoint T_{sp} , and the sensible HVAC load $\dot{Q}_{HVAC}$. In some embodiments, the HVAC load model describes the dynamics of HVAC equipment **308** when providing heating or cooling to building zone **310**. For example, the HVAC load model may represent HVAC equipment **308** as a proportional-integral (PI) control system that provides the sensible HVAC load $\dot{Q}_{HVAC}$ as a function of the indoor air temperature T_{ia} and the temperature setpoint T_{sp} .

[0186] Estimator gain identifier **516** can be configured to calculate the state/disturbance estimation gain (i.e., the estimator gain K). The estimator gain K may be referred to in other contexts as the disturbance matrix or Kalman filter gain. The operation of thermal mass storage model generator **512**, HVAC load model generator **514**, and estimator gain identifier **516** are described in greater detail below.

Thermal Mass Storage Model Generator

[0187] Thermal mass storage model generator **512** can be configured to estimate or identify the parameters of a thermal mass storage model that describes the relationship between indoor air temperature T_{ia} , building mass temperature T_m , outdoor air temperature T_{oa} , internal load $\dot{Q}_{other}$, and sensible HVAC load $\dot{Q}_{HVAC}$. In some embodiments, thermal mass storage model generator **512** performs a multi-stage system identification process to generate the parameters of the thermal mass storage model. The stages of the system identification process may include input-output data collection, generating a parameterized building thermal model, and estimating (i.e., fitting) the parameters of the parameterized building thermal model using a parameter identification algorithm.

[0188] The first stage of the system identification process is input-output data collection. The type of input-output data collected to fit the model parameters may have a significant impact in the usefulness and accuracy of the resulting model. In some embodiments, thermal mass storage model generator **512** manipulates the temperature setpoint T_{sp} in an open-loop fashion as to sufficiently and persistently excite the system. For example, thermal mass storage model generator **512** can persistently excite (PE) a signal $\{s(k); k=0, 1, 2, \dots\}$ with spectrum $\phi_s(\omega)$ if $\phi_s(\omega) > 0$ for almost all ω where:

$$\phi_s(\omega) = \sum_{m=-\infty}^{\infty} \left(\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N s(k)s(k-m) \right) \quad (\text{Equation 12})$$

[0189] Input-output data collected during closed-loop operation may not provide an information-rich dataset with respect to system identification because the inputs tend to be correlated to themselves as well as to the unmeasured disturbances as a result of closing the loop with a controller. As a result input-output data obtained via closed-loop operation may not meet the PE condition without adding perturbations to the inputs similar to that used in extremum seeking control.

[0190] Instead of closed-loop identification, thermal mass storage model generator **512** can perform a system identification experiment to meet the PE condition. In some embodiments, thermal mass storage model generator **512** performs the system identification experiment while building zone **310** is occupied. Accordingly, it may be desirable to design the system identification experiment to be as short as possible in order to prevent or minimize disruptions with building occupants. In some embodiments, thermal mass storage model generator **512** makes small changes to the temperature setpoints T_{sp} (e.g., changes of 3-5° F.) in order to have minimal impact on the comfort of the building occupants.

[0191] In some embodiments, thermal mass storage model generator **512** generates the excitation signals using a pseudorandom binary sequence (PRBS). FIG. 7 shows a graph **700** of an example excitation temperature setpoint signal **702** over a one day period. Thermal mass storage model generator **512** can be configured to generate and provide temperature setpoint signal **702** to smart thermostat **100** during the system identification experiment. Temperature setpoint signal **702** can be generated a priori and tailored to meet the comfort constraints. The minimum and maximum

temperature values of temperature setpoint signal **702** may be different depending on whether HVAC equipment **308** are operating in a heating dominated or cooling dominated mode. The values of temperature setpoint signal **702** may define the values of temperature setpoint variable T_{sp} in the input data set used to fit the model parameters.

[0192] For variable speed systems, excitation signals generated from a PRBS may not be suitable in practice owing to concerns over saturation effects leading to nonlinearities in the experimental dataset. This issue can be monitored and analyzed via high-fidelity building simulations and setpoint experiments on buildings. Modifications to the excitation signal generation methodology can be evaluated first in simulation to understand the technical trade-offs and limitations. Experiments on buildings can also be performed to help assess the use of other types of excitation signals.

[0193] Thermal mass storage model generator **512** can generate a set of input data by modulating the temperature setpoint T_{sp} and recording the values of the temperature setpoint T_{sp} and the outside air temperature T_{oa} at each time step during the system identification period. The set of input data may include a plurality of values of the temperature setpoint T_{sp} and the outside air temperature T_{oa} . Thermal mass storage model generator **512** can generate a set of output data by monitoring and recording zone conditions at each time step of the system identification period. The zone conditions may include the zone air temperature T_{ia} and the heating/cooling load $\dot{Q}_{HVAC}$. The set of output data may include a plurality of values of the zone air temperature T_{ia} and the heating/cooling load $\dot{Q}_{HVAC}$.

[0194] The second stage of the system identification process is generating a parameterized building thermal model. The parameterized building thermal model may describe the relationship between zone temperature T_{ia} and sensible HVAC load $\dot{Q}_{HVAC}$. In some embodiments, the parameterized building thermal model is based on a thermal circuit representation of building zone **310**. An example a thermal circuit representing the heat transfer characteristics of building zone **310** is shown in FIG. 8.

[0195] Referring now to FIG. 8, a thermal circuit **800** representing the heat transfer characteristics of building zone **310** is shown, according to some embodiments. Thermal circuit **800** is a two thermal resistance, two thermal capacitance (2R2C) control-oriented thermal mass model. The thermal energy balance within thermal circuit **800** is given by the following linear differential equations:

$$C_{ia} \dot{T}_{ia} = \frac{1}{R_{mi}} (T_m - T_{ia}) + \frac{1}{R_{oi}} (T_{oa} - T_{ia}) + \dot{Q}_{HVAC} + \dot{Q}_{other} \quad (\text{Equation 13})$$

$$C_m \dot{T}_m = \frac{1}{R_{mi}} (T_{ia} - T_m)$$

where T_{ia} represents the indoor air temperature, T_{oa} represents the outdoor air temperature, T_m represents the thermal mass temperature (i.e., the average temperature of solid objects in building zone **310**), C_m represents thermal capacitance of the thermal mass, C_{ia} represents the indoor air thermal capacitance, R_{mi} represents the thermal resistance between the indoor air and the thermal mass, R_{oi} represents the thermal resistance between the indoor air and the outdoor air, $\dot{Q}_{HVAC}$ represents the sensible heat added to building zone **310** by HVAC equipment **308** (or removed from building zone **310** if cooling is provided), and $\dot{Q}_{other}$

represents the heat load disturbance (e.g., internal heat generation within building zone **310** via solar radiation, occupancy, electrical equipment, etc.).

[0196] In some embodiments, heat transfer between adjacent zones is modeled as part of the heat load disturbance $\dot{Q}_{other}$ and thus, not explicitly accounted for in the model. In other embodiments, thermal interactions between adjacent zones can be accounted for by modeling the temperature of adjacent zones as additional temperature nodes and adding a thermal resistor between each adjacent zone and building zone **310**. For buildings that have significant thermal coupling between zones (e.g., large stores, open floorplan buildings, etc.), it may be desirable to model the thermal interactions between adjacent zones.

[0197] The thermal dynamic model of Equation 13 describes the heat transfer between the indoor air and the thermal mass, the heat transfer through the building walls, the sensible heat/cooling duty of HVAC equipment **308** $\dot{Q}_{HVAC}$, and the internal heat load $\dot{Q}_{other}$ on the space generated by the occupants, electrical plug and lighting, and solar irradiation. Throughout this disclosure, the model of Equation 13 is referred to as the thermal model. The thermal model has been demonstrated to yield acceptable prediction accuracy while managing the trade-off between model complexity (e.g., number of model parameters) and model prediction accuracy. In some embodiments, the internal heat load $\dot{Q}_{other}$ cannot be measured and is therefore treated as an unmeasured, time-varying disturbance.

HVAC Load Model Generator

[0198] HVAC load model generator **514** can be configured to generate a HVAC load model that describes the relationship between indoor air temperature T_{ia} , the temperature setpoint T_{sp} , and the sensible HVAC load $\dot{Q}_{HVAC}$. In some embodiments, the HVAC load model describes the dynamics of HVAC equipment **308** when providing heating or cooling to building zone **310**. For example, the HVAC load model may represent HVAC equipment **308** as a proportional-integral (PI) control system that provides the sensible HVAC load $\dot{Q}_{HVAC}$ as a function of the indoor air temperature T_{ia} and the temperature setpoint T_{sp} .

[0199] In some embodiments, the sensible load $\dot{Q}_{HVAC}$ provided by HVAC equipment **308** is a function of the temperature setpoint T_{sp} as well as other factors including the heat gained/lost through the building walls, the internal heat load $\dot{Q}_{other}$, and the heat transferred to/removed from the building thermal mass. To provide or remove heat to a particular zone, various actions may be performed by HVAC equipment **308**. For example, for a chilled water cooled building with a central air handling unit (AHU) serving several variable air volume (VAV) terminal boxes, a chiller may generate chilled water. The chilled water can be pumped to the AHU cooling coil and air can be forced over the cooling coils of the AHU to cool the air to a supply air temperature setpoint (e.g., 55° F.). The cooled air can be delivered through the supply air duct to terminal VAV boxes at the location of building zone **310**. The flow rate of cooled air can be adjusted via a damper such that building zone **310** reaches and/or stays at the temperature setpoint T_{sp} .

[0200] In some embodiments, the dynamics of the entire process of providing cooling or heating to building zone **310** is non-negligible on the time-scale of interest for utilizing the energy storage of a building mass. To account for the dynamics of the heating/cooling process, HVAC load model

generator **514** can model the dynamics between the temperature setpoint T_{sp} and the HVAC sensible load $\dot{Q}_{HVAC}$. An example of a model that can be model that can be generated by HVAC load model generator **514** is the following proportional-integral (PI) controller model:

$$\begin{aligned} \dot{Q}_{HVAC,j} &= K_{p,j}\epsilon_{sp} + K_{I,j} \int_0^t \epsilon_{sp}(s) ds \\ \epsilon_{sp} &= T_{sp,j} - T_{ia} \end{aligned} \quad (\text{Equation 14})$$

where $j \in \{\text{clg}, \text{hlg}\}$ is the index that is used to denote either heating or cooling mode. In some embodiments, different sets of parameters are generated for the heating and cooling modes. In other words, HVAC load model generator **514** can generate a first set of HVAC load model parameters for the cooling mode and a second set of HVAC load model parameters for the heating mode. In some embodiments, the heating and cooling load is constrained to the following set: $\dot{Q}_{HVAC,clg} \in [-Q_{clg,cap}, 0]$ and $\dot{Q}_{HVAC,hlg} \in [0, Q_{hlg,cap}]$ where $\dot{Q}_{HVAC} < 0$ indicates that HVAC equipment **308** are providing cooling and $\dot{Q}_{HVAC} > 0$ indicates that HVAC equipment **308** are providing heating

[0201] The model of Equation 14 includes a proportional gain term (i.e., $K_{p,j}\epsilon_{sp}$) that accounts for the error ϵ_{sp} between the indoor air temperature T_{ia} and the setpoint $T_{sp,j}$. The model of Equation 14 also includes an integrator term (i.e., $K_{I,j} \int_0^t \epsilon_{sp}(s) ds$) to account for the time-varying nature of the HVAC load $\dot{Q}_{HVAC,j}$ required to force the indoor temperature T_{ia} to its setpoint $T_{sp,j}$. The integrator value may be correlated with the internal heat load disturbance $\dot{Q}_{other}$ and thus, may play the role of an integrating disturbance model. In some embodiments, the model of Equation 14 does not represent a particular PI controller, but rather is a lumped dynamic model describing the relationship between the HVAC load $\dot{Q}_{HVAC,j}$ and the temperature setpoint $T_{sp,j}$. In some embodiments, the HVAC load model defines the HVAC load $\dot{Q}_{HVAC,j}$ as a function of the zone air temperature T_{ia} , the temperature setpoint $T_{sp,j}$, the zone air humidity H_{ia} , a zone air humidity setpoint $H_{sp,j}$, zone heat load $\dot{Q}_{other}$, and/or any of a variety of factors that affect the amount of heating or cooling provided by HVAC equipment. It is contemplated that the HVAC load model can include any combination of these or other variables in various embodiments.

[0202] In some embodiments, HVAC load model generator **514** accounts for staged HVAC equipment in the HVAC load model. For staged equipment, the sensible heating and cooling load $\dot{Q}_{HVAC,j}$ can be modeled to take values in the countable set $\dot{Q}_{HVAC,j} = \{\dot{Q}_{s1}, \dot{Q}_{s2}, \dots, \dot{Q}_{sn}\}$, where $\dot{Q}_{si} \in \{1, \dots, n\}$ is the HVAC load with the first through i th stage on. The staging down and staging up dynamics can be assumed to be negligible. Alternatively, the staging down and staging up dynamics can be captured using a low-order linear model.

[0203] The model of Equation 14 allows for continuous values of $\dot{Q}_{HVAC}$, and may not be directly applicable to staged HVAC equipment. However, HVAC load model generator **514** can compute the time-averaged HVAC load $\dot{Q}_{HVAC,j}$ over a time period, which gives a continuous version of the HVAC load $\dot{Q}_{HVAC,j}$ needed to meet the temperature setpoint $T_{sp,j}$. For example, HVAC load model generator **514** can apply a filtering technique to the equipment stage on/off signal to compute a time-averaged version of the HVAC load

$\dot{Q}_{HVAC,j}$. The filtered version of the HVAC load $\dot{Q}_{HVAC,j}$ can then be used as an input to the model of Equation 14 to describe the relationship between the HVAC load $\dot{Q}_{HVAC,j}$ and the temperature setpoint $T_{sp,j}$. Throughout this disclosure, the variable $\dot{Q}_{HVAC,j}$ is used to represent both the continuous HVAC load when HVAC equipment **308** have a variable speed and to represent the filtered HVAC load when HVAC equipment **308** are staged.

[0204] Incorporating the thermal and the HVAC load models together and writing the system of equations as a linear system of differential equations gives the following state space representation:

$$\begin{aligned} \begin{bmatrix} \dot{T}_{ia} \\ \dot{T}_m \\ \dot{I} \end{bmatrix} &= \begin{bmatrix} \frac{1}{C_{ia}} \left(K_{p,j} - \frac{1}{R_{mi}} - \frac{1}{R_{oi}} \right) & \frac{1}{C_{ia} R_{mi}} & -\frac{K_{p,j} K_{I,j}}{C_{ia}} \\ \frac{1}{C_m R_{mi}} & -\frac{1}{C_m R_{mi}} & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} T_{ia} \\ T_m \\ I \end{bmatrix} + \\ &\quad \begin{bmatrix} -\frac{K_{p,j}}{C_{ia}} & \frac{1}{C_{ia} R_{oi}} \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} T_{sp,j} \\ T_{oa} \end{bmatrix} + \begin{bmatrix} \frac{1}{C_{ia}} \\ 0 \\ 0 \end{bmatrix} \dot{Q}_{other} \\ \begin{bmatrix} T_{ia} \\ \dot{Q}_{HVAC,j} \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ -K_{p,j} & 0 & K_{I,j} \end{bmatrix} \begin{bmatrix} T_{ia} \\ T_m \\ I \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ K_{p,j} & 0 \end{bmatrix} \begin{bmatrix} T_{sp,j} \\ T_{oa} \end{bmatrix} \end{aligned}$$

or more compactly:

$$\begin{aligned} \dot{x} &= A_c(\theta)x + B_c(\theta)u + B_d d \\ y &= C_c(\theta)x + D_c(\theta)u \end{aligned} \quad (\text{Equation 15})$$

where $x^T = [T_{ia}, T_m, I]$, $u^T = [T_{sp,j}, T_{oa}]$, $d = \dot{Q}_{other}/C_{ia}$, $y^T = [T_{ia}, \dot{Q}_{HVAC,j}]$, θ is a parameter vector containing all non-zero entries of the system matrices, and

$$\begin{aligned} A_c(\theta) &= \begin{bmatrix} -\theta_5(\theta_1 + \theta_2 + \theta_4) & \theta_5\theta_2 & \theta_3\theta_4\theta_5 \\ \theta_6\theta_2 & -\theta_6\theta_2 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \\ B_c(\theta) &= \begin{bmatrix} \theta_1\theta_5 & \theta_4\theta_5 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}, B_d(\theta) = \begin{bmatrix} \theta_5 \\ 0 \\ 0 \end{bmatrix}, \\ C_c(\theta) &= \begin{bmatrix} 1 & 0 & 0 \\ -\theta_4 & 0 & \theta_3\theta_4 \end{bmatrix}, C_d(\theta) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, D_c(\theta) = \begin{bmatrix} 0 & 0 \\ \theta_4 & 0 \end{bmatrix} \end{aligned}$$

[0205] The third stage of the system identification process includes fitting the parameters θ of the parameterized building thermal model (i.e., the model of Equation 15) to the input-output data using a parameter identification algorithm. FIG. 9 is a flowchart of a process **900** which can be performed by system identifier **510** to fit the parameters θ of the parameterized building thermal model. If HVAC equipment **308** is staged, system identifier **510** can filter input-output data to remove the high frequency dither in the indoor air temperature T_{ia} and can compute a time-averaged version of the HVAC load $\dot{Q}_{HVAC}$ from the discrete HVAC staging trajectory (step **902**). System identifier **510** can then fit the thermal model parameters and HVAC load model parameters θ to the input-output data (or filtered input-output data)

obtained during the first stage of the system identification process (step **904**). System identifier **510** can augment the resulting state-space model with another integrating disturbance model and can estimate the Kalman filter gain for the resulting model (step **906**). Using data obtained in a secondary experiment or under normal operation, system identifier **510** can validate the model through the use of statistics that capture the multi-step prediction accuracy of the resulting model (step **908**). Each of these steps is described in detail below.

[**0206**] Step **902** may include filtering the input-output data to remove a high frequency dither in the indoor air temperature T_{ia} and computing a time-averaged version of the HVAC load $\dot{Q}_{HVAC}$ from the discrete HVAC staging trajectory. When HVAC equipment **308** are staged, the indoor air temperature trajectory may have a high frequency dither caused by HVAC equipment **308** staging up and down. For example, when operating in a cooling mode, HVAC equipment **308** can be staged up (e.g., by activating discrete chillers) to drive the indoor air temperature T_{ia} to the temperature setpoint T_{sp} minus a deadband. When the indoor air temperature T_{ia} reaches the temperature setpoint T_{sp} minus the deadband, HVAC stages can be deactivated, which results in an increase in the temperature T_{ia} . The indoor air temperature T_{ia} may increase until it reaches the temperature setpoint T_{sp} plus the deadband, at which time the HVAC stages can be activated again. The observed effect may be a high frequency dither or oscillation of the temperature T_{ia} around the setpoint T_{sp} (shown in FIG. **10**). Also, the HVAC load trajectory may take discrete values as stages are turned on and off. The duty cycle of the on/off trajectory of the stages may depend on the heat transfer to/from building zone **310**, the error between the setpoint and indoor air temperature, and the setpoint. Thus, a good indication of the HVAC load $\dot{Q}_{HVAC}$ on a continuous scale is some time-average of the HVAC load trajectory or stage on/off trajectories.

[**0207**] To remove the high frequency dither and to perform the time-averaging on the HVAC load trajectory, system identifier **510** can filter the input-output data obtained from the SI experiment. In some embodiments, system identifier **510** uses a first-order Savitzky-Golay filter (SGF) to filter the input-output data. The SGF involves fitting a polynomial to a set of data samples and evaluating the resulting polynomial at a single point within the approximation interval. The SGF is equivalent to performing discrete convolution with a fixed impulse response. However, the SGF is a non-casual filter. To address this issue, system identifier **510** can use the nearest past filtered measurements in the system identification process. For example, let Δt_{filter} be the filtering window of the SGF. For the filter to compute its smoothed estimate of the data at a given time step t , it can use data from $t - \Delta t_{filter}/2$ to $t + \Delta t_{filter}/2$. To provide the filtered measurements (indoor air temperature T_{ia} and HVAC load $\dot{Q}_{HVAC}$) at a sample time t_k , the filtered measurement from $t_k - \Delta t_{filter}/2$ can be used.

[**0208**] FIG. **10** is a pair of graphs **1000** and **1050** illustrating an example of the indoor air temperature T_{ia} trajectory and the HVAC load $\dot{Q}_{HVAC}$ trajectory before filtering with the SGF. Line **1002** represents the outdoor air temperature T_{oa} and line **1004** represents the unfiltered indoor air temperature T_{ia} . In steady-state, the indoor air temperature T_{ia} may vary around the temperature setpoint within a region

that is plus and minus the control deadband **1106**. Line **1052** represents the unfiltered sensible load $\dot{Q}_{HVAC}$ for staged HVAC equipment.

[**0209**] FIG. **11** is a pair of graphs **1100** and **1150** illustrating an example of the indoor air temperature T_{ia} trajectory and the HVAC load $\dot{Q}_{HVAC}$ trajectory after filtering with the SGF. Line **1102** represents the outdoor air temperature T_{oa} and line **1004** represents the filtered indoor air temperature T_{ia} . In some embodiments, the filtering removes the variation of temperature about the setpoint to smooth the temperature trajectory. Line **1152** represents the filtered sensible load $\dot{Q}_{HVAC}$ for staged HVAC equipment and may be a time-averaged version of the unfiltered sensible load **1052** $\dot{Q}_{HVAC}$ shown in FIG. **10**.

[**0210**] Referring again to FIG. **9**, step **904** may include fitting the thermal model parameters and HVAC load model parameters to the input-output data (or filtered input-output data) obtained during the first stage of the system identification process. In step **904**, the time-varying heat load disturbance d may be omitted. System identifier **510** can identify the parameters for the model given by:

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} \theta_1 & \theta_2 & \theta_3 \\ \theta_4 & -\theta_4 & 0 \\ -1 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} \theta_5 \theta_7 & \theta_5 \theta_6 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 1 & 0 & 0 \\ -\theta_7 & 0 & \theta_8 \end{bmatrix} x(t) + \begin{bmatrix} 0 & 0 \\ \theta_7 & 0 \end{bmatrix} u(t) \end{aligned} \quad (\text{Equation 16})$$

where the effect of the time-varying disturbance d is accounted for via the integrator model.

[**0211**] Using the prediction error method (PEM) described previously, system identifier **510** can obtain an estimate of the model parameters θ . In some embodiments, system identifier **510** estimates the model parameters θ using the functions provided by the Matlab System Identification toolbox. For example, the identification can be performed using the function `greyest` with the initial state option set to estimate, the disturbance model option set to none, and the estimation focus option is set to simulation. Other options may not be specifically set and therefore default values can be used.

[**0212**] The initial condition of the model of Equation 16 may be unknown, but can be from the data. The disturbance component (i.e., the Kalman gain estimate K) may not be included in the model of Equation 16. The estimation focus can be set to prediction so that the algorithm automatically computes the weighting function of the prediction errors. The stability estimation focus option may perform the same weighting as the prediction option, but also enforces model stability. Under the parameterization of the model of Equation 16, the values of the resistance, capacitance, and PI parameters may be computed from the parameter vector θ if necessary. Given that the resulting PEM optimization problem is a nonlinear, non-convex problem, the initial guess on the parameter values may have a significant impact on the parameter values obtained, and multiple executions of the PEM solver provided a different initial guess each time may converge to different local minima.

[**0213**] To perform step **904** autonomously, model verification can be performed before process **900** continues to **906**. If any of the verification steps are not satisfied, the PEM solver can be reinitialized with a different random initial guess. The model verification steps that can be performed in

step **904** may include ensuring stability by checking the eigenvalues of the obtained A. In some embodiments, the model verification steps include ensuring that the identified parameters are greater than zero. This is a condition arising from the physical meaningfulness of the model. The model verification steps may include ensuring that the thermal capacitance of the air C_{ia} is less than the thermal capacitance C_m of the building mass. In some embodiments, the model verification steps include verifying that the matrix A is well conditioned in order to avoid prediction problems and model sensitivity to noise. If the condition number of A is less than 10500, the conditioning of A may be deemed to be acceptable.

[0214] Once the obtained model satisfies all the checks, process **900** may proceed to step **906**. It is possible that the conditions described above are not ever satisfied. However, this could be due to a number of reasons that may be difficult to autonomously diagnose and potentially, be associated with a problem beyond the limitations of the SI procedure. To address this problem, a limit can be set for the number of times through the steps above. If the parameter acceptability criteria is not satisfied after the maximum number of iterations through step **904**, system identifier **510** can report an error for a human operator to investigate and address.

Estimator Gain Identifier

[0215] As discussed above, the model of Equation 16 may only account for the main thermal dynamics of the system and may not identify an appropriate estimator gain. In step **906**, the model of Equation 16 can be augmented with another integrating disturbance model and the estimator gain can be identified. In some embodiments, step **906** is performed by estimator gain identifier **516**. Estimator gain identifier **516** can be configured to augment the model of Equation 16 to produce the augmented model:

$$\begin{bmatrix} \dot{\hat{x}}(t) \\ \dot{\hat{d}}(t) \end{bmatrix} \begin{bmatrix} A_c & B_d \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{x}(t) \\ \hat{d}(t) \end{bmatrix} + \begin{bmatrix} B_c \\ 0 \end{bmatrix} u(t) + \underbrace{\begin{bmatrix} K_x(\phi) \\ K_d(\phi) \end{bmatrix}}_{=:K(\phi)} (y(t) - \hat{y}(t))$$

$$\hat{y}(t) = \begin{bmatrix} C_c & C_d \end{bmatrix} \begin{bmatrix} \hat{x}(t) \\ \hat{d}(t) \end{bmatrix} D_c u(t)$$

where the parameters of A_c , B_c , C_c and D_c are the same as the parameters θ determined in step **904**. The disturbance model can be selected such that $B_d=1$ and $C_d=0$, which is an input integrating disturbance model. $K(\phi)$ is the identified estimator gain parameterized with the parameter vector ϕ as shown in the following equation:

$$K_x(\phi) = \begin{bmatrix} \phi_1 & \phi_2 \\ 0 & 0 \\ \phi_3 & \phi_4 \end{bmatrix}, K_d(\phi) = [\phi_5 \quad \phi_6] \quad (\text{Equation 17})$$

[0216] where the elements corresponding to the mass temperature T_m (i.e., the middle row of the matrix $K_x(\phi)$) are set to zero. If these elements are allowed to take non-zero values, poor estimation performance may result. Accordingly, estimator gain identifier **516** may set the elements corresponding to the mass temperature T_m to zero. The values for the parameters ϕ can be obtained using the

function greyest on the augmented model, using the same input-output data as that used in step **904**.

[0217] Estimator gain identifier **516** can be configured to generate the following augmented matrices:

$$A_{aug} = \begin{bmatrix} A_c & B_d \\ 0 & 0 \end{bmatrix}, B_{aug} = \begin{bmatrix} B_c \\ 0 \end{bmatrix}, C_{aug} = [C_c \quad C_d], D_{aug} = D_c$$

[0218] Similar to the verifications performed in step **904**, estimator gain identifier **516** can check for the stability of the observer system $A_{aug} - K C_{aug}$, and its conditioning number. In this case, the conditioning number of $A_{aug} - K C_{aug}$ may be deemed acceptable if it is less than 70,000.

[0219] Step **908** may include validating the augmented model generated in step **906**. As discussed above, system identifier **510** may select the optimal model parameters that fit the input-output data with respect to an objective function that depends on the one-step error prediction. In step **904**, the PEM is the one-step ahead prediction without estimation (i.e., open-loop estimation), whereas step **906** represents the one-step ahead closed-loop estimation problem. However, within the context of MPC, multi-step open-loop predictions can be used to select the optimal input trajectory over the prediction horizon. Thus, the model validation performed in step **908** may ensure that the estimator is tuned appropriately and that the multi-step open-loop prediction performed with the identified model provides sufficient accuracy.

[0220] In some embodiments, system identifier **510** collects another input-output data set for model validation. This data may be collected during normal operation or from another SI experiment. Using this validation data set, system identifier **510** can generate one or more metrics that quantify the multi-step prediction accuracy. One example of a metric which can be generated in step **908** is the Coefficient of Variation Weighted Mean Absolute Prediction Error (CVWMAPE). CVWMAPE is an exponentially weighted average of the absolute prediction error at each time step that quantifies the prediction error over time. System identifier **510** can calculate the CVWMAPE as follows:

$$CVWMAPE(k) = \quad (\text{Equation 18})$$

$$\frac{\sum_{i=k}^{k+N_h-1} e^{-i/N_h} |y(i) - \hat{y}(i|k)|}{\sum_{i=k}^{k+N_h-1} e^{-i/N_h} |y(i)|}, k = 0, 1, 2, \dots$$

where $N_h \in \mathbb{N}_{>0}$ is the prediction horizon, $y(i)$ is the measured output at time step i , and $\hat{y}(i|k)$ is the predicted output with the identified model given a measurement at time step k and the input sequence $u(k), u(k+1), \dots, u(i-1)$. For notational simplicity the variable y is used to denote one of the two outputs (i.e., in this subsection, y is a scalar). The CVWMAPE can be computed for both outputs.

[0221] Another prediction error to consider is prediction error with respect to the q -step ahead prediction ($q \in \mathbb{N}_{\geq 0}$). System identifier **510** can calculate the Coefficient of Variation Root Mean Squared Prediction Error (CVRMSPE) to evaluate the q -step ahead prediction error. For example, system identifier **510** can calculate the CVRMSPE for a range of values of q from zero-step ahead prediction up to

N_h -step ahead prediction. Given a set of measured output values $\{y(0), \dots, y(M)\}$ for $M \in \mathbb{I}_{\geq 0}$, the CVRMSPE is given by:

$$\text{CVRMSPE}(q) = \frac{\text{RMSPE}(q)}{\bar{y}(q)} \quad (\text{Equation 19})$$

$$\text{RMSPE}(q) = \sqrt{\frac{\sum_{i=q+1}^M (y(i) - \hat{y}(i|q))^2}{M - q}}, \quad (\text{Equation 20})$$

$$\bar{y}(q) = \frac{\sum_{i=q}^M y(i)}{M - q}, \quad (\text{Equation 21})$$

for all $q \in \{0, \dots, N_h - 1\}$. The CVRMSPE helps identify the prediction error over the duration of the optimization period. [0222] Once the system identification process is complete, system identifier 510 can provide the system matrices A, B, C, and D with identified parameters θ and the estimator gain K with identified parameters ϕ to state disturbance estimator 520 and predictive optimizer 522. State disturbance estimator 520 can use the system matrices A, B, C, and D and the estimator gain K to compute an estimate of the current state $\hat{x}(k)$ and unmeasured disturbance $\hat{d}(k)$ at each sample time.

Load/Rate Predictor

[0223] Referring again to FIG. 5, model predictive controller 302 is shown to include a load/rate predictor 518. Load/rate predictor 518 can be configured to predict the heat load disturbance $\hat{Q}_{other}(k)$ and the utility rates $\hat{r}_{elec}(k)$ for each time step k (e.g., $k=1 \dots n$) of an optimization period. Load/rate predictor 518 can provide the load and rate predictions to predictive optimizer 522 for use in determining the optimal temperature setpoints T_{sp} .

[0224] In some embodiments, load/rate predictor 518 predicts the value of $\hat{Q}_{other}(k)$ at each time step k using a history of disturbance estimates $\hat{d}_{hist}$ generated by state/disturbance estimator 520. As described above, the state/disturbance estimation problem is given by:

$$\begin{bmatrix} \hat{T}_{ia}(k+1|k) \\ \hat{T}_m(k+1|k) \\ \hat{I}(k+1|k) \\ \hat{d}(k+1|k) \end{bmatrix} = A_{aug} \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{T}_m(k|k-1) \\ \hat{I}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + B_{aug} \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} + \begin{bmatrix} K_x \\ K_d \end{bmatrix} \left(\begin{bmatrix} T_{ia}(k) \\ \dot{Q}_{HVAC}(k) \end{bmatrix} - \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \dot{Q}_{HVAC}(k|k-1) \end{bmatrix} \right) \quad (\text{Equation 22})$$

$$\begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \dot{Q}_{HVAC}(k|k-1) \end{bmatrix} = C_{aug} \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{T}_m(k|k-1) \\ \hat{I}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + D \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix}$$

where

$$A_{aug} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix}, B_{aug} = \begin{bmatrix} B \\ 0 \end{bmatrix},$$

$C_{aug} = [C \ 0]$. State/disturbance estimator 520 can perform the state/disturbance estimation for each time step k to generate a disturbance estimate $\hat{d}(k)$ for each time step k. State/disturbance estimator 520 can record a history of disturbance estimates $\hat{d}_{hist} = \{\hat{d}(k|k-1)\}_{k=1}^{N_{hist}}$ and provide the history of disturbance estimates $\hat{d}_{hist}$ to load/rate predictor 518.

[0225] Load/rate predictor 518 can use the history of disturbance estimates $\hat{d}_{hist}$ to predict the value of $\hat{Q}_{other}(k)$ at each time step k. In some embodiments, the heat load disturbance $\hat{Q}_{other}(k)$ is a function of the time of day, the day type (e.g., weekend, weekday, holiday), the weather forecasts, building occupancy, and/or other factors which can be used by load/rate predictor 518 to predict the heat load disturbance $\hat{Q}_{other}(k)$. For example, load/rate predictor 518 can predict the heat load disturbance using the equation:

$$\hat{Q}_{other}(k) = f_{pred}(k\Delta + t_0, \hat{d}_{hist}) \quad (\text{Equation 23})$$

where $\Delta > 0$ is the sample period and t_0 is an initial time. In some embodiments, load/rate predictor 518 uses weather forecasts from a weather service 526 and/or load history data from historical data 528 to predict the heat load disturbance $\hat{Q}_{other}(k)$ at each time step.

[0226] In some embodiments, load/rate predictor 518 uses a deterministic plus stochastic model trained from historical load data to predict the heat load disturbance $\hat{Q}_{other}(k)$. Load/rate predictor 518 may use any of a variety of prediction methods to predict the heat load disturbance $\hat{Q}_{other}(k)$ (e.g., linear regression for the deterministic portion and an AR model for the stochastic portion). In some embodiments, load/rate predictor 518 makes load/rate predictions using the techniques described in U.S. patent application Ser. No. 14/717,593.

[0227] Load/rate predictor 518 is shown receiving utility rates from utilities 524. Utility rates may indicate a cost or price per unit of a resource (e.g., electricity, natural gas, water, etc.) provided by utilities 524 at each time step k in the prediction window. In some embodiments, the utility rates are time-variable rates. For example, the price of electricity may be higher at certain times of day or days of the week (e.g., during high demand periods) and lower at other times of day or days of the week (e.g., during low demand periods). The utility rates may define various time periods and a cost per unit of a resource during each time period. Utility rates may be actual rates received from utilities 524 or predicted utility rates estimated by load/rate predictor 518.

[0228] In some embodiments, the utility rates include demand charges for one or more resources provided by utilities 524. A demand charge may define a separate cost imposed by utilities 524 based on the maximum usage of a particular resource (e.g., maximum energy consumption) during a demand charge period. The utility rates may define various demand charge periods and one or more demand charges associated with each demand charge period. In some instances, demand charge periods may overlap partially or completely with each other and/or with the prediction window. Predictive optimizer 522 may be configured to account for demand charges in the high level optimization process

performed by predictive optimizer **522**. Utilities **524** may be defined by time-variable (e.g., hourly) prices, a maximum service level (e.g., a maximum rate of consumption allowed by the physical infrastructure or by contract) and, in the case of electricity, a demand charge or a charge for the peak rate of consumption within a certain period. Load/rate predictor **518** may store the predicted heat load disturbance $\hat{Q}_{other}(k)$ and the utility rates in memory **508** and/or provide the predicted heat load disturbance $\hat{Q}_{other}(k)$ and the utility rates to predictive optimizer **522**.

Predictive Optimizer

[0229] Still referring to FIG. 5, model predictive controller **302** is shown to include a predictive optimizer **522**. Predictive optimizer **522** can manipulate the temperature setpoint T_{sp} when operating in the operational mode to minimize the economic cost of operating HVAC equipment **308** over the duration of the optimization period. To determine the temperature setpoints T_{sp} , predictive optimizer **522** can optimize an objective function (i.e., a cost function) that accounts for the cost of operating HVAC equipment **308** over the duration of the optimization period. The costs of operating HVAC equipment **308** can include, for example, the costs of resources consumed by HVAC equipment **308** during operation (e.g., electricity, natural gas, water, etc.), demand charges imposed by an electric utility, peak load contribution charges, equipment degradation/replacement costs, and/or other costs associated with the operation of HVAC equipment **308**.

[0230] An example of an objective function which can be optimized by predictive optimizer **522** is shown in the following equation:

$$\min_{T_{sp}, \epsilon_T, \delta T_{sp}} \sum_{k=0}^{N-1} (r_{elec}(k) \hat{\ell}_{elec}(k) + p_{\epsilon_T} \epsilon_T(k) + p_{\delta T} \delta T_{sp}(k)) \quad (\text{Equation 24})$$

where $r_{elec}(k)$ is the price of electricity at time step k , $\hat{\ell}_{elec}(k)$ is the predicted electricity consumption of HVAC equipment **308** at time step k , $\epsilon_T(k)$ is a number of degrees by which the zone temperature constraints are violated at time step k , p_{ϵ_T} is a zone air temperature penalty coefficient applied to $\epsilon_T(k)$, $\delta T_{sp}(k)$ is a number of degrees by which the zone temperature setpoint T_{sp} changes between time step $k-1$ and time step k , and $p_{\delta T}$ is a temperature setpoint change penalty coefficient applied to $\delta T_{sp}(k)$. Equation 24 accounts for the cost of electricity consumption ($r_{elec}(k) \hat{\ell}_{elec}(k)$), penalizes violations of the indoor air temperature bounds ($p_{\epsilon_T} \epsilon_T(k)$), and penalizes changes in the temperature setpoint ($p_{\delta T} \delta T_{sp}(k)$).

[0231] If additional resources (other than electricity) are consumed by HVAC equipment **308**, additional terms can be added to Equation 24 to represent the cost of each resource consumed by HVAC equipment **308**. For example, if HVAC equipment **308** consume natural gas and water in addition to electricity, Equation 24 can be updated to include the terms $r_{gas}(k) \hat{\ell}_{gas}(k)$ and $r_{water}(k) \hat{\ell}_{water}(k)$ in the summation, where $r_{gas}(k)$ is the cost per unit of natural gas at time step k , $\hat{\ell}_{gas}(k)$ is the predicted natural gas consumption of HVAC equipment **308** at time step k , $r_{water}(k)$ is the cost per unit of

water at time step k , and $\hat{\ell}_{water}(k)$ is the predicted water consumption of HVAC equipment **308** at time step k .

[0232] Predictive optimizer **522** can be configured to estimate the amount of each resource consumed by HVAC equipment **308** (e.g., electricity, natural gas, water, etc.) as a function of the sensible heating or cooling load $\hat{Q}_{HVAC}$. In some embodiments, a constant efficiency model is used to compute the resource consumption of HVAC equipment **308** as a fixed multiple heating or cooling load $\hat{Q}_{HVAC}$ (e.g., $\hat{\ell}_{elec}(k) = \eta \hat{Q}_{HVAC}(k)$). In other embodiments, equipment models describing a relationship between resource consumption and heating/cooling production can be used to approximate the resource consumption of the HVAC equipment **308**. The equipment models may account for variations in equipment efficiency as a function of load and/or other variables such as outdoor weather conditions (e.g., $\hat{\ell}_{elec}(k) = f(\hat{Q}_{HVAC}(k), T_{oa}(k), \eta_{base})$).

Predictive optimizer **522** can convert the estimated heating/cooling load $\hat{Q}_{HVAC}(k)$ at each time step to one or more resource consumption values (e.g., $\hat{\ell}_{elec}(k)$, $\hat{\ell}_{gas}(k)$, $\hat{\ell}_{water}(k)$, etc.) for inclusion in the objective function.

[0233] In some embodiments, predictive optimizer **522** is configured to modify the objective function to account for various other costs associated with operating HVAC equipment **308**. For example, predictive optimizer **522** can modify the objective function to account for one or more demand charges, peak load contribution charges, equipment degradation costs, equipment purchase costs, revenue generated from participating in incentive-based demand response programs, economic load demand response, and/or any other factors that can contribute to the cost incurred by operating HVAC equipment **308** or revenue generated by operating HVAC equipment **308**. As used herein, the term “energy cost” can include any such charges, costs, revenues, incentives, etc., including combinations thereof, in various embodiments. Predictive optimizer **522** can add one or more additional terms to the objective function to account for these or other factors. Several examples of such functionality are described in U.S. patent application Ser. No. 15/405,236 filed Jan. 12, 2017, U.S. patent application Ser. No. 15/405,234 filed Jan. 12, 2017, U.S. patent application Ser. No. 15/426,962 filed Feb. 7, 2017, U.S. patent application Ser. No. 15/473,496 filed Mar. 29, 2017, and U.S. patent application Ser. No. 15/616,616 filed Jun. 7, 2017. The entire disclosure of each of these patent applications is incorporated by reference herein. In some embodiments, model predictive controller **302** includes some or all of the functionality of the controllers and/or control systems described in these patent applications.

[0234] Predictive optimizer **522** can be configured to automatically generate and impose constraints on the optimization of the objective function. The constraints may be based on the system model provided by system identifier **510**, the state estimates provided by state/disturbance estimator **520**, the heat load disturbance predictions provided by load/rate predictor **518**, and constraints on the zone air temperature T_{ia} . For example, predictive optimizer **522** can generate and impose the following constraints for each time step $k \in \{0, \dots, N-1\}$:

$$\begin{bmatrix} T_{ia}(k+1) \\ T_m(k+1) \\ I(k+1) \end{bmatrix} = A \begin{bmatrix} T_{ia}(k) \\ T_m(k) \\ I(k) \end{bmatrix} + B \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} + B_d \dot{Q}_{other}(k) \quad (\text{Equation 25})$$

$$\dot{Q}_{HVAC}(k) = C \begin{bmatrix} T_{ia}(k) \\ T_m(k) \\ I(k) \end{bmatrix} + D \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} \quad (\text{Equation 26})$$

$$\hat{\ell}_{elec}(k) = \eta \dot{Q}_{HVAC}(k) \quad (\text{Equation 27})$$

$$T_{ia}(k) \leq T_{max}(k) + T_{deadband} + \epsilon_T(k) \quad (\text{Equation 28})$$

$$T_{ia}(k) \geq T_{min}(k) - T_{deadband} - \epsilon_T(k) \quad (\text{Equation 29})$$

$$T_{sp}(k) \in [T_{sp,min}(k), T_{sp,max}(k)] \quad (\text{Equation 30})$$

$$T_{sp}(k+1) - T_{sp}(k) \leq \delta T_{sp}(k) \quad (\text{Equation 31})$$

$$-T_{sp}(k+1) + T_{sp}(k) \leq \delta T_{sp}(k) \quad (\text{Equation 32})$$

$$\epsilon_T(k) \geq 0 \quad (\text{Equation 33})$$

$$T_{ia}(0) = \hat{T}_{ia,0}, T_m(0) = \hat{T}_{m,0}, I(0) = \hat{I}_0 \quad (\text{Equation 34})$$

[0235] Equation 25 is based on the thermal mass storage model provided by thermal mass storage model generator **512**. The thermal mass storage model predicts the system states at each time step $k+1$ as a function of the system states $x(k)$, inputs $u(k)$, and heat load disturbance at the previous time step k . Specifically, Equation 25 defines the relationship between the system states $T_{ia}(k+1)$, $T_m(k+1)$, and $I(k+1)$ at time step $k+1$, the system states $T_{ia}(k)$, $T_m(k)$, and $I(k)$ at time step k , the controlled or measured inputs $T_{sp}(k)$ and $T_{oa}(k)$ at time step k , and the estimated heat load disturbance $\dot{Q}_{other}(k)$ at time step k . The parameters in the matrices A and B can be determined by system identifier **510**, as previously described. The value of $\dot{Q}_{other}(k)$ at each time step k can be determined by load/rate predictor **518**.

[0236] Equation 26 is based on the HVAC load model provided by HVAC load model generator **514**. The HVAC load model defines the heating/cooling load $\dot{Q}_{HVAC}(k)$ at each time step k as a function of the system states $T_{ia}(k)$, $T_m(k)$, and $I(k)$ at time step k and the controlled or measured inputs $T_{sp}(k)$ and $T_{oa}(k)$ at time step k . The parameters in the matrices C and D can be determined by system identifier **510**, as previously described.

[0237] Equation 27 defines the relationship between the predicted electric consumption $\hat{\ell}_{elec}(k)$ at time step k and the heating/cooling load $\dot{Q}_{HVAC}(k)$ at time step k . If additional resources (other than electricity) are consumed by HVAC equipment **308**, additional constraints can be added to define the relationship between $\dot{Q}_{HVAC}(k)$ and the amount of each resource consumed by HVAC equipment **308**.

[0238] Equations 28 and 29 constrain the zone air temperature $T_{ia}(k)$ at each time step k between the minimum temperature threshold $T_{min}(k)$ and the maximum temperature threshold $T_{max}(k)$ at that time step k , plus or minus the temperature deadband $T_{deadband}$ and a temperature error $\epsilon_T(k)$. In some embodiments, the values of the minimum temperature threshold $T_{min}(k)$ and the maximum temperature threshold $T_{max}(k)$ can vary over the duration of the optimization period. For example, the minimum temperature threshold $T_{min}(k)$ and the maximum temperature threshold $T_{max}(k)$ may define a first zone temperature range (i.e., $T_{min}(k)$ to $T_{max}(k)$) during time step k and a second zone temperature range (i.e., $T_{min}(k+1)$ to $T_{max}(k+1)$) during time step $k+1$. The values of T_{min} and T_{max} at each time step can

vary as a function of the time of day, day of the week, building occupancy, or other factors. In some embodiments, the values of T_{min} and T_{max} are input by a user, can be adjusted by a user, or calculated based on another user input or behavior. Predictive optimizer **522** may be allowed to violate the temperature constraints if necessary, but any deviation from the defined temperature range (i.e., between $T_{min}(k) - T_{deadband}$ and $T_{max}(k) + T_{deadband}$) may be penalized in the objective function by imposing a penalty per on the temperature error $\epsilon_T(k)$. Equation 33 requires the temperature error $\epsilon_T(k)$ to be non-negative at each time step k .

[0239] Equation 30 limits the temperature setpoint $T_{sp}(k)$ at each time step between a minimum temperature setpoint $T_{sp,min}(k)$ and a maximum temperature setpoint $T_{sp,max}(k)$, whereas Equations 31-32 define the changes $\delta T_{sp}(k)$ in the temperature setpoint $T_{sp}(k)$ between consecutive time steps. The minimum temperature setpoint $T_{sp,min}(k)$ and the maximum temperature setpoint $T_{sp,max}(k)$ may be different from the minimum temperature threshold $T_{min}(k)$ and the maximum temperature threshold $T_{max}(k)$ used to constrain the zone air temperature $T_{ia}(k)$. In some embodiments, the values of the minimum temperature setpoint $T_{sp,min}(k)$ and the maximum temperature setpoint $T_{sp,max}(k)$ can vary over the duration of the optimization period. For example, the minimum temperature setpoint $T_{sp,min}(k)$ and the maximum temperature setpoint $T_{sp,max}(k)$ may define a first temperature setpoint range (i.e., $T_{sp,min}(k)$ to $T_{sp,max}(k)$) during time step k and a second temperature setpoint range (i.e., $T_{sp,min}(k+1)$ to $T_{sp,max}(k+1)$) during time step $k+1$. The values of $T_{sp,min}$ and $T_{sp,max}$ at each time step can vary as a function of the time of day, day of the week, building occupancy, or other factors. In some embodiments, the values of $T_{sp,min}$ and $T_{sp,max}$ are input by a user, can be adjusted by a user, or calculated based on another user input or behavior. Changes in the temperature setpoint T_{sp} may be penalized in the objective function by imposing a penalty $p_{\delta T_{sp}}$ on the temperature change $\delta T_{sp}(k)$.

[0240] Equation 34 sets the initial system states for the zone air temperature $T_{ia}(0)$, the zone mass temperature $T_m(0)$, and the integrated disturbance $I(0)$ at the first time step $k=0$ of the optimization period. The initial system states $T_{ia}(0)$, $T_m(0)$, and $I(0)$ may be set to the initial state estimates $\hat{T}_{ia,0}$, $\hat{T}_{m,0}$, and $\hat{I}_0$ provided by state/disturbance estimator **520**. In other words, the predictive model shown in Equations 25-26 may be initialized using the state estimates provided by state/disturbance estimator **520**.

[0241] Predictive optimizer **522** can be configured to optimize the objective function (Equation 24) subject to the optimization constraints (Equations 25-34) to determine optimal values of the temperature setpoint $T_{sp}(k)$ at each time step k of the optimization period. Predictive optimizer **522** can use any of a variety of optimization techniques to perform the optimization. For example, the optimization problem can be formulated as a linear program (Equations 24-34) and solved using a linear optimization technique (e.g., a basis exchange algorithm, an interior point algorithm, a cutting-plane algorithm, a branch and bound algorithm, etc.) using any of a variety solvers and/or programming languages. In some embodiments, predictive optimizer **522** performs the optimization at the beginning of each time step k to determine optimal temperature setpoints $T_{sp}(k)$ for the next N time steps.

[0242] Once the optimal temperature setpoints T_{sp} have been determined, predictive optimizer 522 can provide the optimal temperature setpoint $T_{sp}(k)$ for the current time step k to smart thermostat 100 and/or equipment controller 406. State/disturbance estimator 520 can use the temperature setpoint $T_{sp}(k)$ for the current time step k along with feedback received from HVAC equipment 308 and building zone 310 during time step k (e.g., measurements of $T_{ia}(k)$, $T_{oa}(k)$, $\dot{Q}_{HVAC}(k)$, etc.) to update the state/disturbance estimates the next time the optimization is performed.

Model Predictive Control Processes

[0243] Referring now to FIGS. 12-14, several flowcharts of model predictive control processes 1200-1400 are shown, according to some embodiments. Processes 1200-1400 can be performed by one or more components of model predictive control systems 300-400, as described with reference to FIGS. 3-11. For example, processes 1200-1400 can be performed by model predictive controller 302, smart thermostat 100, equipment controller 406, and/or HVAC equipment 308 to control the temperature of building zone 310.

[0244] Referring specifically to FIG. 12, process 1200 is shown to include operating a model predictive controller in system identification mode to identify parameters of a thermal mass storage model and a HVAC load model for a building zone (step 1202). In some embodiments, step 1202 is performed by system identifier 510. Step 1202 can include generating a thermal mass storage model by modeling the heat transfer characteristics of the building zone using a thermal circuit 800, as described with reference to FIG. 8. Step 1202 may also include generating a HVAC load model by modeling the amount of heating/cooling $\dot{Q}_{HVAC}$ provided by HVAC equipment 308 as a function of the temperature setpoint T_{sp} and the zone air temperature T_{ia} . The thermal mass storage model and the HVAC load model can be used to generate a system of differential equations that define the relationship between zone air temperature T_{ia} , zone mass temperature T_m , outdoor air temperature T_{oa} , the amount of heating or cooling $\dot{Q}_{HVAC}$ provided by HVAC equipment 308, the heat load disturbance $\dot{Q}_{other}$, and the temperature setpoint T_{sp} .

[0245] The unknown parameters in the thermal mass storage model and the HVAC load model (e.g., R_{oi} , R_{mi} , C_m , C_{ia}) can be organized into parameter matrices A, B, C, and D, as shown in the following equations:

$$\begin{bmatrix} \dot{T}_{ia} \\ \dot{T}_m \\ I \end{bmatrix} = \begin{bmatrix} \frac{1}{C_{ia}} \left(K_{p,j} - \frac{1}{R_{mi}} - \frac{1}{R_{oi}} \right) & \frac{1}{C_{ia} R_{mi}} & -\frac{K_{p,j} K_{I,j}}{C_{ia}} \\ \frac{1}{C_m R_{mi}} & -\frac{1}{C_m R_{mi}} & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} T_{ia} \\ T_m \\ I \end{bmatrix} + \begin{bmatrix} -\frac{K_{p,j}}{C_{ia}} & \frac{1}{C_{ia} R_{oi}} \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} T_{sp,j} \\ T_{oa} \end{bmatrix} + \begin{bmatrix} \frac{1}{C_{ia}} \\ 0 \\ 0 \end{bmatrix} \dot{Q}_{other}$$

$$\begin{bmatrix} T_{ia} \\ \dot{Q}_{HVAC,j} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -K_{p,j} & 0 & K_{I,j} \end{bmatrix} \begin{bmatrix} T_{ia} \\ T_m \\ I \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ K_{p,j} & 0 \end{bmatrix} \begin{bmatrix} T_{sp,j} \\ T_{oa} \end{bmatrix}$$

or more compactly:

$$\begin{aligned} \dot{x} &= A_c(\theta)x + B_c(\theta)u + B_d d \\ y &= C_c(\theta)x + D_c(\theta)u \end{aligned}$$

where $x^T = [T_{ia}, T_m, I]$, $u^T = [T_{sp,j}, T_{oa}]$, $d = \dot{Q}_{other}/C_{ia}$, $y^T = [T_{ia}, \dot{Q}_{HVAC,j}]$, θ is a parameter vector containing all non-zero entries of the system matrices, and

$$A_c(\theta) = \begin{bmatrix} -\theta_5(\theta_1 + \theta_2 + \theta_4) & \theta_5\theta_2 & \theta_3\theta_4\theta_5 \\ \theta_6\theta_2 & -\theta_6\theta_2 & 0 \\ -1 & 0 & 0 \end{bmatrix},$$

$$B_c = \begin{bmatrix} \theta_1\theta_5 & \theta_4\theta_5 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}, B_d = \begin{bmatrix} \theta_5 \\ 0 \\ 0 \end{bmatrix},$$

$$C_c(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ -\theta_4 & 0 & \theta_3\theta_4 \end{bmatrix}, C_d(\theta) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, D_c(\theta) = \begin{bmatrix} 0 & 0 \\ \theta_4 & 0 \end{bmatrix}$$

[0246] The unknown parameters θ in the thermal mass storage model and the HVAC load model (i.e., the parameters in matrices A, B, C, and D) can be identified by fitting the parameters θ to a set of input-output data (e.g., sets of values T_{ia} , T_{oa} , T_{sp} , and $\dot{Q}_{HVAC}$) collected from the HVAC system when operating in the system identification mode. In some embodiments, the input-output data are collected by modulating the temperature setpoint T_{sp} and observing the corresponding values of T_{ia} , T_{oa} , and $\dot{Q}_{HVAC}$ for a plurality of time steps during the system identification mode. The process of fitting the model parameters θ to the set of input-output data can be accomplished by performing process 900, as described with reference to FIG. 9.

[0247] If HVAC equipment 308 are staged, step 1202 can include filtering input-output data to remove the high frequency dither in the indoor air temperature T_{ia} and can compute a time-averaged version of the HVAC load $\dot{Q}_{HVAC}$ from the discrete HVAC staging trajectory. The thermal model parameters and HVAC load model parameters θ can then be fit to the input-output data (or filtered input-output data). In some embodiments, step 1202 includes augmenting the resulting state-space model with another integrating disturbance model and estimating the Kalman filter gain for the resulting model. Using data obtained in a secondary experiment or under normal operation, step 1202 can include validating the model through the use of statistics that capture the multi-step prediction accuracy of the resulting model.

[0248] Still referring to FIG. 12, process 1200 is shown to include operating the model predictive controller in operational mode to determine temperature setpoints T_{sp} for the building zone (step 1204). In some embodiments, step 1204 is performed by predictive optimizer 522, as described with reference to FIG. 5. To determine the temperature setpoints T_{sp} , step 1204 can include optimizing an objective function (i.e., a cost function) that accounts for the cost of operating HVAC equipment 308 over the duration of the optimization period. The costs of operating HVAC equipment 308 can include, for example, the costs of resources consumed by HVAC equipment 308 during operation (e.g., electricity, natural gas, water, etc.), demand charges imposed by an electric utility, peak load contribution charges, equipment

degradation/replacement costs, and/or other costs associated with the operation of HVAC equipment **308**. The objective function can be optimized subject to constraints provided by the thermal mass storage model and the HVAC load model, as well as constraints on the zone temperature T_{ia} . Step **1204** can be accomplished by performing process **1300**, described in greater detail with reference to FIG. **13**.

[0249] In some embodiments, process **1200** includes automatically switching between the system identification mode and the operational mode based on a prediction error of the thermal mass storage model and/or the HVAC load model. For example, process **1200** can include comparing the value of the zone temperature $T_{ia}(k|k-1)$ predicted by the thermal mass storage model to the actual measured value of the zone temperature $T_{ia}(k)$. If the difference between the predicted value $T_{ia}(k|k-1)$ and the actual value $T_{ia}(k)$ exceeds an error threshold, process **1200** may automatically return to step **1202** and repeat the system identification process. Similarly, process **1200** can include comparing the value of the HVAC equipment load $\dot{Q}_{HVAC}(k|k-1)$ predicted by the HVAC load model to the actual value of the HVAC load $\dot{Q}_{HVAC}(k)$. If the difference between the predicted value $\dot{Q}_{HVAC}(k|k-1)$ and the actual value $\dot{Q}_{HVAC}(k)$ exceeds an error threshold, process **1200** may automatically return to step **1202** and repeat the system identification process.

[0250] Process **1200** is shown to include providing the optimal temperature setpoints T_{sp} to an equipment controller (step **1206**) and using the equipment controller to operate HVAC equipment to drive the temperature of the building zone T_{ia} to the optimal temperature setpoints T_{sp} (step **1208**). Step **1206** can include sending the optimal temperature setpoints T_{sp} from model predictive controller **302** to an equipment controller **406**. In some embodiments, both model predictive controller **302** and equipment controller **406** are components of a smart thermostat **100** (as shown in FIG. **4**). In other embodiments, model predictive controller **302** is separate from smart thermostat **100** and configured to send the optimal temperature setpoints to smart thermostat **100** via a communications network **304** (as shown in FIG. **3**). Smart thermostat **100** and/or equipment controller **406** can use the optimal temperature setpoints T_{sp} to generate control signals for HVAC equipment **308** which operate to drive the temperature of the building zone T_{ia} to the optimal temperature setpoints T_{sp} .

[0251] Referring now to FIG. **13**, a flowchart of a process **1300** for generating optimal temperature setpoints for a building zone is shown, according to some embodiments. In some embodiments, process **1300** is performed by model predictive controller **302**, as described with reference to FIGS. **5-11**. Process **1300** can be performed to accomplish step **1204** of process **1200**.

[0252] Process **1300** is shown to include predicting a heat load disturbance $\dot{Q}_{other}$ experienced by a building zone at each time step ($k=0 \dots N-1$) of an optimization period (step **1302**). In some embodiments, step **1302** is performed by load/rate predictor **518** and state/disturbance predictor **520**. Step **1302** can include generating a disturbance estimate $\hat{d}(k)$ for each time step k in the optimization period. In some embodiments, step **1302** includes estimating the disturbance state $\hat{d}(k)$ at each time step k in the optimization period using the following state/disturbance model:

$$\begin{bmatrix} \hat{T}_{ia}(k+1|k) \\ \hat{T}_m(k+1|k) \\ \hat{l}(k+1|k) \\ \hat{d}(k+1|k) \end{bmatrix} = A_{aug} \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{T}_m(k|k-1) \\ \hat{l}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + B_{aug} \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} + \begin{bmatrix} K_x \\ K_d \end{bmatrix} \left(\begin{bmatrix} T_{ia}(k) \\ \dot{Q}_{HVAC}(k) \end{bmatrix} - \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \dot{Q}_{HVAC}(k|k-1) \end{bmatrix} \right)$$

$$\begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \dot{Q}_{HVAC}(k|k-1) \end{bmatrix} = C_{aug} \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{T}_m(k|k-1) \\ \hat{l}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + D \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix}$$

where

$$A_{aug} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix}, B_{aug} = \begin{bmatrix} B \\ 0 \end{bmatrix},$$

$C_{aug}=[C \ 0]$. Step **1302** can include recording a history of disturbance estimates $\hat{d}_{hist}=\{\hat{d}(k|k-1)\}_{k=1}^{N_{hist}}$ predicted by the state/disturbance model.

[0253] Step **1302** can include using the history of disturbance estimates $\hat{d}_{hist}$ to predict the value of $\hat{Q}_{other}(k)$ at each time step k . In some embodiments, the heat load disturbance $\hat{Q}_{other}(k)$ is a function of the time of the day, the day type (e.g., weekend, weekday, holiday), the weather forecasts, building occupancy, and/or other factors which can be used by load/rate predictor **518** to predict the heat load disturbance $\hat{Q}_{other}(k)$. For example, step **1302** can include predicting the heat load disturbance using the equation:

$$\hat{Q}_{other}(k) = f_{pred}(k\Delta + t_0, \hat{d}_{hist})$$

[0254] where $\Delta>0$ is the sample period and t_0 is an initial time. In some embodiments, step **1302** includes using weather forecasts from a weather service **526** and/or load history data from historical data **528** to predict the heat load disturbance $\hat{Q}_{other}(k)$ at each time step.

[0255] In some embodiments, step **1302** includes using a deterministic plus stochastic model trained from historical load data to predict the heat load disturbance $\hat{Q}_{other}(k)$. Step **1302** can include using any of a variety of prediction methods to predict the heat load disturbance $\hat{Q}_{other}(k)$ (e.g., linear regression for the deterministic portion and an AR model for the stochastic portion). In some embodiments, step **1302** includes predicting the heat load disturbance $\hat{Q}_{other}(k)$ using the techniques described in U.S. patent application Ser. No. 14/717,593.

[0256] Still referring to FIG. **13**, process **1300** is shown to include estimating initial system states $\hat{x}(0)$ for a first time step ($k=0$) of the optimization period (step **1304**). The initial system states $\hat{x}(0)$ may include the estimated zone air temperature $\hat{T}_{ia}(0)$, the estimated zone mass temperature $\hat{T}_m(0)$, and the estimated integrating disturbance $\hat{l}(0)$ at the beginning of the optimization period. In some embodiments,

step **1304** is performed by state/disturbance estimator **520**. The initial system states $\hat{x}(0)$ can be estimated using same the state/disturbance estimation model used in step **1302**. For example, step **1304** can include estimating the system states $\hat{x}(k)$ at each time step k using the following equation:

$$\begin{aligned}\hat{x}(k+1|k) &= A\hat{x}(k|k-1) + Bu(k) + K(y(k) - \hat{y}(k|k-1)) \\ \hat{y}(k|k-1) &= C\hat{x}(k|k-1) + Du(k)\end{aligned}$$

where $\hat{x}(k+1|k)$ is the estimated/predicted state at time step $k+1$ given the measurement at time step k , $\hat{y}(k|k-1)$ is the predicted output at time step k given the measurement at time step $k-1$, and K is the estimator gain.

[0257] In some embodiments, the estimated state vector $\hat{x}(k+1|k)$, the output vector $\hat{y}(k|k-1)$, and the input vector $u(k)$ are defined as follows:

$$\begin{aligned}\hat{x}(k+1|k) &= \begin{bmatrix} \hat{T}_{ia}(k+1|k) \\ \hat{T}_m(k+1|k) \\ \hat{I}(k+1|k) \end{bmatrix}, \\ \hat{y}(k|k-1) &= \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{Q}_{HVAC}(k|k-1) \end{bmatrix}, u(k) = \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix}\end{aligned}$$

where $\hat{T}_{ia}$ is an estimate of the zone air temperature T_{ia} , $\hat{T}_m$ is an estimate of the zone mass temperature T_m , $\hat{I}$ is an estimate of the integrating disturbance I , $\hat{Q}_{HVAC}$ is an estimate of the heating or cooling load provided by HVAC equipment **308**, T_{sp} is the temperature setpoint, and T_{oa} is the outdoor air temperature. The estimated system states $\hat{x}(0)$ at the first time step $k=0$ can be used as the initial system states.

[0258] Still referring to FIG. **13**, process **1300** is shown to include using a thermal mass storage model and a heat load model to predict the temperature of the building zone T_{ia} and the HVAC equipment load $\hat{Q}_{HVAC}$ at each time step as a function of the estimated system states x , temperature setpoint trajectory T_{sp} , and estimated heat load disturbance $\hat{Q}_{other}$ (step **1306**). The thermal mass storage model may be defined as follows:

$$\begin{bmatrix} \hat{T}_{ia}(k+1) \\ \hat{T}_m(k+1) \\ \hat{I}(k+1) \end{bmatrix} = A \begin{bmatrix} \hat{T}_{ia}(k) \\ \hat{T}_m(k) \\ \hat{I}(k) \end{bmatrix} + B \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} + B_d \hat{Q}_{other}(k)$$

where $\hat{T}_{ia}(k+1)$ is the predicted temperature of the building zone at time step $k+1$, $\hat{T}_m(k+1)$ is the predicted temperature of the building mass at time step $k+1$, $\hat{I}(k+1)$ is the predicted value of the integrating disturbance at time step $k+1$, $\hat{T}_{ia}(k)$ is the predicted temperature of the building zone at time step k , $\hat{T}_m(k)$ is the predicted temperature of the building mass at time step k , $\hat{I}(k)$ is the predicted value of the integrating disturbance at time step k , $T_{sp}(k)$ is the temperature setpoint at time step k , $T_{oa}(k)$ is the outdoor air temperature (measured) at time step k , and $\hat{Q}_{other}(k)$ is the estimated heat load disturbance at time step k .

[0259] The HVAC load model may be defined as follows:

$$\begin{bmatrix} \hat{Q}_{HVAC}(k) \end{bmatrix} = C \begin{bmatrix} \hat{T}_{ia}(k) \\ \hat{T}_m(k) \\ \hat{I}(k) \end{bmatrix} + D \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix}$$

where $\hat{Q}_{HVAC}(k)$ is the predicted HVAC equipment load at time step k and $\hat{T}_{ia}(k)$, $\hat{T}_m(k)$, $\hat{I}(k)$, $T_{sp}(k)$, and $T_{oa}(k)$ are the same as the corresponding variables in the thermal mass storage model.

[0260] Process **1300** is shown to include optimizing an economic cost function subject to constraints on the predicted temperature of the building zone $\hat{T}_{ia}$ to determine optimal temperature setpoints T_{sp} for the building zone at each time step of the optimization period (step **1308**). An example of an objective function which can be optimized in step **1308** is shown in the following equation:

$$\min_{T_{sp}, \epsilon_T, \delta T_{sp}} \sum_{k=0}^{N-1} (r_{elec}(k) \hat{\ell}_{elec}(k) + p_{\epsilon_T} \epsilon_T(k) + p_{\delta T} \delta T_{sp}(k))$$

where $r_{elec}(k)$ is the price of electricity at time step k , $\hat{\ell}_{elec}(k)$ is the predicted electricity consumption of HVAC equipment **308** at time step k , $\epsilon_T(k)$ is a number of degrees by which the zone temperature constraints are violated at time step k , p_{ϵ_T} is a zone air temperature penalty coefficient applied to $\epsilon_T(k)$, $\delta T_{sp}(k)$ is a number of degrees by which the zone temperature setpoint T_{sp} changes between time step $k-1$ and time step k , and $p_{\delta T}$ is a temperature setpoint change penalty coefficient applied to $\delta T_{sp}(k)$. Equation 24 accounts for the cost of electricity consumption ($r_{elec}(k) \hat{\ell}_{elec}(k)$), penalizes violations of the indoor air temperature bounds ($p_{\epsilon_T} \epsilon_T(k)$), and penalizes changes in the temperature setpoint ($p_{\delta T} \delta T_{sp}(k)$).

[0261] If additional resources (other than electricity) are consumed by HVAC equipment **308**, additional terms can be added to the objective function to represent the cost of each resource consumed by HVAC equipment **308**. For example, if HVAC equipment **308** consume natural gas and water in addition to electricity, the objective function can be updated to include the terms $r_{gas}(k) \hat{\ell}_{gas}(k)$ and $r_{water}(k) \hat{\ell}_{water}(k)$ in the summation, where $r_{gas}(k)$ is the cost per unit of natural gas at time step k , $\hat{\ell}_{gas}(k)$ is the predicted natural gas consumption of HVAC equipment **308** at time step k , $r_{water}(k)$ is the cost per unit of water at time step k , and $\hat{\ell}_{water}(k)$ is the predicted water consumption of HVAC equipment **308** at time step k .

[0262] In some embodiments, step **1308** includes estimating the amount of each resource consumed by HVAC equipment **308** (e.g., electricity, natural gas, water, etc.) as a function of the sensible heating or cooling load $\hat{Q}_{HVAC}$. In some embodiments, a constant efficiency model is used to compute the resource consumption of HVAC equipment **308** as a fixed multiple heating or cooling load $\hat{Q}_{HVAC}$ (e.g.,

$\hat{\ell}_{elec}(k) = n \hat{Q}_{HVAC}(k)$). In other embodiments, equipment models describing a relationship between resource consumption and heating/cooling production can be used to approximate the resource consumption of the HVAC equip-

ment **308**. The equipment models may account for variations in equipment efficiency as a function of load and/or other variables such as outdoor weather conditions (e.g., $\hat{\ell}_{elec}(k)$) = $f(\hat{Q}_{HVAC}(k), T_{oa}(k), \eta_{base})$. Step **1308** can include converting the estimated heating/cooling load $\hat{Q}_{HVAC}(k)$ at each time step to one or more resource consumption values (e.g., $\hat{\ell}_{elec}(k)$, $\hat{\ell}_{gas}(k)$, $\hat{\ell}_{water}(k)$, etc.) for inclusion in the objective function.

[0263] Step **1308** can include optimizing the objective function to determine optimal values of the temperature setpoint $T_{sp}(k)$ for each time step in the optimization period. The objective function can be optimized subject to a set of constraints (e.g., Equations 25-34). The constraints may include the thermal mass storage model and the HVAC load model, which define the relationship between $\hat{Q}_{HVAC}$, the temperature setpoints T_{sp} , and the zone air temperature T_{ia} at each time step. The constraints may also include constraints on the zone air temperature $T_{ia}(k)$ and constraints that define the penalty terms in the objective function.

[0264] In some embodiments, step **1308** includes performing the optimization at the beginning of each time step k to determine optimal temperature setpoints $T_{sp}(k)$ for the next N time steps. Once the optimal temperature setpoints T_{sp} have been determined, the optimal temperature setpoint $T_{sp}(k)$ for the current time step k can be provided to smart thermostat **100** and/or equipment controller **406** and the optimization period can be shifted forward in time by one time step. The temperature setpoint $T_{sp}(k)$ for the current time step k along with feedback received from HVAC equipment **308** and building zone **310** during time step k (e.g., measurements of $T_{ia}(k)$, $T_{oa}(k)$, $\hat{Q}_{HVAC}(k)$, etc.) can be used to update the state/disturbance estimates the next time the optimization is performed. Steps **1304-1308** can be repeated at the beginning of each time step to determine the optimal setpoint trajectory T_{sp} for the new (i.e., time shifted) optimization period.

[0265] Referring now to FIG. 14, a flowchart of a process **1400** for controlling the temperature of a building zone using model predictive control is shown, according to some embodiments. Process **1400** can be performed by one or more components of model predictive control systems **300-400**, as described with reference to FIGS. 3-11. For example, process **1400** can be performed by model predictive controller **302**, smart thermostat **100**, equipment controller **406**, and/or HVAC equipment **308** to control the temperature of building zone **310**.

[0266] Process **1400** is shown to include identifying parameters of a thermal mass storage model and a HVAC load model for a building zone (step **1402**). In some embodiments, step **1202** is performed by system identifier **510**. Step **1402** can include generating a thermal mass storage model by modeling the heat transfer characteristics of the building zone using a thermal circuit **800**, as described with reference to FIG. 8. Step **1402** may also include generating a HVAC load model by modeling the amount of heating/cooling $\hat{Q}_{HVAC}$ provided by HVAC equipment **308** as a function of the temperature setpoint T_{sp} and the zone air temperature T_{ia} . The thermal mass storage model and the HVAC load model can be used to generate a system of differential equations that define the relationship between zone air temperature T_{ia} , zone mass temperature T_m , outdoor air temperature T_{oa} , the amount of heating or cooling $\hat{Q}_{HVAC}$

provided by HVAC equipment **308**, the heat load disturbance $\hat{Q}_{other}$, and the temperature setpoint T_{sp} .

[0267] The unknown parameters in the thermal mass storage model and the HVAC load model (e.g., R_{oi} , R_{mi} , C_m , C_{ia}) can be organized into parameter matrices A, B, C, and D, as shown in the following equations:

$$\begin{bmatrix} \dot{T}_{ia} \\ \dot{T}_m \\ \dot{I} \end{bmatrix} = \begin{bmatrix} \frac{1}{C_{ia}} \left(K_{p,j} - \frac{1}{R_{mi}} - \frac{1}{R_{oi}} \right) & \frac{1}{C_{ia} R_{mi}} & -\frac{K_{p,j} K_{I,j}}{C_{ia}} \\ \frac{1}{C_m R_{mi}} & -\frac{1}{C_m R_{mi}} & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} T_{ia} \\ T_m \\ I \end{bmatrix} + \begin{bmatrix} -\frac{K_{p,j}}{C_{ia}} & \frac{1}{C_{ia} R_{oi}} \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} T_{sp,j} \\ T_{oa} \end{bmatrix} + \begin{bmatrix} \frac{1}{C_{ia}} \\ 0 \\ 0 \end{bmatrix} \hat{Q}_{other}$$

$$\begin{bmatrix} T_{ia} \\ \hat{Q}_{HVAC,j} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -K_{p,j} & 0 & K_{I,j} \end{bmatrix} \begin{bmatrix} T_{ia} \\ T_m \\ I \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ K_{p,j} & 0 \end{bmatrix} \begin{bmatrix} T_{sp,j} \\ T_{oa} \end{bmatrix}$$

or more compactly:

$$\dot{x} = A_c(\theta)x + B_c(\theta)u + B_d d$$

$$y = C_c(\theta)x + D_c(\theta)u$$

where $x^T = [T_{ia}, T_m, I]$, $u^T = [T_{sp,j}, T_{oa}]$, $d = \hat{Q}_{other}/C_{ia}$, $y^T = [T_{ia}, \hat{Q}_{HVAC,j}]$, θ is a parameter vector containing all non-zero entries of the system matrices, and

$$A_c(\theta) = \begin{bmatrix} -\theta_5(\theta_1 + \theta_2 + \theta_4) & \theta_5\theta_2 & \theta_3\theta_4\theta_5 \\ \theta_6\theta_2 & -\theta_6\theta_2 & 0 \\ -1 & 0 & 0 \end{bmatrix}, B_c(\theta) = \begin{bmatrix} \theta_1\theta_5 & \theta_4\theta_5 \\ 0 & 0 \\ 1 & 0 \end{bmatrix},$$

$$B_d = \begin{bmatrix} \theta_5 \\ 0 \\ 0 \end{bmatrix}, C_c(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ -\theta_4 & 0 & \theta_3\theta_4 \end{bmatrix}, C_d(\theta) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, D_c(\theta) = \begin{bmatrix} 0 & 0 \\ \theta_4 & 0 \end{bmatrix}$$

[0268] The unknown parameters θ in the thermal mass storage model and the HVAC load model (i.e., the parameters in matrices A, B, C, and D) can be identified by fitting the parameters θ to a set of input-output data (e.g., sets of values T_{ia} , T_{oa} , T_{sp} , and $\hat{Q}_{HVAC}$) collected from the HVAC system when operating in the system identification mode. In some embodiments, the input-output data are collected by modulating the temperature setpoint T_{sp} and observing the corresponding values of T_{ia} , T_{oa} , and $\hat{Q}_{HVAC}$ for a plurality of time steps during the system identification mode. The process of fitting the model parameters θ to the set of input-output data can be accomplished by performing process **900**, as described with reference to FIG. 9.

[0269] If HVAC equipment **308** are staged, step **1402** can include filtering input-output data to remove the high frequency dither in the indoor air temperature T_{ia} and can compute a time-averaged version of the HVAC load $\hat{Q}_{HVAC}$ from the discrete HVAC staging trajectory. The thermal model parameters and HVAC load model parameters θ can then be fit to the input-output data (or filtered input-output data). In some embodiments, step **1402** includes augmenting the resulting state-space model with another integrating disturbance model and estimating the Kalman filter gain for

the resulting model. Using data obtained in a secondary experiment or under normal operation, step **1402** can include validating the model through the use of statistics that capture the multi-step prediction accuracy of the resulting model.

[0270] Still referring to FIG. **14**, process **1400** is shown to include predicting a heat load disturbance $\hat{Q}_{other}$ experienced by a building zone at each time step ($k=0 \dots N-1$) of an optimization period (step **1404**). In some embodiments, step **1404** is performed by load/rate predictor **518** and state/disturbance predictor **520**. Step **1404** can include generating a disturbance estimate $\hat{d}(k)$ for each time step k in the optimization period. In some embodiments, step **1404** includes estimating the disturbance state $\hat{d}(k)$ at each time step k in the optimization period using the following state/disturbance model:

$$\begin{bmatrix} \hat{T}_{ia}(k+1|k) \\ \hat{T}_m(k+1|k) \\ \hat{T}(k+1|k) \\ \hat{d}(k+1|k) \end{bmatrix} = A_{aug} \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{T}_m(k|k-1) \\ \hat{T}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + B_{aug} \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} + \begin{bmatrix} K_s \\ K_d \end{bmatrix} \left(\begin{bmatrix} T_{ia}(k) \\ \hat{Q}_{HVAC}(k) \end{bmatrix} - \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{Q}_{HVAC}(k|k-1) \end{bmatrix} \right)$$

$$\begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{Q}_{HVAC}(k|k-1) \end{bmatrix} = C_{aug} \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{T}_m(k|k-1) \\ \hat{T}(k|k-1) \\ \hat{d}(k|k-1) \end{bmatrix} + D \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix}$$

where

$$A_{aug} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix}, B_{aug} = \begin{bmatrix} B \\ 0 \end{bmatrix},$$

$C_{aug}=[C \ 0]$. Step **1404** can include recording a history of disturbance estimates $\hat{d}_{hist}=\{\hat{d}(k|k-1)\}_{k=1}^{N_{hist}}$ predicted by the state/disturbance model.

[0271] Step **1404** can include using the history of disturbance estimates $\hat{d}_{hist}$ to predict the value of $\hat{Q}_{other}(k)$ at each time step k . In some embodiments, the heat load disturbance $\hat{Q}_{other}(k)$ is a function of the time of day, the day type (e.g., weekend, weekday, holiday), the weather forecasts, building occupancy, and/or other factors which can be used by load/rate predictor **518** to predict the heat load disturbance $\hat{Q}_{other}(k)$. For example, step **1404** can include predicting the heat load disturbance using the equation:

$$\hat{Q}_{other}(k) = f_{pred}(k\Delta + t_0, \hat{d}_{hist})$$

[0272] where $\Delta>0$ is the sample period and t_0 is an initial time. In some embodiments, step **1404** includes using weather forecasts from a weather service **526** and/or load history data from historical data **528** to predict the heat load disturbance $\hat{Q}_{other}(k)$ at each time step.

[0273] In some embodiments, step **1404** includes using a deterministic plus stochastic model trained from historical

load data to predict the heat load disturbance $\hat{Q}_{other}(k)$. Step **1404** can include using any of a variety of prediction methods to predict the heat load disturbance $\hat{Q}_{other}(k)$ (e.g., linear regression for the deterministic portion and an AR model for the stochastic portion). In some embodiments, step **1404** includes predicting the heat load disturbance $\hat{Q}_{other}(k)$ using the techniques described in U.S. patent application Ser. No. 14/717,593.

[0274] Still referring to FIG. **14**, process **1400** is shown to include generating an economic cost function that accounts for a cost of resources consumed by HVAC equipment at each time step ($k=0 \dots N-1$) of an optimization period (step **1406**). An example of an objective function which can be generated in step **1406** is shown in the following equation:

$$\min_{T_{sp}, \epsilon_T, \delta T_{sp}} \sum_{k=0}^{N-1} (r_{elec}(k) \hat{\ell}_{elec}(k) + p_{\epsilon_T} \epsilon_T(k) + p_{\delta T} \delta T_{sp}(k))$$

where $r_{elec}(k)$ is the price of electricity at time step k , $\hat{\ell}_{elec}(k)$ is the predicted electricity consumption of HVAC equipment **308** at time step k , $\epsilon_T(k)$ is a number of degrees by which the zone temperature constraints are violated at time step k , p_{ϵ_T} is a zone air temperature penalty coefficient applied to $\epsilon_T(k)$, $\delta T_{sp}(k)$ is a number of degrees by which the zone temperature setpoint T_{sp} changes between time step $k-1$ and time step k , and $p_{\delta T}$ is a temperature setpoint change penalty coefficient applied to $\delta T_{sp}(k)$. Equation 24 accounts for the cost of electricity consumption ($r_{elec}(k) \hat{\ell}_{elec}(k)$), penalizes violations of the indoor air temperature bounds ($p_{\epsilon_T} \epsilon_T(k)$), and penalizes changes in the temperature setpoint ($p_{\delta T} \delta T_{sp}(k)$).

[0275] If additional resources (other than electricity) are consumed by HVAC equipment **308**, additional terms can be added to the objective function to represent the cost of each resource consumed by HVAC equipment **308**. For example, if HVAC equipment **308** consume natural gas and water in addition to electricity, the objective function can be updated to include the terms $r_{gas}(k) \hat{\ell}_{gas}(k)$ and $r_{water}(k) \hat{\ell}_{water}(k)$ in the summation, where $r_{gas}(k)$ is the cost per unit of natural gas at time step k , $\hat{\ell}_{gas}(k)$ is the predicted natural gas consumption of HVAC equipment **308** at time step k , $r_{water}(k)$ is the cost per unit of water at time step k , and $\hat{\ell}_{water}(k)$ is the predicted water consumption of HVAC equipment **308** at time step k .

[0276] In some embodiments, step **1406** includes estimating the amount of each resource consumed by HVAC equipment **308** (e.g., electricity, natural gas, water, etc.) as a function of the sensible heating or cooling load $\hat{Q}_{HVAC}$. In some embodiments, a constant efficiency model is used to compute the resource consumption of HVAC equipment **308** as a fixed multiple heating or cooling load $\hat{Q}_{HVAC}$ (e.g.,

$\hat{\ell}_{elec}(k)=\eta \hat{Q}_{HVAC}(k)$). In other embodiments, equipment models describing a relationship between resource consumption and heating/cooling production can be used to approximate the resource consumption of the HVAC equipment **308**. The equipment models may account for variations in equipment efficiency as a function of load and/or other

variables such as outdoor weather conditions (e.g., $\hat{\ell}_{elec}(k)$ $= f(\hat{Q}_{HVAC}(k), T_{oa}(k), \eta_{base})$). Step **1406** can include converting the estimated heating/cooling load $\hat{Q}_{HVAC}(k)$ at each time step to one or more resource consumption values (e.g., $\hat{\ell}_{elec}(k)$, $\hat{\ell}_{gas}(k)$, $\hat{\ell}_{water}(k)$, etc.) for inclusion in the objective function.

[0277] Still referring to FIG. 14, process **1400** is shown to include estimating initial system states $\hat{x}(0)$ for a first time step ($k=0$) of the optimization period (step **1408**). The initial system states $\hat{x}(0)$ may include the estimated zone air temperature $\hat{T}_{ia}(0)$, the estimated zone mass temperature $\hat{T}_m(0)$, and the estimated integrating disturbance $\hat{I}(0)$ at the beginning of the optimization period. In some embodiments, step **1408** is performed by state/disturbance estimator **520**. The initial system states $\hat{x}(0)$ can be estimated using same the state/disturbance estimation model used in step **1404**. For example, step **1408** can include estimating the system states $\hat{x}(k)$ at each time step k using the following equation:

$$\begin{aligned}\hat{x}(k+1|k) &= A\hat{x}(k|k-1) + Bu(k) + K(y(k) - \hat{y}(k|k-1)) \\ \hat{y}(k|k-1) &= C\hat{x}(k|k-1) + Du(k)\end{aligned}$$

where $\hat{x}(k+1|k)$ is the estimated/predicted state at time step $k+1$ given the measurement at time step k , $\hat{y}(k|k-1)$ is the predicted output at time step k given the measurement at time step $k-1$, and K is the estimator gain.

[0278] In some embodiments, the estimated state vector $\hat{x}(k+1|k)$, the output vector $\hat{y}(k|k-1)$, and the input vector $u(k)$ are defined as follows:

$$\begin{aligned}x(k+1|k) &= \begin{bmatrix} \hat{T}_{ia}(k+1|k) \\ \hat{T}_m(k+1|k) \\ \hat{I}(k+1|k) \end{bmatrix}, \\ \hat{y}(k|k-1) &= \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{Q}_{HVAC}(k|k-1) \end{bmatrix}, u(k) = \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix}\end{aligned}$$

where $\hat{T}_{ia}$ is an estimate of the zone air temperature T_{ia} , $\hat{T}_m$ is an estimate of the zone mass temperature T_m , $\hat{I}$ is an estimate of the integrating disturbance I , $\hat{Q}_{HVAC}$ is an estimate of the heating or cooling load provided by HVAC equipment **308**, T_{sp} is the temperature setpoint, and T_{oa} is the outdoor air temperature. The estimated system states $\hat{x}(0)$ at the first time step $k=0$ can be used as the initial system states.

[0279] Still referring to FIG. 14, process **1400** is shown to include using a thermal mass storage predictive model to constrain system states $\hat{x}(k+1|k)$ at time step $k+1$ to the system states $\hat{x}(k)$ and the temperature setpoint $T_{sp}(k)$ at time step k (step **1410**). An example of a constraint which can be generated based on the thermal mass storage predictive model is:

$$\begin{bmatrix} \hat{T}_{ia}(k+1) \\ \hat{T}_m(k+1) \\ \hat{I}(k+1) \end{bmatrix} = A \begin{bmatrix} \hat{T}_{ia}(k) \\ \hat{T}_m(k) \\ \hat{I}(k) \end{bmatrix} + B \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} + B_d \hat{Q}_{other}(k)$$

where $\hat{T}_{ia}(k+1)$ is the predicted temperature of the building zone at time step $k+1$, $\hat{T}_m(k+1)$ is the predicted temperature of the building mass at time step $k+1$, $\hat{I}(k+1)$ is the predicted value of the integrating disturbance at time step $k+1$, $\hat{T}_{ia}(k)$ is the predicted temperature of the building zone at time step k , $\hat{T}_m(k)$ is the predicted temperature of the building mass at time step k , $\hat{I}(k)$ is the predicted value of the integrating disturbance at time step k , $T_{sp}(k)$ is the temperature setpoint at time step k , $T_{oa}(k)$ is the outdoor air temperature (measured) at time step k , and $\hat{Q}_{other}(k)$ is the estimated heat load disturbance at time step k .

[0280] Process **1400** is shown to include using a HVAC load predictive model to constrain the HVAC equipment load $\hat{Q}_{HVAC}(k)$ at time step k to systems states $\hat{x}(k)$ and the temperature setpoint $T_{sp}(k)$ at time step k (step **1412**). An example of a constraint which can be generated based on the HVAC load model is:

$$\begin{bmatrix} \hat{Q}_{HVAC}(k) \end{bmatrix} = C \begin{bmatrix} \hat{T}_{ia}(k) \\ \hat{T}_m(k) \\ \hat{I}(k) \end{bmatrix} + D \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix}$$

where $\hat{Q}_{HVAC}(k)$ is the predicted HVAC equipment load at time step k and $\hat{T}_{ia}(k)$, $\hat{T}_m(k)$, $\hat{I}(k)$, $T_{sp}(k)$, and $T_{oa}(k)$ are the same as the corresponding variables in the thermal mass storage model.

[0281] Still referring to FIG. 14, process **1400** is shown to include optimizing the economic cost function subject to constraints defined by the predictive models and constraints on the predicted zone temperature $\hat{T}_{ia}(k)$ to determine optimal temperature setpoints T_{sp} for the building zone for each time step ($k=0 \dots N-1$) of the optimization period (step **1414**). The objective function can be optimized subject to a set of constraints (e.g., Equations 25-34). The constraints may include the thermal mass storage model and the HVAC

load model, which define the relationship between $\hat{Q}_{HVAC}$, the temperature setpoints T_{sp} , and the zone air temperature T_{ia} at each time step. The constraints may also include constraints on the zone air temperature $T_{ia}(k)$ and constraints that define the penalty terms in the objective function. The optimization performed in step **1414** may generate a temperature setpoint $T_{sp}(k)$ for each time step in the optimization period.

[0282] Process **1400** is shown to include providing the temperature setpoint T_{sp} for the first time step ($k=0$) of the optimization period to an equipment controller (step **1416**). Step **1416** can include sending the optimal temperature setpoints T_{sp} from model predictive controller **302** to an equipment controller **406**. In some embodiments, both model predictive controller **302** and equipment controller **406** are components of a smart thermostat **100** (as shown in FIG. 4). In other embodiments, model predictive controller **302** is separate from smart thermostat **100** and configured to send the optimal temperature setpoints to smart thermostat **100** via a communications network **304** (as shown in FIG. 3). In some embodiments, only the temperature setpoint for the first time step (i.e., $T_{sp}(0)$) is provided to the equipment controller in step **1416**. The remaining temperature setpoints are for future time steps and may be updated the next time the optimization is performed (e.g., at the beginning of the next time step).

[0283] In some embodiments, step 1416 includes providing the entire set of temperature setpoints T_{sp} (i.e., a temperature setpoint $T_{sp}(k)$ for each time step k) to the equipment controller. In the event that communication between model predictive controller 302 and equipment controller 406 is lost (e.g., network connectivity is disrupted), equipment controller 406 can use the set of temperature setpoints provided in step 1416 until communications between model predictive controller 302 and equipment controller 406 are restored. For example, equipment controller 406 can use each temperature setpoint $T_{sp}(k)$ received in step 1416 to control HVAC equipment 308 during the corresponding time step k until communications between model predictive controller 302 and equipment controller 406 are restored.

[0284] Still referring to FIG. 14, process 1400 is shown to include using feedback collected from the HVAC system during the first time step ($k=0$) to update the thermal mass storage predictive model (step 1418). The feedback collected from the HVAC system may include measurements of the zone air temperature T_{ia} and/or measurements of the HVAC equipment load $\dot{Q}_{HVAC}$. Step 1418 may include calculating the difference between the value of the zone air temperature $\hat{T}_{ia}(k|k-1)$ predicted in step 1408 and the actual measured value of the zone air temperature $T_{ia}(k)$ collected in step 1418. Similarly, step 1418 may include calculating the difference between the value of the HVAC equipment load $\hat{\dot{Q}}_{HVAC}(k|k-1)$ predicted in step 1408 and the actual measured value of the HVAC equipment load $\dot{Q}_{HVAC}(k)$ collected in step 1418.

[0285] The updating performed in step 1418 may include multiplying the Kalman gain matrix K generated in step 1402 by the vector of differences, as shown in the following equation:

$$\begin{bmatrix} \hat{T}_{ia}(k+1|k) \\ \hat{T}_m(k+1|k) \\ \hat{\dot{Q}}_{HVAC}(k+1|k) \\ \hat{\dot{Q}}_{HVAC}(k+1|k) \end{bmatrix} = A_{aug} \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{T}_m(k|k-1) \\ \hat{\dot{Q}}_{HVAC}(k|k-1) \\ \hat{\dot{Q}}_{HVAC}(k|k-1) \end{bmatrix} + B_{aug} \begin{bmatrix} T_{sp}(k) \\ T_{oa}(k) \end{bmatrix} + \begin{bmatrix} K_x \\ K_d \end{bmatrix} \left(\begin{bmatrix} T_{ia}(k) \\ \dot{Q}_{HVAC}(k) \end{bmatrix} - \begin{bmatrix} \hat{T}_{ia}(k|k-1) \\ \hat{\dot{Q}}_{HVAC}(k|k-1) \end{bmatrix} \right)$$

[0286] In some embodiments, the updating performed in step 1418 is equivalent to the updating step of a Kalman filter in which the predicted states $\hat{x}(k+1|k)$ are calculated based on the error between the predicted system output $\hat{y}(k|k-1)$ at time step k and the actual/measured values of the predicted output variables $y(k)$.

[0287] Process 1400 is shown to include shifting the optimization forward in time by one time step (step 1420) and returning to step 1408. Steps 1408-1420 can be repeated at the beginning of each time step to generate a set of temperature setpoints T_{sp} for each time step in the shifted optimization period. The first temperature setpoint $T_{sp}(0)$ generated for the shifted optimization period can be provided to the equipment controller and used to control the HVAC equipment.

Performance Graphs

[0288] Referring now to FIGS. 15-18, several graphs 1500-1800 illustrating the performance of model predictive

control (MPC) systems 300-400 are shown, according to some embodiments. Graph 1500 compares the energy consumption of MPC systems 300-400 to the energy consumption of a baseline temperature control system without MPC. Graph 1600 indicates the cost of electricity (i.e., \$/kWh) which varies over time. Line 1502 represents the energy consumption of the baseline system, whereas line 1504 represents the energy consumption of MPC systems 300-400. MPC systems 300-400 use more energy than the baseline system during an off-peak period 1506 when energy prices are low. However, MPC systems 300-400 use less energy than the baseline system during a peak period 1508 when energy prices are high. Advantageously, this results in a cost savings relative to the baseline system.

[0289] Graph 1700 compares the building mass temperature trajectories of the baseline system and MPC systems 300-400. Line 1702 represents the temperature T_m of the solid mass within building zone 310 when the baseline system is used to provide temperature control. Since the baseline system does not store energy in the building mass, the temperature T_m remains relatively constant over the duration of the optimization period. Line 1704 represents the temperature T_m of the solid mass within building zone 310 when MPC systems 300-400 are used to provide temperature control. During the off-peak period 1506, building zone 310 is pre-cooled, which results in a decrease in the temperature T_m of the building mass. During the peak period 1508, thermal energy from the air within building zone 310 is moved into the building mass, which increases the temperature T_m of the building mass and provides cooling for the air within building zone 310.

[0290] Graph 1800 compares the zone air temperature trajectories of the baseline system and MPC systems 300-400. Line 1802 represents the temperature T_{ia} of the air within building zone 310 when the baseline system is used to provide temperature control. The temperature T_{ia} remains relatively constant over the duration of the optimization period. Line 1704 represents the zone air temperature T_{ia} within building zone 310 when MPC systems 300-400 are used to provide temperature control. During the off-peak period 1506, building zone 310 is pre-cooled to the minimum comfortable zone temperature T_{min} (e.g., 70° F.). The zone air temperature T_{ia} remains relatively constant during the off-peak period 1506, but the temperature of the building mass T_m decreases as heat is removed from building zone 310. During the peak period 1508, the building zone temperature T_{ia} is allowed to increase to the maximum comfortable zone temperature T_{max} (e.g., 76° F.). Heat from the air within building zone 310 flows into the building mass, which provides cooling for the air within building zone 310 and increases the temperature of the building mass T_m .

System Architecture

[0291] Referring now to FIG. 19, a block diagram of a system 1900 is shown, according to an exemplary embodiment. The system 1900 provides for multiple smart thermostats 100 (and corresponding sets of HVAC equipment 308) to be enabled with model predictive control functionality via communication with a shared model predictive control server 1902. The system 1900 of FIG. 19 is shown to include multiple sets of HVAC equipment 308, multiple smart thermostats 100, a provider computing system 1904, multiple user mobile devices 1906, a utility provider computing system 1908, and a weather service 1910. The provider

computing system **1904** is shown to include a model predictive controller **1902**, a security and authorization system **1912**, a model predictive control (MPC) application programming interface (API) **1914** and a thermostat API **1916**.

[0292] As shown in FIG. 19, the system **1900** includes multiple smart thermostats **100**. Each smart thermostat **100** may be associated with (e.g., installed at) a different space, for example a different house, commercial building, building zone (e.g., different apartment or office in a shared building), etc. In the example shown, each smart thermostat **100** is communicable with and configured to control a set of HVAC equipment **308** which serves the space associated with the smart thermostat **100**. For example, in some embodiments each smart thermostat **100** is located in and associated with a separate house and is configured to control the HVAC equipment **308** (home furnace, air conditioning system, etc.) for that house. Each smart thermostat **100** may be associated with a different end user (e.g., owners of the various houses). In various implementations, the system **1900** may thereby serve many (e.g., several, dozens, hundreds, thousands, etc.) spaces (e.g., homes, apartments, condos, buildings, offices, retail spaces, etc.) which, in some cases, are owned, leased, and/or occupied by many different people.

[0293] The system **1900** is also shown to include a provider computing system **1904**. The provider computing system **1904** may be implemented using servers, cloud-based computing resources, and/or other computing platforms. The provider computing system **1904** may be associated with a provider of smart thermostat services. The provider computing system **1904** is communicable with the smart thermostats **100** via a network (e.g., network **304** as described with reference to FIG. 3). The provider computing system **1904** is also shown as being communicable with multiple user mobile devices **1906**, a utility provider computing system **1908**, and a weather service **1910**, for example via the network **304**.

[0294] The thermostat API **1916** facilitates interoperability of the smart thermostats **100** with the provider computing system **1904** (e.g., the MPC API **1914** and the security and authorization system **1912**), the user mobile devices **1906**, and, in some embodiments, the utility provider computing system **1908**. The MPC API **1914** facilitates interoperability of the model predictive controller **1902** with other elements of the provider computing system **1904** (e.g., the security and authorization system **1912** and the thermostat API **1916**), the weather service **1910**, and the utility provider computing system **1908**.

[0295] The security and authorization system **1912** is configured to manage access of the multiple smart thermostats **100** and user mobile devices **1906** to the features provided by the provider computing system **1904**. The security and authorization system **1912** is configured to provide a tokenized authentication process. For example, it may be the case that only a subset of the smart thermostats **100** are authorized (e.g., due to subscription models, regulatory compliance issues, etc.) to interoperate with the model predictive controller **1902**. In such an example, the security and authorization system **1912** may receive a user name and password combination from a user mobile device **1906** and/or from a smart thermostat **100**, and may use the user name and password combination to verify authorization of that user and the corresponding smart thermostat **100** to access the features of the model predictive controller **1902**. As another example, the security and authorization system

1912 may be configured to facilitate secure (e.g., encrypted and authenticated) communications between the smart thermostats **100** and the model predictive controller **1902** by applying a data authentication process and/or encryption/decryption process at the MPC API **1914** and thermostat API **1916**.

[0296] The weather service **1910** is configured to provide weather reports and/or forecasts to the MPC API **1914**. The weather service **1910** may be a governmental or non-governmental meteorological system capable of providing weather data corresponding to the location(s) of the smart thermostats **100**. In some cases, the smart thermostats **100**, the HVAC equipment **308**, and the spaces served thereby may be located at geographically dispersed locations such that the weather is different at the various spaces. In such cases, the weather data provided by the weather service **1910** may include a set of data corresponding to each geographic location associated with a smart thermostat **100**. The geographic locations of the smart thermostats **100** may be automatically determined, for example using IP addresses associated with the smart thermostats **100**, or input by a user via a smart thermostat **100** or a user mobile device **1906**. Multiple weather services **1910** may be used to provide sufficient geographic coverage.

[0297] The system **1900** may include one or more utility provider computing systems **1912** associated with the one or more utility providers that may provide utility resources (e.g., electricity, natural gas, etc.) consumed by the HVAC equipment **308**. The utility provider computing system(s) **1912** are configured to provide utility rate information to the provider computing system **1904** (e.g., to the MPC API **1914** and/or the thermostat API **1916**). Utility rates may be time-varying, for example such that the price of a unit of a utility resource (e.g., one kilowatt) may change each hour, each day, every 15 minutes, etc. The utility provider computing systems **1912** may provide the current rate and/or historical or future rates.

[0298] The utility provider computing system(s) **1908** may also provide signals relating to demand response programs, for example incentive-based demand response programs. In one example herein, a utility provider may provide incentives (e.g., monetary credit, redeemable rewards points, prizes, etc.) for customers who agree to reduce energy consumption during peak periods defined by the utility provider, for example by lowering a temperature setpoint by a predefined amount over the peak period or otherwise loosening constraints on inside air temperature over the peak period. In such an example, a utility provider computing system **1908** may be configured transmit an indication of an upcoming peak period to the provider computing system **1904** to allow the provider computing system **1904** to control one or more smart thermostat(s) **100** to participate in demand response program as described in detail below. The MPC API **1914** and/or the thermostat API **1916** may be configured to provide data back to the utility provider computing system(s) **1908**, for example indicating whether or not a particular smart thermostat **100** participated in the demand response program for a particular peak period. The system architecture of FIG. 19 provides for various such electronic communications to facilitate participation in various demand response programs.

[0299] The model predictive controller **1902** is configured as described above for the model predictive controller **302**, adapted to provide model predictive control functionality for

multiple smart thermostats **100**. For example, the model predictive controller **1902** may be configured to execute a separate parameter identification (system identification) process for each building or space served by a smart thermostat **100** enrolled in an MPC mode. The model predictive controller **1902** may thereby obtain, for each smart thermostat **100**, a predictive model of the thermal dynamics of the space served by the corresponding HVAC equipment **308**. The model predictive controller **1902** may then use each predictive model to generate temperature setpoints for the corresponding smart thermostat **100** as described in detail above.

[0300] In some embodiments, the model predictive controller **1902** includes a database in addition to the various elements shown for model predictive controller **302** in FIG. 5. The database may be configured to store, for each smart thermostat **100** served by the model predictive controller, profile information for the smart thermostat **100** (e.g., a network address or other fully-qualified reference, a geographic location, an associated utility provider, etc.), temperature setpoint schedules or ranges, and identified parameters for a thermal model for the space corresponding to the smart thermostat. In some embodiments, the database of the model predictive controller **1902** stores historical data (e.g., loads, weather, rates) for each smart thermostat **100**.

[0301] The model predictive controller **1902** may be configured to pull data from the database and use that data in a model predictive control process, for example as described with reference above to the model predictive controller **302**. In various embodiments, the model predictive controller **1902** is configured to perform model predictive control processes for multiple smart thermostats **100** in series or in parallel. In either case, providing model predictive control functionality for many smart thermostats **100** via unified remote computing resource (i.e., via the model predictive controller **1902**) may reduce the total amount of computational resources required to provide such functionality, as compared to implementing the model predictive controller **302** separately for all spaces and smart thermostats **100**. Furthermore, complex computations may be shifted to a computing resource remote from the smart thermostat **100** and HVAC equipment **308**, thereby reducing the hardware requirements at the smart thermostat **100**.

[0302] The architecture of system **1900** may also facilitate updates and improvements to the model predictive control process over time via software updates to the model predictive controller **1902**. Additionally, the architecture of system **1900** may facilitate learning based on aggregation of thermal models for many spaces. For example, in some embodiments, the model predictive controller **1902** can use the multiple models to improve the efficiency of model training/identification processes, for example by using a model from a similar space to provide initial guesses for parameters of the predictive model for a new space.

Training Data Collection for Smart Thermostats

[0303] Referring now to FIGS. 20-24, a process for collecting building data for use identifying predictive models for a use in model predictive control is illustrated. FIG. 20 shows a flowchart of a process **2000** for obtaining training data, according to an exemplary embodiment. FIGS. 21-24 illustrate user interfaces that may be presented on a smart thermostat **100** to facilitate process **2000**, according to exemplary embodiments. Process **2000** may be executed by the system **1900** of FIG. 19, the system **300** of FIG. 3, or the

system **400** of FIG. 4 in various embodiments, and the present disclosure contemplates various distributions of the functions of process **2000** across the computing resources of the system **1900** of FIG. 19, the system **300** of FIG. 3, or the system **400** of FIG. 4.

[0304] At step **2002** of process **2000**, a user is prompted to select an option to enable a model predictive control (MPC) mode. For example, the smart thermostat **100** may display a graphical user interface that includes a button, toggle, etc. selectable by a user to enable a model predictive control mode. FIG. 21 shows an example of such a user interface, according to some embodiments. As shown in FIG. 21, the display screen **102** of the smart thermostat **100** displays text **2102** explaining an energy aid (MPC) mode to a user and a toggle **2100**. The toggle **2100** is selectable by a user to enable (or disable) an MPC mode (e.g., by touching the toggle **2100** on the display screen **102**). In some embodiments, the user may be prompted to select an option to enable the MPC mode via a user mobile device **1906**.

[0305] In response to a user selecting the option to enable the MPC mode (e.g., via the smart thermostat **100**, at step **2004** a determination is made regarding whether sufficient historical training data are available for model identification. For example, in some embodiments, building and weather data (e.g., indoor and outdoor air temperatures, temperature setpoints, equipment operating points, etc.) are collected and stored during normal operation of the smart thermostat **100**. For example, the data may be stored within a memory device of the smart thermostat **100**, at a model predictive controller **302**, at a model predictive controller **1906**, or in some other computer memory resource. To determine whether the historical training data is sufficient at step **2004**, in some embodiments the amount of available historical training data is assessed by determining whether a number of time steps or a duration of time represented in the historical training data exceeds a threshold value. In other embodiments, to determine whether the historical training data is sufficient at step **2004**, a system identification process is executed and the prediction error of the identified model is compared to a maximum acceptable error (e.g., a one-step ahead or multi-step ahead prediction error), such that the training data is considered insufficient if the resulting model does not generate predictions within the acceptable error.

[0306] If the sufficient historical training data are determined to be available at step **2004**, process **2000** proceeds to step **2016** wherein a system model is identified using the historical training data, for example as described in detail above. If sufficient historical training data are determined to not be available, process **2000** proceeds to step **2006** where the user is prompted for permission to initiate a system identification experiment. FIG. 22 shows an example of a user interface on the display device **102** of the smart thermostat **100** that may be provided at step **2006**. As shown in FIG. 22, text **2200** provides information regarding the system identification experiment, for example to inform the user that a test is needed to allow energy and cost savings to be optimized and that the user will lose control of the temperature setpoint during the test. The display **102** is also shown as providing a button **2202** selectable by the user to choose to start the system identification experiment at a present time. In some embodiments, the smart thermostat **100** can provide the user with an option to schedule the system identification experiment for a future time.

[0307] In response to a user selection to initiate a system identification experiment, the process 2000 proceeds to step 2008, step 2010, and step 2012, which may be executed substantially simultaneously. At step 2008, the system identification experiment is executed by varying temperature setpoints over the duration of a system identification experiment to induce a dynamic response of indoor air temperature. For example, setpoints may be varied over a duration of the system identification experiment using an algorithm designed to increase the amount of robust training data acquired over the time period, for example as described in U.S. patent application Ser. No. 16/516,928 filed Jun. 19, 2019, the entire disclosure of which is incorporated by reference herein. In some embodiments, the duration of the system identification experiment is twenty-four hours (i.e., one day), allowing for capture of system dynamics across a full day. In other embodiments, a different duration is used (e.g., twelve hours, two days, one week, one month, etc.).

[0308] At step 2010, during execution of step 2008, an indication is provided to the user that the system identification experiment is ongoing. FIG. 23 shows an example user interface that can be displayed at step 2010, in which a test indicator 2300 is provided on the display 102 of the smart thermostat 100 to inform a user that the system identification experiment is ongoing. As shown in FIG. 23, the display 102 also shows a current indoor air temperature 2302, a heating setpoint 2304, and a cooling setpoint 2306. The heating setpoint 2304 and/or the cooling setpoint 2306 may show the current setpoint being applied by the system identification experiment executed at step 2008. For example, as shown in FIG. 23, a cooling setpoint of seventy degrees is less than an indoor air temperature of seventy two degrees, such that the setpoint used by the system identification experiment is causing excitation of system dynamics that may drive the indoor air temperature from seventy two degrees towards seventy degrees. By displaying information regarding the ongoing system identification experiment via the display 102, an occupant of a space is allowed to easily see and understand the cause of any temperature fluctuations and/or discomfort caused by the system identification while verifying that the smart thermostat 100 and the HVAC equipment 308 are operating properly.

[0309] At step 2012, a user is prevented from modifying temperature setpoints during execution of the system identification experiment of step 2008. Preventing user modification of setpoints allows for the system identification experiment to be executed consistently and in an automated manner configured to generate robust training data without interruptions caused by user adjustments. FIG. 24 shows a message 2400 that may be provided on the display 102 of the smart thermostat 100 in response to a user's attempt to modify a temperature setpoint during execution of a system identification experiment. The message 2400 informs the user that the system identification experiment is ongoing and that a purpose of the system identification experiment is to enable the smart thermostat 100 to provide energy and cost savings for the user. The message 2400 also includes a timer 2402 that indicates when the system identification experiment will end, after which point the user may be allowed to modify setpoints for the smart thermostat 100. Preventing a user from overriding or prematurely terminating the system identification experiment may improve the reliability of process 2000 and reduce the amount of incomplete/insufficient data stored by the system.

[0310] At step 2014, training data is collected from execution of the system identification experiment (i.e., the execution of step 2008). For example, temperature setpoints, indoor air temperature, weather data, equipment setpoints, etc. may be obtained and saved at the model predictive controller 1902. At step 2016, a system model is identified based on the training data, for example as described above with reference to FIG. 9. In some embodiments, step 2016 is executed using approaches described in U.S. patent application Ser. No. 15/963,324 filed Apr. 13, 2018, U.S. patent application Ser. No. 15/900,459, filed Feb. 20, 2018, U.S. patent application Ser. No. 16/447,724, filed Jun. 20, 2019, and/or U.S. patent application Ser. No. 16/418,715, filed May 21, 2019, all of which are incorporated by reference herein in their entireties. The HVAC equipment 308 can then be controlled based on the identified system model as described in detail above.

Model Predictive Controller with Demand Response

[0311] Referring now to FIG. 25-30, flowcharts and example user interfaces illustrating a demand response mode for the smart thermostat 100 are shown, according to exemplary embodiments. FIGS. 25-30 contemplate a demand response program in which a utility provider offers incentives (e.g., monetary incentives, discounts/credit on utility bills, redeemable rewards points, prizes, etc.) to customers who agree to allow their thermostats to be automatically adjusted during utility-provider-designated peak hours. For example, in some implementations the utility provider may provide an incentive when a cooling setpoint is increased by a designated amount (e.g., four degrees) during a peak period (e.g., mid-afternoon on a hot summer day). In other examples, the utility provider may provide an incentive if some other energy-savings mode, comfort range, etc. is applied by a thermostat during the designated peak hours. With reference to the architecture of FIG. 19, these types of demand response programs are made possible by real-time or ahead-of-time communication from a utility provider computing system 1908 that indicates that the current time is being treated as a peak period or specifying a future time period that will be treated as a peak period.

[0312] FIG. 25 shows a flowchart of a process 2500 for enrolling a thermostat in a demand response program, while FIGS. 26-27 show user interfaces on the display 102 of the smart thermostat 100 that can be used in executing process 2500. Process 2500 may be executed by the system 1900 of FIG. 19, the system 300 of FIG. 3, or the system 400 of FIG. 4 in various embodiments, and the present disclosure contemplates various distributions of the functions of process 2500 across the computing resources of the system 1900 of FIG. 19, the system 300 of FIG. 3, or the system 400 of FIG. 4.

[0313] At step 2502, a particular utility provider associated with a particular smart thermostat 100 is identified. As described with reference to FIG. 19, a provider computing system 1904 may serve many smart thermostats 100 that may be located in different geographic regions and, accordingly, may control HVAC equipment 308 that consumes utility resources supplied by a set of multiple utility providers. Step 2502 may include selecting the utility provider that is relevant for a particular smart thermostat 100 from the set of multiple utility providers. For example, the provider computing system 1904 (e.g., the security and authorization system 1912) may store a mapping between geographic locations (e.g., zip codes, GPS coordinates, city/state/coun-

try names, etc.) and the utility providers that serve those geographic regions. In this example, step **2502** includes determining the geographic location of the smart thermostat **308** (and/or the HVAC equipment **308**), for example based on user input of a geographic location to a smart thermostat **100** or a user mobile device **1906**, GPS data collected by a smart thermostat **100** or a user mobile device **1906**, or based on an IP address associated with a smart thermostat **100** or a user mobile device **1906**. In most instances, a utility provider can be selected based on the geographic location (e.g., due to a monopoly held by the utility provider). In other embodiments, a user is prompted (e.g., via the smart thermostat **100**, via a user mobile device **1906**) to directly select a utility company from a list of utility companies (e.g., an alphabetical list, a text-searchable list).

[**0314**] At step **2504**, a determination is made regarding whether a demand response program is available from the relevant utility provider. For example, a look-up table of utility providers and associated incentive programs may be used to find the available programs based on the utility provider identified at step **2504**. In some embodiments, a query is sent to a utility provider computing system **1908** to prompt the utility provider computing system **1908** for confirmation of whether a demand response program is available for a particular smart thermostat **100**.

[**0315**] If a demand response program is not available from the relevant utility provider (e.g., due to a policy or pricing scheme of the utility provider), at step **2506** an indication is provided to a user that a demand response program is not currently available. For example, a message stating that the user's utility provider does not offer a demand response program may be displayed on the display **102** of the smart thermostat **100**.

[**0316**] If a demand response program is determined to be available for the relevant utility provider at step **2504**, at step **2508** the user is prompted to opt-in to peak hours savings, i.e., to enroll/participate in the demand response program. For example, FIG. **26** shows a selectable toggle **2600** presented on the display **102** of the smart thermostat **100** that can be selected to enable (or disable) a peak savings mode. The user interface of FIG. **26** also shows fields **2602** that provide information regarding temperature bounds that will be used during peak hours.

[**0317**] In response to receiving a user selection of the option to opt-in to peak hours savings, at step **2510** peak savings temperature bounds are obtained which will be applicable during peak hours. For example, as shown in FIG. **27**, a user interface may be presented on the display **102** of the smart thermostat **100** that allows a user to input a comfortable cooling (or heating) range for a space served by the HVAC equipment **308** controlled by the smart thermostat **100**. As shown in FIG. **27**, a temperature selection widget **2702** allows a user to scroll through available numbers (e.g., integers) to select a lower bound and an upper bound on a comfort range. The view in FIG. **27** may be reached by selecting field **2602** in FIG. **26**.

[**0318**] In other embodiments, step **2510** includes obtaining the peak savings temperature bounds by automatically assigning the peak savings temperature bounds based on a policy of the utility provider. For example, in some embodiments, the terms of the demand response program may require use of a utility-provider-defined set of peak savings temperature bounds in exchange for the incentive offered by the demand response program. As another example, the

utility provider may require at least a minimum difference between the temperature setpoint/bounds used during normal operation and peak savings temperature bounds used for participation in the demand response program. In such an example, the peak savings temperature bounds for a particular smart thermostat **100** can be calculated using the differential required by the utility provider and temperature bounds from a normal operating mode or schedule. The temperature bounds obtained at step **2510** may be stored by the model predictive controller **1902** or the mode predictive controller **302**.

[**0319**] A smart thermostat **100** may thereby be enrolled in a demand response program. The smart thermostat **100** can then be controlled to participate in the demand response program, for example as shown in FIG. **28**. FIG. **28** shows a flowchart of a process **2800** for participating in a demand response program is shown, according to an exemplary embodiment. FIGS. **29-30** show example user interfaces that may be shown by the display **102** of the smart thermostat **100** as part of process **2800**. Process **2800** may be executed by the system **1900** of FIG. **19**, the system **300** of FIG. **3**, or the system **400** of FIG. **4** in various embodiments, and the present disclosure contemplates various distributions of the functions of process **2800** across the computing resources of the system **1900** of FIG. **19**, the system **300** of FIG. **3**, or the system **400** of FIG. **4**.

[**0320**] At step **2802**, an indication is received from the utility provider of a peak period. In the example of FIG. **19**, a utility provider computing system **1908** provides an indication (e.g., message, signal, electronic communication, etc.) of a peak period to the MPC API **1914** and/or the thermostat API **1916**. The indication may include details of the timing of the peak period (e.g., start time, end time, duration). In some cases, the peak period indicated by the utility provider computing system **1908** may begin at approximately the same time the indication is transmitted to the provider computing system **1904**. In other cases, the indication may provide information of an upcoming peak period, for example provided one or more hours before the peak period.

[**0321**] At step **2804**, an opportunity is provided for the user of a smart thermostat **100** to opt-out of participating in the demand response program for the particular peak period (i.e., for the peak period indicated in step **2802**). FIG. **29** shows an example of a user interface that may be presented on the display **102** of the smart thermostat **100** to allow the user to opt-out (or opt-in) to participating in the demand response program. In the example shown, the display **102** shows text **2900** describing the peak period (e.g., start and end times) and explaining that the thermostat **100** will be controlled to participate in the demand response program for that peak period if the user does not opt-out. The display **102** also provides an opt-out button **2902** that can be selected to opt-out of the identified peak period, and a continue button **2904** that can be selected to opt-in to the identified peak period. In the embodiment shown, the smart thermostat **100** participates in the demand response program for the peak period if no response from the user is received (e.g., before the beginning of the peak period, after a threshold amount of time). In other embodiments, affirmative opt-in is required for participation in the demand response program for a particular peak period.

[**0322**] In some embodiments, two optimizations are performed over the peak period, including a first optimization

executed using a normal temperature setpoint or comfort range and a second optimization executed using a peak saving temperature bounds. The optimizations may be performed as described in detail above. As a result of the optimizations, two predicted costs of operating the HVAC equipment over the peak period are generated, including a predicted cost of operating the HVAC equipment subject to the normal temperature setpoint/bounds and a predicted cost of operating the HVAC equipment subject to the peak saving temperature bounds. Step **2804** may include comparing the two costs and displaying a predicted savings with the prompt to opt-out of peak savings for the particular peak period. The predicted savings can help a user determine whether to opt-out of the peak savings mode for the particular peak period.

[0323] Step **2806** asks whether an opt-out is received. If an opt-out is received from the user (e.g., if the opt-out button **2902** is selected by a user), then process **2800** proceeds to step **2808** where the HVAC equipment **308** is control as per a normal operational mode. For example, a temperature setpoint or comfort range may remain as defined by a normal operating schedule. In some cases, a model predictive control process operates as described above based on the normal schedule to provide temperature setpoints to the smart thermostat **100**. Accordingly, at step **2808**, the demand response program is not participated in for the peak period identified at step **2802**. In some embodiments, the smart thermostat **100** provides a message via the thermostat API to the utility provider computing system **1908** indicating that smart thermostat **100** is not participating in the demand response program for the identified peak period.

[0324] If the user does not opt-out of the demand response program for the peak period (or an affirmative opt-in is received, for example by selection of continue button **2904**), the process **2800** proceeds to step **2810**. At step **2810**, constraints for the model predictive control process over the peak period are defined using peak savings temperature bounds, for example as obtained in process **2500**. For example, the values of T_{max} , T_{min} and/or $T_{sp,min}$, $T_{sp,max}$ in equations 28-30 above may be defined over the peak period using the peak savings temperature bounds. The model predictive controller **1902** or model predictive controller **302** can thereby update the constraints on the optimization of the objective function to transition from a normal operating mode (in which a first set of values of T_{max} , T_{min} and/or $T_{sp,min}$, $T_{sp,max}$ is used) and the peak savings mode to participate in the demand response program (in which a second set of values of T_{max} , T_{min} and/or $T_{sp,min}$, $T_{sp,max}$ is used).

[0325] As noted above, the values of $T_{max}(k)$, $T_{min}(k)$ and/or $T_{sp,min}(k)$, $T_{sp,max}(k)$ may be time-variant. Accordingly, the constraints may be defined over a time horizon that includes at least a portion of a peak period in addition to time steps outside of a peak period. In a situation where advanced notice of a peak period is obtained (i.e., the indication of step **2802** identifies a future time step as the start of the peak period), each of $T_{max}(k)$, $T_{min}(k)$ and/or $T_{sp,min}(k)$, $T_{sp,max}(k)$ may be defined to have different values before $k=s$ and after $k=s$, where $k=s$ indicates the starting time step of the peak period. As one example, taking an illustrative scenario where the peak temperature bounds are offset from the normal temperature bounds by six degrees, the minimum temperature setpoint may be defined as

$$T_{sp,min}(k) = \begin{cases} 70^\circ \text{ F.}, & k = 1, \dots, s-1 \\ 64^\circ \text{ F.}, & k = s, \dots, h \end{cases}$$

while the maximum temperature setpoint may be defined as

$$T_{sp,max}(k) = \begin{cases} 74^\circ \text{ F.}, & k = 1, \dots, s-1 \\ 80^\circ \text{ F.}, & k = s, \dots, h \end{cases}$$

The constraints can then be used to optimize over a time horizon that includes time steps both within the peak period and outside the peak period (e.g., before the peak period, after the peak period).

[0326] At step **2812**, an optimization is performed subject to the constraints defined at step **2810** to determine temperature setpoints for the smart thermostat over a time horizon that includes at least a portion of a peak period. The optimization and other details of a model predictive control process are described in detail above with reference to at least FIGS. **12-14**. The optimization may be performed over a horizon that includes time steps both inside and outside the peak period, in which case step **2812** facilitates transition from a normal operating mode to the peak savings operating mode. Step **2812** can be performed multiple times (e.g., as explained with reference to step **1420**) as the peak period elapses to generate temperature setpoints for the smart thermostat during the peak period. Step **2812** may be performed by the model predictive controller **1902** or the model predictive controller **302**.

[0327] At step **2814**, equipment (e.g., HVAC equipment **308**) is controlled in accordance with the temperature setpoints generated as a result of the optimization of step **2812**. In the architecture of system **1900** of FIG. **19**, the model predictive controller **1902** may perform an optimization subject to the constraints (at step **2812**) and provide the resulting temperature setpoints to the corresponding smart thermostat **100** via the MPC API **1914** and/or the thermostat API **1916**. The smart thermostat **100** may then control the HVAC equipment **308** associated with the smart thermostat **100** to drive the indoor air temperature at the space to the temperature setpoint identified by the model predictive controller **1902**. For example, equipment commands (e.g., on/off decisions, operating capacities, damper/valve positions, fan speeds) can be generated by smart thermostat **100**, for example using a proportional-integral (PI) controller, a proportional-integral-derivative (PID) controller, or some other approach.

[0328] The HVAC equipment **308** is thereby controlled to participate in the demand response program for a peak period. In some cases, communications between the smart thermostat **100** and the utility provider computing system **1908** are used to confirm that the smart thermostat **100** participated in the demand response program for the peak period and to confirm that the utility provider computing system **1908** recorded the participation and that the incentive will be awarded to the user. In some embodiments, a notification (e.g., via the smart thermostat **100**, via a user mobile device **1906**) is provided to a user congratulating the user on successful participation in the demand response program.

[0329] In some embodiments, for example as shown in FIG. **30**, the smart thermostat **100** may provide a user with an option to opt-out of an ongoing peak period (i.e., to cancel

participation in the demand response program for a current peak period). As shown in FIG. 30, the display 102 of the smart thermostat 100 includes an opt-out button 3000 that a user can select to escape participation. In response to a user selection of opt-out button 3000, the temperature constraints may be reset to those corresponding to a normal schedule and the optimization may be rerun. The user may not receive the incentive associated with the demand response program.

Energy Savings Analytics and Visualization

[0330] Referring now to FIG. 31, a graphical user interface 3100 is shown, according to an exemplary embodiment. The graphical user interface 3100 may be displayed on a user mobile device 1906 and/or the display 102 of a smart thermostat 100. The data visualized on the graphical user interface 3100 may be generated by the model predictive controller 1902 or the model predictive controller 302.

[0331] The graphical user interface 3100 is shown to include a graph 3102 of energy usage versus time. For example, the graph 3102 shows hours used (e.g., total time for which HVAC equipment is on) for each day in the last week. Other units of energy consumption, power consumption, equipment utilization, etc. may be used in other embodiments, as may other time scales. The graph 3102 includes a standard usage line 3104 that represents the amount of energy that would have been consumed by the equipment if controlled under a standard approach, for example if controlled according to a predefined setpoint or schedule. The model predictive controller may be configured to generate the data used to plot the standard usage line 3104. For example, the predictive thermal models described above may be used with the standard predefined setpoint or schedule to estimate energy consumption over the plotted time period.

[0332] The graph 3102 also includes an actual usage line 3106 that shows the actual energy consumption of the equipment, as controlled by the model predictive controller 1902 or model predictive controller 302. The actual usage line 3106 may represent measurements (e.g., energy meter data) or estimates, for example estimates based on control signals (e.g., equipment commands) or optimal temperature setpoints generated by the optimization processes described herein. Icons 3108 may indicate energy-usage-related events, for example peak periods for which the smart thermostat 100 participated in a demand response program.

[0333] By showing the actual usage line 3106 the standard usage line 3104 together on the graph 3102, a user can easily see the benefits provided by the model predictive control processes and system architectures described herein. User trust in the autonomous optimization processes described herein may thereby be substantially increased, increasing user adoption of the technological advantages described herein.

[0334] As can be seen from the graph 3102, substantial reductions in energy usage by the equipment 308 can be achieved using the systems and methods described herein, both on the individual level (e.g., for individual homeowners, residents, or building managers) and in aggregate across many users of many smart thermostats 100 as in FIG. 19. Energy costs savings, improved sustainability, reduced carbon emissions, reduced peak demands on energy grids, etc. are all technical advantages of the present disclosure.

Configuration of Exemplary Embodiments

[0335] The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements can be reversed or otherwise varied and the nature or number of discrete elements or positions can be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps can be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions can be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

[0336] The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure can be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0337] Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps can be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

What is claimed is:

1. A system comprising:

a processor for controlling temperature within a space and provide information for a display, the processor configured to:

define a first temperature range for a normal operating mode;

receive an indication from a source of an occurrence of a peak period;
 provide an indication of the peak period prior to the peak period on the display; and
 operate in a peak savings mode for the peak period and perform preheating or precooling prior to the peak period.

2. The system of claim 1, wherein the processor is configured to:

obtain the first temperature range from user selections.

3. The system of claim 2, wherein the processor is configured provide the indication of the peak period as a time period for the peak period.

4. The system of claim 1, wherein the processor is configured to allow a user to exit the peak savings mode using a user actuated input.

5. The system of claim 1, wherein the peak period defines a second temperature range comprising expanding the first temperature range.

6. The system of claim 5, wherein defining the second temperature range comprises expanding the first temperature range by an amount prescribed by the source.

7. The system of claim 1, wherein the processor is configured to perform an optimization in the normal operating mode.

8. The system of claim 1 wherein the processor is coupled to a wireless interface and the the indication of the occurrence is received from the wireless interface.

9. The system of claim 1 wherein the indication of the peak period comprises a demand response icon.

10. A method of managing temperature in a space, the method comprising:

using first temperature setpoints for a normal operating mode;

receiving an indication from a source of an occurrence of a peak period; and

operating in a peak savings mode for the peak period and performing preheating or precooling prior to the peak period.

11. The method of claim 10, further comprising:

providing an indication of the peak period prior to the peak period on a display for a user.

12. The method of claim 11, further comprising:

defining a temperature range for the by prompting the user to input a maximum and minimum temperature for the peak savings mode.

13. The method of claim 12, wherein defining the temperature range comprises expanding another temperature range provided for the normal operating mode.

14. The method of claim 11, wherein the display is part of a smart thermostat.

15. The method of claim 14, further comprising:

exiting the peak savings mode in response to a user actuated input on the smart thermostat.

16. The method of claim 11, wherein the indication of the occurrence is received wirelessly.

17. The method of claim 11, wherein prompting the user to opt-out of applying the peak savings mode for the peak period comprises providing an option to opt-out on a graphical user interface of a smartphone.

18. A thermostat for equipment that operate to affect a temperature of a space, the thermostat comprising:

a display;

a processing circuit configured to:

define a first temperature setpoints for a normal operating mode;

receive an indication from a source of an occurrence of a peak period;

provide an indication of the peak period prior to the peak period on the display; and

operate in a peak savings mode for the peak period and perform preheating or precooling prior to the peak period.

19. The thermostat of claim 18, wherein the processing circuit is configured to wirelessly receive the indication of the occurrence.

20. The thermostat of claim 18, wherein the processing circuit is configured to wirelessly receive weather information and in the normal operating mode is configured to define the first temperature setpoints in response to the weather information.

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